Thermal Hall Effect in Topological Superconductor

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Abstract

We illustrate in [1] that the anomalous Hall conductivity does not jump discontinuously at the phase transition between the mother metal phase and the superconducting phase, and it offers us an experimentally accessible way to track the topological superconductivity. However, this quantity can not be directly measured in the superconducting phase, while the thermal Hall conductivity, which is proportional to the quasi-particle anomalous Hall conductivity, can be measured in both metal and superconducting phases. We link this quantity with the deformation process [1].

Contents

I. Introduction 3

II. Theoretical Framework of Electro-Thermal Effect 4

A. Static Linear Response 4

B. Time-dependent Linear Response/Kubo Formula 6

C. A Basic of Quantum Thermodynamics 13

D. Fourier Transform of Bloch States 15

E. Sommerfeld Expansion 16

F. Thermodynamics of Magnetizations 18

G. Linear Transport Response 27

H. Formulation of Current Operators 31

I. Thermal-Hall Coefficients from Microscopic States 36

J. Thermoelectric-Hall Coefficients as a Direct Analogy 50

III. Numerical Results 51

A. 2DEG 51

B. TI Film 53

IV. Discussion and Conclusion 54

References 54

I. Introduction

The electro-thermal effect is a phenomenon where applied either electric field or temperature gradient induces either an electric current or a heat current, thus generating electro, thermal, or thermoelectric effects. Each effect can be either longitudinal or transverse. The framework is established by linear response theory and Onsager relations[2, 3], where the latter ingredient enters through the thermodynamic properties. The two obstacles in the sense of philosophy are:

• To account for thermal response, one needs to translate the temperature gradient into its mechanical equivalent, i.e., a fictitious gravitational field which couples to the energy density, but not the electron mass, of the system[4]. In this way, the thermal current can be well-defined with thermal continuity equation.

• Aside from a usual Kubo formula contribution by current-current correlation, the thermal response requires a subtraction of the circulating (heat) current, the energy magnetization that does not participate in transport and is experimentally unobservable[5–7].

The main result is the formula of thermal Hall conductivity

$$\begin{aligned}\kappa_{xy} &= \omega \frac{1}{hT} \int \mathrm{d}\varepsilon \ (\varepsilon - \mu)^2 N(\varepsilon) f'(\varepsilon) \\ &\approx -\omega \frac{\pi^2 k_B^2}{3h} N(\mu) T,\end{aligned} \tag{1}$$

where the reduced quasi-particle anomalous Hall conductivity is defined as

$$N(\varepsilon) = \frac{2\pi}{V} \sum_{\varepsilon_{nk}<\varepsilon} \Omega_{nk} = \frac{4\pi}{V} \sum_{\varepsilon_{nk}<\varepsilon} \mathrm{Im} \left\langle \partial_{k_x} u_{nk} | \partial_{k_y} u_{nk} \right\rangle , \tag{2}$$

$f(\varepsilon)$ is the Fermi-Dirac distribution, and $\omega = 1$ for the metal phase and $\omega = 1/2$ for the superconducting phase. μ is the chemical potential, and in superconductor $\mu = 0$, giving $N(0)$ the superconducting Chern number. The second line is the low-temperature approximation by Sommerfeld expansion, where k_B is the Boltzmann constant and T is the temperature. The range of application of the expansion is mainly determined by the lowest energy scale of the system, i.e., magnetic or superconducting gap.

II. Theoretical Framework of Electro-Thermal Effect

A. Static Linear Response

We start from introducing the tool of linear response theory by the Kubo canonical correlation function[8]. Let

$$\langle \hat{B}; \hat{A} \rangle = \frac{1}{\beta} \mathrm{Tr}\left(\hat{\rho}_{eq} \int_0^\beta \mathrm{d}\lambda \; e^{\lambda \hat{H}} \hat{B} e^{-\lambda \hat{H}} \hat{A} \right), \tag{3}$$

where the equilibrium density operator is $\hat{\rho}_{eq} = e^{-\beta \hat{H}}/Z$ with $Z = \mathrm{Tr}\left(e^{-\beta \hat{H}}\right)$, we have several properties as

$$\langle \hat{A}; \hat{B} \rangle = \langle \hat{B}; \hat{A} \rangle \,, \tag{4a}$$

$$\beta \langle \dot{\hat{A}}; \hat{B} \rangle = -\beta \langle \hat{A}; \dot{\hat{B}} \rangle = \frac{1}{i\hbar} \langle [\hat{A}, \hat{B}] \rangle_0 \,, \tag{4b}$$

$$\chi_{BA} = \beta \langle \Delta \hat{B}; \Delta \hat{A} \rangle \,. \tag{4c}$$

Here, the susceptibility χ_{BA} is defined as the static linear response of the operator $\hat{B}$ to the perturbation $\hat{A}$ as

$$\hat{H}' = \hat{H} - \hat{A}\delta x, \; \delta \langle B \rangle = \langle \hat{B} \rangle_{\delta x} - \langle \hat{B} \rangle_0 \equiv \chi_{BA} \delta x, \tag{5}$$

where $\langle \rangle_0$ and $\langle \rangle_{\delta x}$ are the thermal expectation values in the unperturbed and perturbed Hamiltonian, respectively. Note $\Delta \hat{O} = \hat{O} - \langle O \rangle_0$ is the fluctuation of the operator $\hat{O}$. The proof of the first two follows mainly from the cyclic property of trace as

$$\begin{aligned}
\langle \hat{A}; \hat{B} \rangle &= \frac{1}{\beta} \mathrm{Tr}\left(\hat{\rho}_{eq} \int_0^\beta \mathrm{d}\lambda \; e^{\lambda \hat{H}} \hat{A} e^{-\lambda \hat{H}} \hat{B} \right) \\
&= \frac{1}{\beta Z} \mathrm{Tr}\left(\int_0^\beta \mathrm{d}\lambda \; e^{-(\beta-\lambda)\hat{H}} \hat{A} e^{-\lambda \hat{H}} \hat{B} \right) \\
&\overset{s=\beta-\lambda}{=\!=\!=} \frac{1}{\beta Z} \mathrm{Tr}\left(e^{-\beta \hat{H}} \int_0^\beta \mathrm{d}x \; e^{s\hat{H}} \hat{B} e^{-s\hat{H}} \hat{A} \right) \\
&= \langle \hat{B}; \hat{A} \rangle \,.
\end{aligned}$$

$$
\begin{aligned}
\beta \langle \dot{\hat{A}}; \hat{B} \rangle &= \mathrm{Tr}\left(\hat{\rho}_{eq} \int_0^\beta \mathrm{d}\lambda \; e^{\lambda \hat{H}} \dot{\hat{A}} e^{-\lambda \hat{H}} \hat{B} \right) \\
&= \mathrm{Tr}\left(\hat{\rho}_{eq} \int_0^\beta \mathrm{d}\lambda \; e^{\lambda \hat{H}} \frac{1}{i\hbar} [\hat{A}, \hat{H}] e^{-\lambda \hat{H}} \hat{B} \right) \\
&= \frac{1}{i\hbar} \mathrm{Tr}\left(\hat{\rho}_{eq} \int_0^\beta \mathrm{d}\lambda \; e^{\lambda \hat{H}} (\hat{A}\hat{H} - \hat{H}\hat{A}) e^{-\lambda \hat{H}} \hat{B} \right) \\
&\overset{(1)}{=\!=} \frac{1}{i\hbar} \mathrm{Tr}\left(\hat{\rho}_{eq} \int_0^\beta \mathrm{d}\lambda \; e^{\lambda \hat{H}} \hat{A} e^{-\lambda \hat{H}} (\hat{H}\hat{B} - \hat{B}\hat{H}) \right) \\
&= -\mathrm{Tr}\left(\hat{\rho}_{eq} \int_0^\beta \mathrm{d}\lambda \; e^{\lambda \hat{H}} \hat{A} e^{-\lambda \hat{H}} \frac{1}{i\hbar} [\hat{B}, \hat{H}] \right) \\
&= -\beta \langle \hat{A}; \dot{\hat{B}} \rangle \, ; \\
&\overset{(2)}{=\!=} -\frac{1}{i\hbar} \mathrm{Tr}\left(\hat{\rho}_{eq} \int_0^\beta \mathrm{d}\lambda \; \frac{\mathrm{d}}{\mathrm{d}\lambda} (e^{\lambda \hat{H}} \hat{A} e^{-\lambda \hat{H}} \hat{B}) \right) \\
&= -\frac{1}{i\hbar Z} \mathrm{Tr}\Big(e^{-\beta \hat{H}} (e^{\beta \hat{H}} \hat{A} e^{-\beta \hat{H}} \hat{B} - \hat{A}\hat{B}) \Big) \\
&= -\frac{1}{i\hbar Z} \mathrm{Tr}\Big(e^{-\beta \hat{H}} (\hat{B}\hat{A} - \hat{A}\hat{B}) \Big) \\
&= \frac{1}{i\hbar} \langle [\hat{A}, \hat{B}] \rangle_0 \, .
\end{aligned}
$$

To prove the third, we need to prove one more equality by noticing that

$$
\begin{aligned}
& \frac{\mathrm{d}}{\mathrm{d}s} \left(e^{-s\hat{a}} e^{s(\hat{a}+\hat{b})} \right) = e^{-s\hat{a}} \hat{b} e^{s(\hat{a}+\hat{b})} \\
\overset{\int}{\Longrightarrow} \quad & e^{-\beta \hat{a}} e^{\beta(\hat{a}+\hat{b})} - 1 \;\; = \int_0^\beta \mathrm{d}s \; e^{-s\hat{a}} \hat{b} e^{s(\hat{a}+\hat{b})} \\
\Longrightarrow \quad & e^{\beta(\hat{a}+\hat{b})} \qquad\qquad\;\; = e^{\beta \hat{a}} \left(1 + \int_0^\beta \mathrm{d}s \; e^{-s\hat{a}} \hat{b} e^{s(\hat{a}+\hat{b})} \right) .
\end{aligned}
$$

With this, we realize that the Boltzmann factor can be written as

$$
\begin{aligned}
e^{-\beta(\hat{H} - \hat{A}\delta x)} &= e^{-\beta \hat{H}} \left(1 + \int_0^\beta \mathrm{d}\lambda \; e^{\lambda \hat{H}} \hat{A} e^{-\lambda(\hat{H} - \hat{A}\delta x)} \delta x \right) \\
&= e^{-\beta \hat{H}} \left[1 + \int_0^\beta \mathrm{d}\lambda \; e^{\lambda \hat{H}} \hat{A} e^{-\lambda \hat{H}} \left(1 + \int_0^\lambda \mathrm{d}\lambda' \; e^{\lambda' \hat{H}} \hat{A} e^{-\lambda'(\hat{H} - \hat{A}\delta x)} \delta x \right) \delta x \right] \\
&= e^{-\beta \hat{H}} \left(1 + \int_0^\beta \mathrm{d}\lambda \; e^{\lambda \hat{H}} \hat{A} e^{-\lambda \hat{H}} \delta x + O(\delta x^2) \right) ,
\end{aligned}
$$

where one recognizes the integral solution of the Boltzmann factor as a perturbation expansion in δx. This form also appears in the time-dependent perturbation theory in quantum mechanics by Dyson series, and here, the imaginary time λ plays the role of time.

In this way, we can expand the perturbed density operator as

$$\begin{aligned}
\hat{\rho}(\delta x) &= \frac{1}{Z'} e^{-\beta \hat{H}'} = \frac{e^{-\beta(\hat{H}-\hat{A}\delta x)}}{\mathrm{Tr}\Big(e^{-\beta(\hat{H}-\hat{A}\delta x)}\Big)} \\
&= \frac{e^{-\beta\hat{H}}\left(1+\int_0^\beta \mathrm{d}\lambda\ e^{\lambda\hat{H}}\hat{A}e^{-\lambda\hat{H}}\delta x + O(\delta x^2)\right)}{\mathrm{Tr}\Big(e^{-\beta\hat{H}} + e^{-\beta\hat{H}}\int_0^\beta \mathrm{d}\lambda\ e^{\lambda\hat{H}}\hat{A}e^{-\lambda\hat{H}}\delta x + O(\delta x^2)\Big)} \\
&= \frac{e^{-\beta\hat{H}}\left(1+\int_0^\beta \mathrm{d}\lambda\ e^{\lambda\hat{H}}\hat{A}e^{-\lambda\hat{H}}\delta x + O(\delta x^2)\right)}{Z(1+\beta\left\langle A\right\rangle_0\delta x) + O(\delta x^2)} \\
&= \frac{1}{Z}e^{-\beta\hat{H}}\left(1+\int_0^\beta \mathrm{d}\lambda\ e^{\lambda\hat{H}}(\hat{A}-\left\langle\hat{A}\right\rangle_0)e^{-\lambda\hat{H}}\delta x + O(\delta x^2)\right) \\
&\approx \hat{\rho}_{eq}\left(1+\int_0^\beta \mathrm{d}\lambda\ e^{\lambda\hat{H}}\Delta\hat{A}e^{-\lambda\hat{H}}\delta x\right),
\end{aligned}$$

where we have used the Taylor expansion of $\frac{1}{1+\epsilon} = 1-\epsilon+O(\epsilon^2)$. Thus, the change of the expectation value of the operator $\hat{B}$ is

$$\begin{aligned}
\delta\left\langle\hat{B}\right\rangle &= \left\langle\hat{B}\right\rangle_{\delta x} - \left\langle\hat{B}\right\rangle_0 = \mathrm{Tr}\Big(\hat{\rho}(\delta x)\hat{B}\Big) - \mathrm{Tr}\Big(\hat{\rho}_{eq}\hat{B}\Big) \\
&= \mathrm{Tr}\left(\hat{\rho}_{eq}\int_0^\beta \mathrm{d}\lambda\ e^{\lambda\hat{H}}\Delta\hat{A}e^{-\lambda\hat{H}}\hat{B}\delta x\right) \\
&= \mathrm{Tr}\left(\hat{\rho}_{eq}\int_0^\beta \mathrm{d}\lambda\ e^{\lambda\hat{H}}\Delta\hat{A}e^{-\lambda\hat{H}}\Delta\hat{B}\right)\delta x \\
&= \beta\left\langle\Delta\hat{A};\Delta\hat{B}\right\rangle\delta x = \beta\left\langle\Delta\hat{B};\Delta\hat{A}\right\rangle\delta x,
\end{aligned}$$

where one notices that $\langle\Delta\hat{A};\langle\hat{B}\rangle_0\rangle = 0$ so we insert one. The susceptibility then reads $\chi_{BA} = \beta\langle\Delta\hat{B};\Delta\hat{A}\rangle$.

The set of deduction clearly holds when we replace $\hat{H}\to\hat{K} = \hat{H}-\mu\hat{N}$ for grand canonical ensemble.

B. Time-dependent Linear Response/Kubo Formula

In the time-dependent linear response theory, we consider the perturbation of the Hamiltonian as a time-dependent perturbation that is switched on adiabatically. One can use the standard Kubo formula[8], by considering the time-evolution under the interacting picture thus performing the average over the equilibrium state. Here, since our derivation utilizes the density matrix operator, we shall focus instead on the time-dependent perturbation of the density operator and thermal average[9].

Let the Hamiltonian be $\hat{H}(t) = \hat{H}_0 + \hat{H}'(t)$, and the response of current operator should be

$$\delta \boldsymbol{J} = \langle \hat{\boldsymbol{J}} \rangle (t) - \langle \hat{\boldsymbol{J}} \rangle_0 = \mathrm{Tr}\Big(\hat{\rho}(t)\hat{\boldsymbol{J}}\Big) - \mathrm{Tr}\Big(\hat{\rho}^{eq}\hat{\boldsymbol{J}}\Big), \tag{6}$$

where we know clearly that $\hat{\rho}^{eq} = e^{-\beta \hat{H}_0}/Z$ (or in grand canonical ensemble). However, we can not write $\hat{\rho}(t)$ directly in a similar way since the system is not in equilibrium. Instead, we need to consider the full time-evolution of the density operator in the Schrödinger picture as

$$i\hbar \partial_t \hat{\rho}(t) = [\hat{H}(t), \hat{\rho}(t)], \tag{7}$$

with boundary condition $\hat{\rho}(-\infty) = \hat{\rho}^{eq}$. One can imagine that the equation is solved by an expansion in the perturbation $\hat{H}'(t)$ and leads to orders of responses, and for our purpose, we let $\hat{\rho}(t) = \hat{\rho}^{eq} + \hat{\rho}_1(t)$ with $\hat{\rho}_1(t) \sim O(\hat{H}')$, $\hat{\rho}_1(-\infty) = 0$, which leads to the EoM as

$$\begin{aligned} i\hbar \partial_t \hat{\rho}_1(t) &= [\hat{H}_0 + \hat{H}'(t), \hat{\rho}^{eq} + \hat{\rho}_1(t)] \\ &= [\hat{H}_0, \hat{\rho}^{eq}] + [\hat{H}_0, \hat{\rho}_1(t)] + [\hat{H}'(t), \hat{\rho}^{eq}] + [\hat{H}'(t), \hat{\rho}_1(t)] \\ &= [\hat{H}_0, \hat{\rho}_1(t)] + [\hat{H}'(t), \hat{\rho}^{eq}] + O(\hat{H}'^2). \end{aligned} \tag{8}$$

This leads to

$$\begin{aligned} & & i\hbar \partial_t \hat{\rho}_1(t) - [\hat{H}_0, \hat{\rho}_1(t)] &= [\hat{H}'(t), \hat{\rho}^{eq}] \\ \Longrightarrow \quad & e^{-i\hat{H}_0 t/\hbar} \left(i\hbar \frac{\mathrm{d}}{\mathrm{d}t} \left(e^{i\hat{H}_0 t/\hbar} \hat{\rho}_1(t) e^{-i\hat{H}_0 t/\hbar} \right) \right) e^{i\hat{H}_0 t/\hbar} &= [\hat{H}'(t), \hat{\rho}^{eq}] \\ \Longrightarrow \quad & i\hbar \frac{\mathrm{d}}{\mathrm{d}t} \left(e^{i\hat{H}_0 t/\hbar} \hat{\rho}_1(t) e^{-i\hat{H}_0 t/\hbar} \right) &= e^{i\hat{H}_0 t/\hbar} [\hat{H}'(t), \hat{\rho}^{eq}] e^{-i\hat{H}_0 t/\hbar} \\ & & &= [\hat{H}'_I(t), \hat{\rho}^{eq}], \end{aligned}$$

and by a further integration, we have

$$\hat{\rho}_1(t) = \frac{1}{i\hbar} e^{-i\hat{H}_0 t/\hbar} \left(\int_{-\infty}^{t} \mathrm{d}t' \ [\hat{H}'_I(t'), \hat{\rho^{eq}}] \right) e^{i\hat{H}_0 t/\hbar}, \tag{9}$$

where the boundary condition of first order perturbation $\hat{\rho}_1(-\infty) = 0$ is used. We also use lower index I for the interacting picture.

By assuming the equilibrium state is a global equilibrium state without local temperature or chemical potential gradient, we can write the expectation value of the local current

operator as

$$
\begin{aligned}
\langle \hat{\boldsymbol{J}} \rangle (\boldsymbol{r}, t) &= \mathrm{Tr}\Big(\hat{\rho}(t)\hat{\boldsymbol{J}}(\boldsymbol{r})\Big) = \mathrm{Tr}\left(\hat{\rho}_1(t)\hat{\boldsymbol{J}}(\boldsymbol{r})\right) \\
&= \frac{1}{i\hbar}\,\mathrm{Tr}\left(e^{-i\hat{H}_0 t/\hbar}\left(\int_{-\infty}^{t} \mathrm{d}t'\ [\hat{H}'_I(t'), \hat{\rho^{eq}}]\right) e^{i\hat{H}_0 t/\hbar}\hat{\boldsymbol{J}}(\boldsymbol{r})\right) \\
&= \frac{1}{i\hbar}\,\mathrm{Tr}\left(\int_{-\infty}^{t} \mathrm{d}t'\ [\hat{H}'_I(t'), \hat{\rho^{eq}}]\hat{\boldsymbol{J}}_I(\boldsymbol{r}, t)\right) \\
&= \frac{1}{i\hbar}\,\mathrm{Tr}\left(\int_{-\infty}^{t} \mathrm{d}t'\ [\hat{H}'_I(t')\hat{\rho^{eq}}\hat{\boldsymbol{J}}_I(\boldsymbol{r}, t) - \hat{\rho^{eq}}\hat{H}'_I(t')\hat{\boldsymbol{J}}_I(\boldsymbol{r}, t)]\right) \\
&= \frac{1}{i\hbar}\,\mathrm{Tr}\left(\int_{-\infty}^{t} \mathrm{d}t'\ \hat{\rho^{eq}}[\hat{\boldsymbol{J}}_I(\boldsymbol{r}, t), \hat{H}'_I(t')]\right) \\
&= \frac{1}{i\hbar}\int_{-\infty}^{t} \mathrm{d}t'\ \langle[\hat{\boldsymbol{J}}_I(\boldsymbol{r}, t), \hat{H}'_I(t')]\rangle_0 = \frac{1}{i\hbar}\int_{-\infty}^{t} \mathrm{d}t'\ \langle[\hat{\boldsymbol{J}}_I(\boldsymbol{r}, t-t'), \hat{H}'(t')]\rangle_0 \\
&\overset{s=t-t'}{=\!=\!=} \frac{1}{i\hbar}\int_{0}^{+\infty} \mathrm{d}s\ \langle[\hat{\boldsymbol{J}}_I(\boldsymbol{r}, s), \hat{H}'(t-s)]\rangle_0
\end{aligned}
\tag{10}
$$

as exactly the Kubo formula. Note that if the system was not in global equilibrium, then an extra static response is expected in the local equilibrium state for the local current operator counting for the spatially inhomogeneous temperature or chemical potential.

The above form is certainly correct, but when we want to evaluate the DC ($\omega = 0$) case, we need to take the AC calculation and take the limit $\omega \to 0$. If we want to avoid this, we may use some strategy. Say now we consider perturbation

$$
\hat{H}'(t) = e^{\epsilon t}\int \mathrm{d}\boldsymbol{r}\ \psi(\boldsymbol{r})\hat{n}(\boldsymbol{r}), \tag{11}
$$

with ψ being a static potential generated by a homogeneous external field, $\hat{n}$ the corresponding density operator satisfying the continuity equation

$$
\partial_t \hat{n}(\boldsymbol{r}, t) + \nabla \cdot \hat{\boldsymbol{J}}(\boldsymbol{r}, t) = 0, \tag{12}
$$

and ϵ a small positive number counting for the adiabatic switching on of the perturbation. By denoting $\hat{H}'(t) = e^{\epsilon t}\hat{F}$, and decomposing the density matrix as $\hat{\rho}(t) = \hat{\rho}^{eq} + e^{\epsilon t}\hat{\rho}_1(t)$, we similarly obtain

$$
i\hbar\partial_t\left(\hat{U}_0^\dagger(t)e^{\epsilon t}\hat{\rho}_1(t)\hat{U}_0(t)\right) = \hat{U}_0^\dagger(t)[\hat{H}'(t), \hat{\rho}^{eq}]\hat{U}_0(t) = e^{\epsilon t}[\hat{F}_I(t), \hat{\rho}^{eq}],
$$

which leads to

$$\begin{aligned}\hat{\rho}_1(t) &= \frac{1}{i\hbar}\int_{-\infty}^{t} \mathrm{d}t'\ e^{\epsilon(t'-t)}\hat{U}_0(t)[\hat{F}_I(t'),\hat{\rho}^{eq}]\hat{U}_0^{\dagger}(t)\\ &= \frac{1}{i\hbar}\int_{-\infty}^{t} \mathrm{d}t'\ e^{\epsilon(t'-t)}[\hat{F}_I(t'-t),\hat{\rho}^{eq}]\\ &\overset{s=t-t'}{=\!=\!=} \frac{1}{i\hbar}\int_{0}^{+\infty} \mathrm{d}s\ e^{-\epsilon s}[\hat{F}_I(-s),\hat{\rho}^{eq}].\end{aligned} \tag{13}$$

Now we introduce

$$\begin{aligned}& i\hat{\rho}^{eq}\int_0^{\beta}\mathrm{d}\beta'\ \frac{\partial}{\partial t}\hat{F}_I(-t-i\beta'\hbar) = \frac{1}{\hbar}\hat{\rho}^{eq}\int_0^{\beta}\mathrm{d}\beta'\ \frac{\partial}{\partial\beta'}\hat{F}_I(-t-i\beta'\hbar)\\ =&\frac{1}{\hbar}\hat{\rho}^{eq}\hat{F}_I(-t-i\beta'\hbar)\big|_0^{\beta} = \frac{1}{\hbar}\hat{\rho}^{eq}\left[\hat{F}_I(-t-i\beta\hbar)-\hat{F}_I(-t)\right]\\ =&\frac{1}{\hbar}\hat{\rho}^{eq}\left[e^{\beta\hat{H}_0}\hat{F}_I(-t)e^{-\beta\hat{H}_0}-\hat{F}_I(-t)\right] = \frac{1}{\hbar}\left[\hat{F}_I(-t)\hat{\rho}^{eq}-\hat{\rho}^{eq}\hat{F}_I(-t)\right]\\ =&\frac{1}{\hbar}[\hat{F}_I(-t),\hat{\rho}^{eq}],\end{aligned}$$

which brings us to the expression of the first order perturbation as

$$\hat{\rho}_1(t) = \int_0^{+\infty}\mathrm{d}t\ e^{-\epsilon t}\hat{\rho}^{eq}\int_0^{\beta}\mathrm{d}\beta'\ \frac{\partial}{\partial t}\hat{F}_I(-t-i\beta'\hbar). \tag{14}$$

Note that we can ignore the interaction picture index I and treat operator in Heisenberg picture generated by the unperturbed Hamiltonian $\hat{H}_0$. Now since

$$\begin{aligned}\frac{\partial}{\partial t}\hat{F}(-t-i\beta'\hbar) &= \int \mathrm{d}\boldsymbol{r}\ \psi(\boldsymbol{r})\partial_t\hat{n}(\boldsymbol{r},-t-i\beta'\hbar)\\ &= \int \mathrm{d}\boldsymbol{r}\ \psi(\boldsymbol{r})\nabla\cdot\hat{\boldsymbol{J}}(\boldsymbol{r},-t-i\beta'\hbar)\\ &\overset{\text{IBP}}{=\!=} -\int \mathrm{d}\boldsymbol{r}\ (\nabla\psi(\boldsymbol{r}))\cdot\hat{\boldsymbol{J}}(\boldsymbol{r},-t-i\beta'\hbar)\\ &= \hat{\boldsymbol{j}}(-t-i\beta'\hbar)\cdot\boldsymbol{F},\end{aligned}$$

where a constant force $\boldsymbol{F}=-\nabla\psi$ is introduced, we can write the first order perturbation of current as

$$\begin{aligned}\boldsymbol{J}(\boldsymbol{r}) &= \langle\hat{\boldsymbol{J}}(\boldsymbol{r})\rangle = \mathrm{Tr}\Big(\hat{\rho}_1\hat{\boldsymbol{J}}(\boldsymbol{r})\Big)\\ &= \int_0^{+\infty}\mathrm{d}t\ e^{-\epsilon t}\,\mathrm{Tr}\left(\hat{\rho}^{eq}\int_0^{\beta}\mathrm{d}\beta'\ \hat{j}_{\alpha}(-t-i\beta'\hbar)\hat{\boldsymbol{J}}(\boldsymbol{r})\right)F_{\alpha},\end{aligned} \tag{15}$$

and we thus obtain the linear response of the current to the external field as

$$\begin{aligned}L_{\alpha\beta} &= \frac{1}{V}\frac{j_{\alpha}}{F_{\beta}} = \frac{1}{V}\int\mathrm{d}\boldsymbol{r}\ \frac{J_{\alpha}(\boldsymbol{r})}{F_{\beta}}\\ &= \frac{1}{V}\int_0^{+\infty}\mathrm{d}t\ e^{-\epsilon t}\,\mathrm{Tr}\left(\hat{\rho}^{eq}\int_0^{\beta}\mathrm{d}\beta'\ \hat{j}_{\beta}(-t-i\beta'\hbar)\hat{j}_{\alpha}\right).\end{aligned} \tag{16}$$

How do we actually perform the calculation? Especially say over a non-interacting system with band structure? The calculation can be simplified to precisely over the single particle states. Let us generally denote the states of the Hamiltonian $\hat{H}_0$ as $\hat{H}_0 |\lambda\rangle = E_\lambda |\lambda\rangle$, where one notes that even for a non-interacting system, the states can be a many-body one with multiple particles. The key is to rely on the second-quantization formalism and (a twisted) Wick's theorem. Let us firstly represent the current operator in second quantization as

$$\hat{j}_\alpha = \sum_{mnk} j^\alpha_{mn}(k) c^\dagger_{mk} c_{nk},\ j^\alpha_{mn}(k) = \langle mk|\hat{j}_\alpha|nk\rangle\,, \tag{17}$$

with k wavevector and m band index, and we can write the current-current correlation function as

$$\begin{aligned}
L_{\alpha\beta} &= \frac{1}{V}\int_0^{+\infty} \mathrm{d}t\ e^{-\epsilon t}\,\mathrm{Tr}\Big(\hat{\rho}^{eq}\int_0^\beta \mathrm{d}\beta'\ \hat{j}_\beta(-t-i\beta'\hbar)\hat{j}_\alpha\Big) \\
&= \frac{1}{V}\int_0^{+\infty} \mathrm{d}t\ e^{-\epsilon t}\int_0^\beta \mathrm{d}\beta'\ \sum_{mnk}\sum_{opq} j^\beta_{mn}(k) j^\alpha_{op}(q)\,\langle \hat{c}^\dagger_{mk}(-t-i\beta'\hbar)\hat{c}_{nk}(-t-i\beta'\hbar)\hat{c}^\dagger_{oq}\hat{c}_{pq}\rangle_0 \\
&= \frac{1}{V}\int_0^{+\infty} \mathrm{d}t\ e^{-\epsilon t}\int_0^\beta \mathrm{d}\beta'\ \sum_{mnk}\sum_{opq} j^\beta_{mn}(k) j^\alpha_{op}(q)\,\langle \hat{c}^\dagger_{mk}(-t-i\beta'\hbar)\hat{c}_{nk}(-t-i\beta'\hbar)\hat{c}^\dagger_{oq}\hat{c}_{pq}\rangle_0 \\
&= -\frac{1}{V}\int_0^{+\infty} \mathrm{d}t\ e^{-\epsilon t}\int_0^\beta \mathrm{d}\beta'\ \sum_{mnk}\sum_{opq} j^\beta_{mn}(k) j^\alpha_{op}(q)\,\langle \mathcal{T}\hat{c}_{pq}\hat{c}^\dagger_{mk}(-t-i\beta'\hbar)\rangle_0\,\langle \mathcal{T}\hat{c}_{nk}\hat{c}^\dagger_{oq}(t+i\beta'\hbar)\rangle_0\,,
\end{aligned}$$

where we have used the Wick's theorem to decompose the four-operator correlation function into two two-operator correlation functions. The time ordered two-point correlation function can be written as

$$\langle \mathcal{T} c_{k'} c^\dagger_k(t)\rangle_0 = \begin{cases} -\langle c^\dagger_k(t) c_{k'}\rangle_0\,,\ t>0 \\ \langle c_{k'} c^\dagger_k(t)\rangle_0\,,\ t<0 \end{cases},$$

and similarly. For the partition function $Z = \mathrm{Tr}\Big(e^{-\beta\hat{H}_0}\Big) = \sum_\lambda e^{-\beta E_\lambda}$, and since λ can be a many-body state, we can further write it as

$$Z = \sum_{n=1}^\infty \sum_{m_i k_i} \langle m_1 k_1, \cdots, m_n k_n|\, e^{-\beta\hat{H}_0}\, |m_1 k_1, \cdots, m_n k_n\rangle = \sum_{n=1}^\infty \sum_{m_i k_i} e^{-\beta\sum_{i=1}^n \varepsilon_{m_i k_i}}.$$

A hidden requirement is that $m_i k_i \neq m_j k_j$ for any $i \neq j$, due to the Pauli exclusion principle.

In this way, we can expand the partition function as

$$
\begin{aligned}
Z &= \sum_{n=1}^{\infty} \sum_{m_i k_i} e^{-\beta \sum_{i=1}^{n} \varepsilon_{m_i k_i}} \\
&= \sum_{mk} e^{-\beta \varepsilon_{mk}} + \sum_{m_1 k_1 \neq m_2 k_2} e^{-\beta \varepsilon_{m_1 k_1}} e^{-\beta \varepsilon_{m_2 k_2}} + \sum_{m_1 k_1 \neq m_2 k_2 \neq m_3 k_3} e^{-\beta \varepsilon_{m_1 k_1}} e^{-\beta \varepsilon_{m_2 k_2}} e^{-\beta \varepsilon_{m_3 k_3}} + \cdots \\
&= \prod_{mk} (1 + e^{-\beta \varepsilon_{mk}}),
\end{aligned}
$$

which is just the product of the single particle partition functions. Thus, we separately calculate the positive t contribution as

$$
\begin{aligned}
\langle c_k^\dagger(t) c_{k'} \rangle_0 &= \frac{1}{Z} \operatorname{Tr}\left(c_k^\dagger(t) c_{k'} e^{-\beta H_0} \right) \\
&= \frac{1}{Z} \left(\sum_{k_1} \langle k_1 | \, c_k^\dagger(t) c_{k'} e^{-\beta H_0} \, | k_1 \rangle + \frac{1}{2!} \sum_{k_1 k_2} \langle k_1 k_2 | \, c_k^\dagger(t) c_{k'} e^{-\beta H_0} \, | k_1 k_2 \rangle \cdots \right) \\
&= \frac{1}{Z} \left(\sum_{k_1} e^{i\varepsilon_{k_1} t/\hbar} \delta_{k k_1} \delta_{k' k_1} e^{-\beta \varepsilon_{k_1}} + \frac{1}{2} \sum_{k_1 k_2} e^{i(\varepsilon_{k_1} + \varepsilon_{k_2}) t/\hbar} \left[\langle k_1 | \, \delta_{k k_2} e^{-i\varepsilon_{k_1} t/\hbar} \right. \right. \\
&\quad \left. \left. - \langle k_2 | \, \delta_{k k_2} e^{-i\varepsilon_{k_2} t/\hbar} \right] \left[\delta_{k' k_1} \, | k_2 \rangle - \delta_{k' k_2} \, | k_1 \rangle \right] e^{-\beta(\varepsilon_{k_1} + \varepsilon_{k_2})} + \cdots \right) \\
&= \frac{1}{Z} \left(e^{i\varepsilon_k t/\hbar} e^{-\beta \varepsilon_k} \delta_{k k'} + e^{i\varepsilon_k t/\hbar} e^{-\beta \varepsilon_k} \delta_{k k'} \sum_{k_1 \neq k} e^{-\beta \varepsilon_{k_1}} + \cdots \right) \\
&= e^{i\varepsilon_k t/\hbar} \delta_{k k'} \frac{e^{-\beta \varepsilon_k} \left(1 + \sum_{k_1 \neq k} e^{-\beta \varepsilon_{k_1}} + \sum_{k_2 \neq k_1 \neq k} e^{-\beta \varepsilon_{k_1}} e^{-\beta \varepsilon_{k_2}} + \cdots \right)}{\prod_k (1 + e^{-\beta \varepsilon_k})} \\
&= e^{i\varepsilon_k t/\hbar} \delta_{k k'} \frac{e^{-\beta \varepsilon_k} \prod_{k_1 \neq k} (1 + e^{-\beta \varepsilon_{k_1}})}{\prod_k (1 + e^{-\beta \varepsilon_k})} \\
&= e^{i\varepsilon_k t/\hbar} \delta_{k k'} \frac{e^{-\beta \varepsilon_k}}{1 + e^{-\beta \varepsilon_k}} = e^{i\varepsilon_k t/\hbar} \delta_{k k'} f(\varepsilon_k),
\end{aligned}
$$

where one needs to be careful about the completeness relation of nth Fock space as

$$
\frac{1}{n!} \sum_{k_i} |k_1, \cdots, k_n\rangle\langle k_1, \cdots, k_n| = \mathbb{I}_n,
$$

in correspondence to the action of the fermionic creation and annihilation operators with

unit coefficient. Similarly for the negative t part, we have

$$\begin{aligned}
\langle c_{k'} c_k^\dagger(t)\rangle_0 &= \frac{1}{Z}\,\mathrm{Tr}\Big(c_{k'} c_k^\dagger(t) e^{-\beta H_0}\Big) \\
&= \frac{1}{Z}\left(\langle GS|\, c_{k'} c_k^\dagger(t) e^{-\beta H_0}\,|GS\rangle + \sum_{k_1}\langle k_1|\, c_{k'} c_k^\dagger(t) e^{-\beta H_0}\,|k_1\rangle + \cdots\right) \\
&= \frac{1}{Z}\left(e^{i\varepsilon_{k'} t/\hbar}\,\langle k'|k\rangle + \sum_{k_1} e^{i(\varepsilon_{k_1}+\varepsilon_{k'})t/\hbar} e^{-i\varepsilon_{k_1} t/\hbar} e^{-\beta\varepsilon_{k_1}}\,\langle k_1 k'|k k_2\rangle + \cdots\right) \\
&= \frac{1}{Z}\left(e^{i\varepsilon_k t/\hbar}\delta_{k'k} + e^{i\varepsilon_k t/\hbar}\delta_{k'k}\sum_{k_1\neq k} e^{-\beta\varepsilon_{k_1}} + \cdots\right) \\
&= e^{i\varepsilon_k t/\hbar}\delta_{kk'}\frac{1+\sum_{k_1\neq k} e^{-\beta\varepsilon_{k_1}} + \sum_{k_2\neq k_1\neq k} e^{-\beta\varepsilon_{k_1}} e^{-\beta\varepsilon_{k_2}} + \cdots}{\prod_k (1+e^{-\beta\varepsilon_k})} \\
&= e^{i\varepsilon_k t/\hbar}\delta_{kk'}\frac{1}{1+e^{-\beta\varepsilon_k}} = e^{i\varepsilon_k t/\hbar}\delta_{kk'}(1-f(\varepsilon_k)),
\end{aligned}$$

and we thus write

$$\langle \mathcal{T} c_{k'} c_k^\dagger(t)\rangle_0 = \left[-\Theta(t) f(\varepsilon_k) + \Theta(-t)(1-f(\varepsilon_k))\right] e^{i\varepsilon_k t/\hbar}\delta_{kk'}.$$

Note that by imaginary time $\tau = it/\hbar$, we have two contradictory time orders in the correlation, and we shall focus on the imaginary time ordering

$$\langle \mathcal{T} \hat{c}_{k'} \hat{c}_k^\dagger(\tau)\rangle_0 = \left[-\Theta(\tau) f(\varepsilon_k) + \Theta(-\tau)(1-f(\varepsilon_k))\right] e^{\varepsilon_k \tau}\delta_{kk'}.$$

Thus, we can write the current-current correlation function as

$$\begin{aligned}
L_{\alpha\beta} &= -\frac{1}{V}\int_0^{+\infty} \mathrm{d}t\; e^{-\epsilon t}\int_0^\beta \mathrm{d}\beta'\;\sum_{mnk}\sum_{opq} j^\beta_{mn}(k) j^\alpha_{op}(q)\,\langle \mathcal{T}\hat{c}_{pq}\hat{c}^\dagger_{mk}(\tau-\beta')\rangle_0\,\langle \mathcal{T}\hat{c}_{nk}\hat{c}^\dagger_{oq}(-\tau+\beta')\rangle_0 \\
&= -\frac{1}{V}\int_0^{+\infty} \mathrm{d}t\; e^{-\epsilon t}\int_0^\beta \mathrm{d}\beta'\;\sum_{mnk}\sum_{opq} j^\beta_{mn}(k) j^\alpha_{op}(q) \\
&\qquad \Big[-(1-f_m(k))\delta_{mp}\delta_{kq} e^{(\tau-\beta')\varepsilon_{mk}}\Big]\Big[f_n(k)\delta_{no}\delta_{kq} e^{(-\tau+\beta')\varepsilon_{nk}}\Big] \\
&= \frac{1}{V}\sum_{mnk}\int_0^{+\infty}\mathrm{d}t\; e^{[i(\varepsilon_{mk}-\varepsilon_{nk})/\hbar-\epsilon]t}\int_0^\beta \mathrm{d}\beta'\; e^{-\beta'(\varepsilon_{mk}-\varepsilon_{nk})}(1-f_m(k)) f_n(k) j^\beta_{mn}(k) j^\alpha_{nm}(k) \\
&= \frac{1}{V}\sum_{mnk}\frac{1}{i(\varepsilon_{mk}-\varepsilon_{nk})/\hbar-\epsilon}\frac{1}{\varepsilon_{mk}-\varepsilon_{nk}}\left(e^{-\beta(\varepsilon_{mk}-\varepsilon_{nk})}-1\right)(1-f_m(k)) f_n(k) j^\beta_{mn}(k) j^\alpha_{nm}(k) \\
&= \frac{\hbar}{iV}\sum_{mnk}\frac{1}{\varepsilon_{mk}-\varepsilon_{nk}+i\epsilon}\frac{1}{\varepsilon_{mk}-\varepsilon_{nk}}\left[f_m(k)-f_n(k)\right] j^\beta_{mn}(k) j^\alpha_{nm}(k).
\end{aligned}$$

Written more compactly, we have

$$L_{\alpha\beta} = \frac{\hbar}{iV}\sum_{mnk}\frac{f_m(k)-f_n(k)}{(\varepsilon_{mk}-\varepsilon_{nk})^2}\,\langle mk|\hat{j}_\beta(k)|nk\rangle\,\langle nk|\hat{j}_\alpha(k)|mk\rangle\,. \tag{18}$$

For our purpose of lateral calculation, we need to consider the situation where the one-body operator shares a cross-wavevector representation as

$$\hat{A} = \sum_{mnkk'} A_{mn}(k,k') c^\dagger_{mk} c_{nk'}, \tag{19}$$

and evaluation over its Kubo canonical commutator will similarly give

$$\begin{aligned}
\langle \hat{A}; \hat{B} \rangle &= \frac{1}{\beta} \operatorname{Tr}\left(\hat{\rho}^{eq} \int_0^\beta \mathrm{d}\lambda \; e^{\lambda \hat{H}} \hat{A} e^{-\lambda \hat{H}} \hat{B} \right) \\
&= -\frac{1}{\beta} \int_0^\beta \mathrm{d}\lambda \sum_{mnkk'} \sum_{opqq'} A_{mn}(k,k') B_{op}(q,q') \left\langle \mathcal{T} \hat{c}_{pq'} \hat{c}^\dagger_{mk}(-\lambda) \right\rangle_0 \left\langle \mathcal{T} \hat{c}_{nk'} \hat{c}^\dagger_{oq}(\lambda) \right\rangle_0 \\
&= \frac{1}{\beta} \int_0^\beta \mathrm{d}\lambda \sum_{mnkk'} e^{-\lambda(\varepsilon_{mk} - \varepsilon_{nk'})} (1 - f_m(k)) f_n(k') A_{mn}(k,k') B_{nm}(k',k) \\
&= \frac{1}{\beta} \sum_{mnkk'} \frac{f_m(k) - f_n(k')}{\varepsilon_{mk} - \varepsilon_{nk'}} \langle mk | \hat{A} | nk' \rangle \langle nk' | \hat{B} | mk \rangle .
\end{aligned} \tag{20}$$

C. A Basic of Quantum Thermodynamics

Given Hamiltonian $\hat{H}$ in global equilibrium, there are several essential thermodynamic quantities defined as

$$Z = \operatorname{Tr}\left(e^{-\beta \hat{H}} \right), \tag{21a}$$

$$F = -\frac{1}{\beta} \ln Z, \tag{21b}$$

$$\hat{\rho} = \frac{1}{Z} e^{-\beta \hat{H}}, \tag{21c}$$

$$U = \langle \hat{H} \rangle_0 = \operatorname{Tr}\left(\hat{H} \hat{\rho} \right), \tag{21d}$$

$$S = -k_B \operatorname{Tr}(\hat{\rho} \ln \hat{\rho}). \tag{21e}$$

The definitions are self-consistent, in the sense that

$$\begin{aligned}
\operatorname{Tr}(\hat{\rho} \ln \hat{\rho}) &= -\operatorname{Tr}\left(\hat{\rho} \beta \hat{H} + \hat{\rho} \ln Z \right) = -\beta \operatorname{Tr}\left(\hat{\rho} \hat{H} \right) - \ln Z \operatorname{Tr}(\hat{\rho}) \\
&= -\beta U - \ln Z, \\
-\frac{\partial F}{\partial T} &= -k_B \beta^2 \left(-\frac{1}{\beta^2} \ln Z + \frac{1}{\beta Z} \partial_\beta Z \right) \\
&= k_B \ln Z - \frac{k_B \beta}{Z} \left(-\operatorname{Tr}\left(\hat{H} e^{-\beta \hat{H}} \right) \right) \\
&= k_B (\ln Z + \beta U),
\end{aligned}$$

and we recognize the equivalence of the definitions of the entropy as

$$S = -\frac{\partial F}{\partial T} = -k_B \operatorname{Tr}(\hat{\rho} \ln \hat{\rho}).$$

And

$$TS = -\frac{1}{\beta} \operatorname{Tr}(\hat{\rho} \ln \hat{\rho}) = U + \frac{1}{\beta} \ln Z = U - F.$$

The formula can be generalized to the grand canonical ensemble by replacing $\hat{H} \to \hat{K} = \hat{H} - \mu \hat{N}$, and we shall have the grand potential as

$$\Omega = -\frac{1}{\beta} \ln\left(e^{-\beta \hat{K}}\right) = F - \mu N = K - TS, \tag{22a}$$

$$N = \langle \hat{N} \rangle_0 , \tag{22b}$$

$$K = \langle \hat{K} \rangle_0 = U - \mu N. \tag{22c}$$

Note that the expectation can be done either in the grand canonical ensemble or in the canonical ensemble.

For system in global equilibrium with static homogeneous external fields, we expect the grand canonical partition function to contain two equivalent expressions[5], one with field-perturbed Hamiltonian (see below), generalized inverse temperature ξ, generalized chemical potential α, and the other with the unperturbed Hamiltonian and the usual inverse temperature β and chemical potential μ as

$$\Omega = \frac{1}{Z} e^{-\xi(\hat{H}_{\phi,\psi} - \alpha \hat{N})} = \frac{1}{Z} e^{-\beta(\hat{H} - \mu \hat{N})}, \tag{23}$$

which leads us to the equation that

$$\xi \left[(1 + \psi)(\hat{H} + \phi \hat{n}) - \alpha \hat{N} \right] = \beta \left[\hat{H} - \mu \hat{N} \right], \tag{24}$$

requiring coefficients to be equal of arbitrary Hamiltonian and particle number operators, we arrive at the expressions of the generalized inverse temperature and chemical potential

$$\begin{cases} \xi = (1 + \psi)^{-1} \beta \\ \alpha = (1 + \psi)(\mu + \phi) \end{cases}. \tag{25}$$

These quantities are important since they can be generalized to the case of local equilibrium, with local temperature and chemical potential, which are necessary for the definition of local thermodynamic quantities. For instance, the local heat current in the case without

external fields reads $J_Q(r) = J_e(r) - \mu J_c(r)$ (see below), but with external fields, we have the generalized local heat current as

$$\hat{\boldsymbol{J}}_Q^{\phi,\psi}(\boldsymbol{r}) = \hat{\boldsymbol{J}}_e^{\phi,\psi}(\boldsymbol{r}) - \alpha(\boldsymbol{r})\hat{\boldsymbol{J}}_c^{\phi,\psi}(\boldsymbol{r}), \tag{26}$$

as under the situation, we only have the grand partition function in the first form.

D. Fourier Transform of Bloch States

For later convenience, we introduce the Fourier transformation and orthonormal relation of Bloch states. In correspondence with the basic position-momentum relation

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}, \tag{27}$$

we introduce the canonical Fourier transformation of Bloch operators as

$$\hat{c}_r = \frac{1}{\sqrt{V}} \sum_k e^{ik\cdot r} \hat{c}_k, \ \hat{c}_k = \frac{1}{\sqrt{V}} \int \mathrm{d}r e^{-ik\cdot r} \hat{c}_r, \tag{28}$$

with $\hat{c}_k$, $\hat{c}_r$ annihilation operators for basis Wannier states and Bloch states, respectively, and V the volume of the system. This holds since

$$\begin{aligned}
\hat{c}_r &= \frac{1}{\sqrt{V}} \sum_k e^{ik\cdot r} \hat{c}_k = \frac{1}{\sqrt{V}} \sum_k e^{ik\cdot r} \frac{1}{\sqrt{V}} \int \mathrm{d}r' e^{-ik\cdot r'} \hat{c}_{r'} \\
&= \frac{1}{V} \int \mathrm{d}r' \sum_k e^{ik\cdot(r-r')} \hat{c}_{r'} = \int \mathrm{d}r' \delta(r-r') \hat{c}_{r'} = \hat{c}_r,
\end{aligned}$$

where we have used the orthonormal relation

$$\frac{1}{V} \sum_k e^{ik\cdot(r-r')} = \delta(r-r'). \tag{29}$$

This originates from the standard delta function

$$\int \mathrm{d}k e^{ik\cdot r} = (2\pi)^d \delta(r), \tag{30}$$

and in condensed matter systems, we have $\mathrm{d}k_i = 2\pi/L_i$ such that

$$\int \mathrm{d}k = \frac{(2\pi)^d}{V} \sum_k, \tag{31}$$

which leads to the above relation. Meanwhile, the identity

$$f(k) = \sum_{k'} f(k')\delta_{kk'} = \int \mathrm{d}k' f(k')\delta(k-k')$$

leads to that

$$\delta_{kk'} = \frac{(2\pi)^d}{V} \delta(k-k'). \tag{32}$$

E. Sommerfeld Expansion

Here we briefly review the Sommerfeld expansion for evaluating integrals of the form

$$I = \int_{-\infty}^{\infty} \mathrm{d}\varepsilon \; f[\beta(\varepsilon - \mu)] H(\varepsilon), \tag{33}$$

where $f(x) = 1/(e^x + 1)$ is the reduced Fermi-Dirac distribution, and $H(\varepsilon)$ is a smooth function of ε that vanishes sufficiently fast as $\varepsilon \to \pm\infty$. The basic idea is that the physical temperature is small to make strong modification to distribution, such that by changing variables with $\tau = \beta^{-1}$ and $x = \beta(\varepsilon - \mu)$, we have

$$\begin{aligned}
I &= \tau \int_{-\infty}^{\infty} \mathrm{d}x \; f(x) H(\mu + \tau x) \\
&= \tau \int_{-\infty}^{0} \mathrm{d}x \; f(x) H(\mu + \tau x) + \tau \int_{0}^{\infty} \mathrm{d}x \; f(x) H(\mu + \tau x) \\
&= \tau \int_{0}^{\infty} \mathrm{d}x \; f(-x) H(\mu - \tau x) + \tau \int_{0}^{\infty} \mathrm{d}x \; f(x) H(\mu + \tau x) \\
&= \tau \int_{0}^{\infty} \mathrm{d}x \; [1 - f(x)] H(\mu - \tau x) + \tau \int_{0}^{\infty} \mathrm{d}x \; f(x) H(\mu + \tau x) \\
&= \tau \int_{0}^{\infty} \mathrm{d}x \; H(\mu - \tau x) + \tau \int_{0}^{\infty} \mathrm{d}x \; f(x) \left[H(\mu + \tau x) - H(\mu - \tau x) \right] \\
&= \tau \int_{0}^{\infty} \mathrm{d}x \; H(\mu - \tau x) + \tau \int_{0}^{\infty} \mathrm{d}x \; f(x) \sum_{n=0}^{\infty} \frac{H^{(n)}(\mu)}{n!} \left[(\tau x)^n - (-\tau x)^n \right] \\
&= \tau \int_{0}^{\infty} \mathrm{d}x \; H(\mu - \tau x) + 2\tau \int_{0}^{\infty} \mathrm{d}x \; f(x) \sum_{n=0}^{\infty} \frac{H^{(2n+1)}(\mu)}{(2n+1)!} (\tau x)^{2n+1} \\
&= \tau \int_{0}^{\infty} \mathrm{d}x \; H(\mu - \tau x) + 2 \sum_{n=0}^{\infty} \frac{\tau^{2n+2}}{(2n+1)!} H^{(2n+1)}(\mu) \int_{0}^{\infty} \mathrm{d}x \; \frac{x^{2n+1}}{e^x + 1} \\
&= \int_{-\infty}^{\mu} \mathrm{d}\varepsilon \; H(\varepsilon) + 2 \sum_{n=0}^{\infty} \frac{\tau^{2n+2}}{(2n+1)!} H^{(2n+1)}(\mu) I_{2n+2},
\end{aligned}$$

where a Taylor expansion is involved, and we have defined the integrals

$$\begin{aligned}
I_p &= \int_0^\infty \mathrm{d}x\ \frac{x^{p-1}}{e^x+1} \\
&= \int_0^\infty \mathrm{d}x\ x^{p-1}e^{-x}\frac{1}{1+e^{-x}} \\
&= \int_0^\infty \mathrm{d}x\ x^{p-1}e^{-x}\sum_{m=0}^\infty (-1)^m e^{-mx} \\
&= \sum_{m=0}^\infty (-1)^m \int_0^\infty \mathrm{d}x\ x^{p-1}e^{-(m+1)x} \\
&\overset{u=(m+1)x}{=\!=\!=} \sum_{m=0}^\infty (-1)^m \frac{1}{(m+1)^p}\int_0^\infty \mathrm{d}u\ u^{p-1}e^{-u} \\
&= \Gamma(p)\sum_{m=1}^\infty (-1)^{m+1}\frac{1}{m^p} = \Gamma(p)\frac{2^p-2}{2^p}\zeta(p),
\end{aligned}$$

where Gamma function $\Gamma(p)$ and Riemann zeta function $\zeta(p)$ are defined as

$$\begin{aligned}
\Gamma(p) &= \int_0^\infty \mathrm{d}x\ x^{p-1}e^{-x}, \\
\zeta(p) &= \sum_{m=1}^\infty \frac{1}{m^p}.
\end{aligned}$$

A partial summation of the series gives

$$\begin{aligned}
\sum_{n=0}^\infty \frac{1}{(2n+2)^p} &= \frac{1}{2^p}\sum_{n=0}^\infty \frac{1}{(n+1)^p} = \frac{1}{2^p}\zeta(p), \\
\zeta(p) &= \sum_{n=1}^\infty \frac{1}{n^p} = \sum_{n=0}^\infty \frac{1}{(2n+1)^p} + \sum_{n=0}^\infty \frac{1}{(2n+2)^p} \\
&= \sum_{n=0}^\infty \frac{1}{(2n+1)^p} + \frac{1}{2^p}\zeta(p), \\
\Longrightarrow \sum_{n=0}^\infty \frac{1}{(2n+1)^p} &= \left(1-\frac{1}{2^p}\right)\zeta(p), \\
\Longrightarrow \sum_{n=1}^\infty (-1)^{n+1}\frac{1}{n^p} &= 2\sum_{n=0}^\infty \frac{1}{(2n+1)^p} - \sum_{n=1}^\infty \frac{1}{n^p} = \frac{2^p-2}{2^p}\zeta(p).
\end{aligned}$$

The first several integrals are

$$\zeta(2) = \frac{\pi^2}{6}, \quad \zeta(4) = \frac{\pi^4}{90}, \quad \zeta(6) = \frac{\pi^6}{945}, \cdots,$$

then with $\Gamma(n) = (n-1)!$, we obtain the final result

$$\begin{aligned} I &= \int_{-\infty}^{\mu} \mathrm{d}\varepsilon\ H(\varepsilon) + 2\sum_{n=0}^{\infty} \frac{\tau^{2n+2}}{(2n+1)!} H^{(2n+1)}(\mu) I_{2n+2} \\ &= \int_{-\infty}^{\mu} \mathrm{d}\varepsilon\ H(\varepsilon) + 2\frac{\pi^2\tau^2}{6}\frac{2^2-2}{2^2} H'(\mu) + 2\frac{\pi^4\tau^4}{3!\times 90} 3! \frac{2^4-2}{2^4} H'''(\mu) + \cdots \\ &= \int_{-\infty}^{\mu} \mathrm{d}\varepsilon\ H(\varepsilon) + \frac{\pi^2\tau^2}{6} H'(\mu) + \frac{7\pi^4\tau^4}{360} H'''(\mu) + \cdots . \end{aligned} \tag{34}$$

F. Thermodynamics of Magnetizations

The afore mentioned gravitational field is introduced by writing the local Hamiltonian density, with an electric potential coupled to charge density (setting $e = 1$), as

$$\hat{h}_{\phi,\psi}(\boldsymbol{r}) = (1 + \psi(\boldsymbol{r}))(\hat{h}(\boldsymbol{r}) + \phi(\boldsymbol{r})\hat{n}(\boldsymbol{r})), \tag{35}$$

where $\phi(\boldsymbol{r})$ and $\psi(\boldsymbol{r})$ are the electric potential and the fictitious gravitational potential, respectively. The Hamiltonian then reads

$$\hat{H}_{\phi,\psi} = \int \mathrm{d}\boldsymbol{r}\ \hat{h}_{\phi,\psi}(\boldsymbol{r}). \tag{36}$$

The continuity equations for charge and energy densities then lead to the definition of the electric current and energy current densities as

$$\frac{1}{i\hbar}[\hat{n}(\boldsymbol{r}), \hat{H}_{\phi,\psi}] = \frac{\partial \hat{n}(\boldsymbol{r})}{\partial t} = -\nabla \cdot \hat{\boldsymbol{J}}_c^{\phi,\psi}(\boldsymbol{r}), \tag{37a}$$

$$\frac{1}{i\hbar}[\hat{h}_{\phi,\psi}(\boldsymbol{r}), \hat{H}_{\phi,\psi}] = \frac{\partial \hat{h}_{\phi,\psi}(\boldsymbol{r})}{\partial t} = -\nabla \cdot \hat{\boldsymbol{J}}_e^{\phi,\psi}(\boldsymbol{r}), \tag{37b}$$

where in the second equation, we ignore explicit time dependence of the Hamiltonian. The form of the field perturbed Hamiltonian density, the above form of equation and the locality principle (which should stand for system with short range interactions only) suggest the scaling relation of the current densities as

$$\hat{\boldsymbol{J}}_c^{\phi,\psi}(\boldsymbol{r}) = [1 + \psi(\boldsymbol{r})]\hat{\boldsymbol{J}}_c(\boldsymbol{r}), \tag{38a}$$

$$\hat{\boldsymbol{J}}_e^{\phi,\psi}(\boldsymbol{r}) = [1 + \psi(\boldsymbol{r})]^2[\hat{\boldsymbol{J}}_e(\boldsymbol{r}) + \phi(\boldsymbol{r})\hat{\boldsymbol{J}}_c(\boldsymbol{r})], \tag{38b}$$

where $\hat{J}_c(\boldsymbol{r})$ and $\hat{J}_e(\boldsymbol{r})$ are the unperturbed current densities defined from the unperturbed Hamiltonian density $\hat{h}(\boldsymbol{r})$.

For system in equilibrium without external fields, we have the equilibrium current densities constrained by

$$\nabla \cdot \boldsymbol{J}_c^{eq} = \nabla \cdot \boldsymbol{J}_e^{eq} = 0, \tag{39}$$

where $\boldsymbol{J}_c^{eq}$ and $\boldsymbol{J}_e^{eq}$ are the equilibrium current densities by $\boldsymbol{J}_{c/e}^{eq}(\boldsymbol{r}) = \langle \hat{\boldsymbol{J}}_{c/e}(\boldsymbol{r}) \rangle_0$ within grand canonical ensemble thermal average $\langle O \rangle_0 = \mathrm{Tr}\Big(\hat{\rho}_{eq}\hat{O}\Big)/Z_{eq}$ with $\hat{\rho}_{eq} = e^{-\beta\hat{K}}$, $Z_{eq} = \mathrm{Tr}\Big(e^{-\beta\hat{K}}\Big)$, where the grand potential operator $\hat{K} = \hat{H} - \mu\hat{N}$ with local expression $\hat{K} = \int \mathrm{d}\boldsymbol{r}\ \hat{k}(\boldsymbol{r})$ where $\hat{k}(\boldsymbol{r}) = \hat{h}(\boldsymbol{r}) - \mu\hat{n}(\boldsymbol{r})$. The zero-field (energy) magnetization density is then defined as

$$\boldsymbol{J}_{c/e}^{eq}(\boldsymbol{r}) = \nabla \times \boldsymbol{M}_{c/e}(\boldsymbol{r}), \tag{40}$$

and we have the zero-field heat current and heat magnetization density

$$\hat{\boldsymbol{J}}_Q(\boldsymbol{r}) = \hat{\boldsymbol{J}}_e(\boldsymbol{r}) - \mu\hat{\boldsymbol{J}}_c(\boldsymbol{r}), \tag{41a}$$

$$\boldsymbol{J}_Q^{eq}(\boldsymbol{r}) = \langle \hat{\boldsymbol{J}}_Q(\boldsymbol{r}) \rangle_0 = \nabla \times \boldsymbol{M}_Q(\boldsymbol{r}), \tag{41b}$$

$$\boldsymbol{M}_Q(\boldsymbol{r}) = \boldsymbol{M}_e(\boldsymbol{r}) - \mu\boldsymbol{M}_c(\boldsymbol{r}). \tag{41c}$$

Now let us consider the full linear response of the currents of the system to small derivation to external fields, i.e., the electric potential $\phi(\boldsymbol{r}, t)$ and the fictitious gravitational potential $\psi(\boldsymbol{r}, t)$. As we are generally considering local fields, we extend the global equilibrium state to a local equilibrium state, where the local temperature $T(\boldsymbol{r})$ and local chemical potential $\mu(\boldsymbol{r})$ are defined. In the case, we write the total density matrix operator as

$$\hat{\rho}_{\phi,\psi}(t) = \hat{\rho}^{leq} + \hat{\rho}_1(t), \tag{42}$$

where the time-dependent linear response correction, as discussed in the previous section, is responsible for the response to the external fields determined by the Kubo formula, leading to

$$\boldsymbol{J}_i^{\mathrm{Kubo},\phi,\psi}(\boldsymbol{r}, t) = \mathrm{Tr}\Big(\hat{\rho}_1(t)\hat{\boldsymbol{J}}_i^{\phi,\psi}(\boldsymbol{r})\Big). \tag{43}$$

On the other hand, the local equilibrium density matrix is defined as

$$\hat{\rho}^{leq} = \frac{1}{Z^{leq}} \exp\bigg(-\int \mathrm{d}\boldsymbol{r}\beta(\boldsymbol{r}) \Big[\hat{h}(\boldsymbol{r}) - \mu(\boldsymbol{r})\hat{n}(\boldsymbol{r})\Big]\bigg), \tag{44}$$

where the local inverse temperature reads $\beta(\boldsymbol{r}) = 1/k_B T(\boldsymbol{r})$. Different from the global equilibrium state, the local equilibrium state can generate a non-zero expectation value of the current densities, i.e.,

$$\boldsymbol{J}_i^{leq,\phi,\psi}(\boldsymbol{r}) = \mathrm{Tr}\Big(\hat{\rho}^{leq}\hat{\boldsymbol{J}}_i^{\phi,\psi}(\boldsymbol{r})\Big), \tag{45}$$

and the total current density can be written as

$$\boldsymbol{J}_i^{\phi,\psi}(\boldsymbol{r},t) = \boldsymbol{J}_i^{leq,\phi,\psi}(\boldsymbol{r}) + \boldsymbol{J}_i^{\mathrm{Kubo},\phi,\psi}(\boldsymbol{r},t). \tag{46}$$

For our purpose, we consider the local equilibrium with small derivation from global equilibrium as

$$\begin{cases} \mu(\boldsymbol{r}) = \mu + \delta\mu(\boldsymbol{r}) \\ \beta(\boldsymbol{r}) = \beta + \delta\beta(\boldsymbol{r}) \end{cases}, \tag{47}$$

such that the local equilibrium density matrix can be expanded as

$$\begin{aligned} \hat{\rho}^{leq} &= \frac{1}{Z^{leq}} \exp\left(- \int \mathrm{d}\boldsymbol{r} \ (\beta + \delta\beta(\boldsymbol{r})) \left[\hat{h}(\boldsymbol{r}) - (\mu + \delta\mu(\boldsymbol{r}))\hat{n}(\boldsymbol{r}) \right] \right) \\ &= \frac{1}{Z^{leq}} \exp\left(- \int \mathrm{d}\boldsymbol{r} \ \beta \left[\hat{k}(\boldsymbol{r}) - \hat{n}(\boldsymbol{r})\delta\mu(\boldsymbol{r}) + \hat{k}(\boldsymbol{r})\delta\beta(\boldsymbol{r})/\beta \right] + O(\delta^2) \right). \end{aligned}$$

In this way, we identify the linear response up to linear order of $\phi, \psi, \delta\mu$ and $\delta\beta$. We also identify the linear response of the local equilibrium density matrix to be calculated with the static linear response theory as

$$\boldsymbol{J}_i^{leq,\phi,\psi}(\boldsymbol{r}) = \boldsymbol{J}_i^{eq,\phi,\psi}(\boldsymbol{r}) + \sum_{j=1}^{2} \int \mathrm{d}\boldsymbol{r}' \ \boldsymbol{\chi}_{ij}(\boldsymbol{r},\boldsymbol{r}') x_j(\boldsymbol{r}'), \tag{48}$$

where $x_1(\boldsymbol{r}) = \delta\mu(\boldsymbol{r})$ and $x_2(\boldsymbol{r}) = -\delta\beta(\boldsymbol{r})/\beta = -T\delta(1/T(\boldsymbol{r}))$, and $\boldsymbol{J}_i^{eq,\phi,\psi}(\boldsymbol{r}) = \langle \hat{\boldsymbol{J}}_i^{\phi,\psi}(\boldsymbol{r}) \rangle_0$, which contains the same scaling relations with $\boldsymbol{J}_i^{eq}(\boldsymbol{r})$. Note that the local equilibrium current density can be evaluated independently without external fields. Now we introduce the vectorized matrix susceptibility as

$$\boldsymbol{\chi}_{ij}(\boldsymbol{r},\boldsymbol{r}') = \beta \left\langle \Delta\hat{n}_j(\boldsymbol{r}'); \Delta\hat{\boldsymbol{J}}_i(\boldsymbol{r}) \right\rangle, \tag{49}$$

with $i, j = 1, 2$, and

$$\begin{cases} \hat{n}_1(\boldsymbol{r}) = \hat{n}(\boldsymbol{r}) \\ \hat{n}_2(\boldsymbol{r}) = \hat{k}(\boldsymbol{r}) \end{cases}, \quad \begin{cases} \hat{\boldsymbol{J}}_1(\boldsymbol{r}) = \hat{\boldsymbol{J}}_c(\boldsymbol{r}) \\ \hat{\boldsymbol{J}}_2(\boldsymbol{r}) = \hat{\boldsymbol{J}}_Q(\boldsymbol{r}) \end{cases}. \tag{50}$$

Note that this term is not involved with the external fields since it is linear order response to the local equilibrium state, and inclusion of the external fields will lead to a higher order

response. We then have

$$\begin{aligned}
\nabla_{\boldsymbol{r}} \cdot \boldsymbol{\chi}_{ij}(\boldsymbol{r}, \boldsymbol{r}') &= \beta \left\langle \Delta\hat{n}_j(\boldsymbol{r}'); \nabla_{\boldsymbol{r}} \cdot \Delta\hat{\boldsymbol{J}}_i(\boldsymbol{r}) \right\rangle \\
&= \beta \left\langle \Delta\hat{n}_j(\boldsymbol{r}'); \nabla_{\boldsymbol{r}} \cdot \hat{\boldsymbol{J}}_i(\boldsymbol{r}) - \nabla_{\boldsymbol{r}} \cdot \boldsymbol{J}_i^{eq}(\boldsymbol{r}) \right\rangle \\
&= -\beta \left\langle \Delta\hat{n}_j(\boldsymbol{r}'); \dot{\hat{n}}_i(\boldsymbol{r}) \right\rangle \\
&= \frac{1}{i\hbar} \left\langle [\Delta\hat{n}_j(\boldsymbol{r}'), \hat{n}_i(\boldsymbol{r})] \right\rangle_0 \\
&= \frac{1}{i\hbar} \left\langle [\hat{n}_j(\boldsymbol{r}'), \hat{n}_i(\boldsymbol{r})] \right\rangle_0 .
\end{aligned}$$

Note for $i = 2$, the continuity equation holds with homogeneous chemical potential, which is the case for global equilibrium state utilized in the susceptibility definition. Now we introduce the partial Fourier transform of the susceptibility as

$$\boldsymbol{\chi}_{ij}^{\boldsymbol{q}}(\boldsymbol{r}) = \int \mathrm{d}\boldsymbol{r}' \ \boldsymbol{\chi}_{ij}(\boldsymbol{r}, \boldsymbol{r}') e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}, \tag{51}$$

which makes the $(\boldsymbol{q}, \boldsymbol{r})$ labelled susceptibility to contain both positional and momentum dependence. This is much like the wave-packet description of the electronic wavefunction. Note that we have shifted the momentum integral for by a Bloch phase factor. Under the transformation, we have

$$\begin{aligned}
\nabla_{\boldsymbol{r}} \cdot \boldsymbol{\chi}_{ij}^{\boldsymbol{q}}(\boldsymbol{r}) &= \int \mathrm{d}\boldsymbol{r}' \ \nabla_{\boldsymbol{r}} \cdot \left[\boldsymbol{\chi}_{ij}(\boldsymbol{r}, \boldsymbol{r}') e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')} \right] \\
&= \int \mathrm{d}\boldsymbol{r}' \ e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')} \left(\nabla_{\boldsymbol{r}} - i\boldsymbol{q} \right) \cdot \boldsymbol{\chi}_{ij}(\boldsymbol{r}, \boldsymbol{r}') \\
&= \int \mathrm{d}\boldsymbol{r}' \ \frac{1}{i\hbar} \left\langle [\hat{n}_j(\boldsymbol{r}'), \hat{n}_i(\boldsymbol{r})] \right\rangle_0 e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')} - i\boldsymbol{q} \cdot \boldsymbol{\chi}_{ij}^{\boldsymbol{q}}(\boldsymbol{r}),
\end{aligned}$$

i.e., we arrive at the equation

$$(\nabla + i\boldsymbol{q}) \cdot \boldsymbol{\chi}_{ij}^{\boldsymbol{q}}(\boldsymbol{r}) = \frac{1}{i\hbar} \int \mathrm{d}\boldsymbol{r}' \ \left\langle [\hat{n}_j(\boldsymbol{r}'), \hat{n}_i(\boldsymbol{r})] \right\rangle_0 e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}. \tag{52}$$

We shall ignore the lower index $\boldsymbol{r}$ for ∇ for simplicity.

Now let us list again the definitions of the current densities and their scaling relations

$$\begin{aligned}
\frac{1}{i\hbar}[\hat{n}(\boldsymbol{r}), \hat{H}_{\phi,\psi}] &= -\nabla \cdot \hat{\boldsymbol{J}}_c^{\phi,\psi}(\boldsymbol{r}), \\
\frac{1}{i\hbar}[\hat{h}_{\phi,\psi}(\boldsymbol{r}), \hat{H}_{\phi,\psi}] &= -\nabla \cdot \hat{\boldsymbol{J}}_e^{\phi,\psi}(\boldsymbol{r}), \\
\hat{\boldsymbol{J}}_c^{\phi,\psi}(\boldsymbol{r}) &= [1 + \psi(\boldsymbol{r})] \hat{\boldsymbol{J}}_c(\boldsymbol{r}), \\
\hat{\boldsymbol{J}}_e^{\phi,\psi}(\boldsymbol{r}) &= [1 + \psi(\boldsymbol{r})]^2 [\hat{\boldsymbol{J}}_e(\boldsymbol{r}) + \phi(\boldsymbol{r}) \hat{\boldsymbol{J}}_c(\boldsymbol{r})].
\end{aligned}$$

Now we derive the momentum-dependent EoM for the charge and energy current densities.

• We firstly set $\phi(\boldsymbol{r})=0$ and $1+\psi(\boldsymbol{r})=e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$, and the equation for charge current density reads

$$\begin{aligned}
&\frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ [e^{i\boldsymbol{q}\cdot\boldsymbol{r}}\hat{h}(\boldsymbol{r}'),\hat{n}(\boldsymbol{r})]=\nabla\cdot(e^{i\boldsymbol{q}\cdot\boldsymbol{r}}\hat{\boldsymbol{J}}_c(\boldsymbol{r}))=e^{i\boldsymbol{q}\cdot\boldsymbol{r}}(\nabla+i\boldsymbol{q})\cdot\hat{\boldsymbol{J}}_c(\boldsymbol{r})\\
\Longrightarrow\quad &(\nabla+i\boldsymbol{q})\cdot\hat{\boldsymbol{J}}_c(\boldsymbol{r})=\frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ [e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\hat{h}(\boldsymbol{r}'),\hat{n}(\boldsymbol{r})]\\
\overset{\langle\rangle_0}{\Longrightarrow}\quad &i\boldsymbol{q}\cdot(\nabla\times\boldsymbol{M}_c(\boldsymbol{r}))=\frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \langle[\hat{h}(\boldsymbol{r}'),\hat{n}(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\\
&\qquad\qquad=\frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \langle[\hat{k}(\boldsymbol{r}'),\hat{n}(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')},
\end{aligned}$$

where in the second deduction, we have used the fact that $\nabla\cdot\langle\hat{\boldsymbol{J}}_c(\boldsymbol{r})\rangle_0=\nabla\cdot\boldsymbol{J}_c^{eq}(\boldsymbol{r})=0$ and the magnetization expression, while in the last line, we notice that density operator commutes at different positions, and we replace the Hamiltonian density with the grand potential density.

• Then we let $\psi(\boldsymbol{r})=0$, $\phi(\boldsymbol{r})=e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$, and the equation for energy current density reads

$$\begin{aligned}
&\frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ [\hat{h}(\boldsymbol{r}')+e^{i\boldsymbol{q}\cdot\boldsymbol{r}'}\hat{n}(\boldsymbol{r}'),\hat{h}(\boldsymbol{r})+e^{i\boldsymbol{q}\cdot\boldsymbol{r}}\hat{n}(\boldsymbol{r})]=\nabla\cdot(\hat{\boldsymbol{J}}_e(\boldsymbol{r})+e^{i\boldsymbol{q}\cdot\boldsymbol{r}}\hat{\boldsymbol{J}}_e(\boldsymbol{r}))\\
&\qquad\qquad=\nabla\cdot\hat{\boldsymbol{J}}_e(\boldsymbol{r})+e^{i\boldsymbol{q}\cdot\boldsymbol{r}}(\nabla+i\boldsymbol{q})\cdot\hat{\boldsymbol{J}}_c(\boldsymbol{r})\\
\overset{\langle\rangle_0}{\Longrightarrow}\quad &\frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \langle[\hat{h}(\boldsymbol{r}'),\hat{h}(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}+\langle[\hat{n}(\boldsymbol{r}'),\hat{h}(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\\
&\qquad\qquad+\langle[\hat{h}(\boldsymbol{r}'),\hat{n}(\boldsymbol{r})]\rangle_0=i\boldsymbol{q}\cdot(\nabla\times\boldsymbol{M}_c(\boldsymbol{r}))\\
\Longrightarrow\quad &\frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \langle[\hat{n}(\boldsymbol{r}'),\hat{h}(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\cancel{+\frac{1}{i\hbar}\langle[\hat{H},\hat{h}(\boldsymbol{r})e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}+\hat{n}(\boldsymbol{r})]\rangle_0}=i\boldsymbol{q}\cdot(\nabla\times\boldsymbol{M}_c(\boldsymbol{r}))\\
\Longrightarrow\quad &\frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \langle[\hat{n}(\boldsymbol{r}'),\hat{k}(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}=i\boldsymbol{q}\cdot(\nabla\times\boldsymbol{M}_c(\boldsymbol{r})),
\end{aligned}$$

where the crossed term vanishes by trace property, and we also replace the Hamiltonian density with the grand potential density.

• Again we let $\phi(\boldsymbol{r})=0$ and $1+\psi(\boldsymbol{r})=e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$, and the equation for energy current density reads

$$\begin{aligned}
&\frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ [e^{i\boldsymbol{q}\cdot\boldsymbol{r}'}\hat{h}(\boldsymbol{r}'),e^{i\boldsymbol{q}\cdot\boldsymbol{r}}\hat{h}(\boldsymbol{r})]=\nabla\cdot(e^{2i\boldsymbol{q}\cdot\boldsymbol{r}}\hat{\boldsymbol{J}}_e(\boldsymbol{r}))=e^{2i\boldsymbol{q}\cdot\boldsymbol{r}}(\nabla+2i\boldsymbol{q})\cdot\hat{\boldsymbol{J}}_e(\boldsymbol{r})\\
\overset{\langle\rangle_0}{\Longrightarrow}\quad &i\boldsymbol{q}\cdot(\nabla\times 2\boldsymbol{M}_e(\boldsymbol{r}))=\frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \langle[\hat{h}(\boldsymbol{r}'),\hat{h}(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\\
&\qquad\qquad=\frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \left(\langle[\hat{k}(\boldsymbol{r}'),\hat{k}(\boldsymbol{r})]\rangle_0+\mu\langle[\hat{h}(\boldsymbol{r}'),\hat{n}(\boldsymbol{r})]\rangle_0+\mu\langle[\hat{n}(\boldsymbol{r}'),\hat{h}(\boldsymbol{r})]\rangle_0\right)e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\\
&\qquad\qquad=\frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \langle[\hat{k}(\boldsymbol{r}'),\hat{k}(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}+2i\boldsymbol{q}\cdot(\nabla\times\boldsymbol{M}_c(\boldsymbol{r})),
\end{aligned}$$

$$\implies \qquad i\boldsymbol{q}\cdot(\nabla\times 2\boldsymbol{M}_Q(\boldsymbol{r})) = \frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \langle[\hat{k}(\boldsymbol{r}'),\hat{k}(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}.$$

Finally, we can list the above results with an additional trivial equation

$$0 = \frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \langle[\hat{n}(\boldsymbol{r}'),\hat{n}(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}, \tag{53a}$$

$$i\boldsymbol{q}\cdot(\nabla\times\boldsymbol{M}_c(\boldsymbol{r})) = \frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \langle[\hat{k}(\boldsymbol{r}'),\hat{n}(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}, \tag{53b}$$

$$i\boldsymbol{q}\cdot(\nabla\times\boldsymbol{M}_c(\boldsymbol{r})) = \frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \langle[\hat{n}(\boldsymbol{r}'),\hat{k}(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}, \tag{53c}$$

$$i\boldsymbol{q}\cdot(\nabla\times 2\boldsymbol{M}_Q(\boldsymbol{r})) = \frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \langle[\hat{k}(\boldsymbol{r}'),\hat{k}(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}, \tag{53d}$$

and we can re-write them in a compact matrix form as

$$\begin{aligned} i\boldsymbol{q}\cdot(\nabla\times\boldsymbol{M}_{ij}(\boldsymbol{r})) &= \frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\ \langle[\hat{n}_j(\boldsymbol{r}'),\hat{n}_i(\boldsymbol{r})]\rangle_0\, e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')} \\ &= (\nabla + i\boldsymbol{q})\cdot\boldsymbol{\chi}^{\boldsymbol{q}}_{ij}(\boldsymbol{r}), \end{aligned} \tag{54}$$

where the vector-valued magnetization matrix is defined as

$$\overleftrightarrow{\boldsymbol{M}} = \begin{pmatrix} 0 & \boldsymbol{M}_c \\ \boldsymbol{M}_c & 2\boldsymbol{M}_Q \end{pmatrix} = \boldsymbol{M}_c\sigma_x + \boldsymbol{M}_Q(\sigma_0 - \sigma_z). \tag{55}$$

The above equation can be re-written as

$$\nabla\cdot\boldsymbol{\chi}^{\boldsymbol{q}}_{ij}(\boldsymbol{r}) + i\boldsymbol{q}\cdot[\boldsymbol{\chi}^{\boldsymbol{q}}_{ij}(\boldsymbol{r}) - \nabla\times\boldsymbol{M}_{ij}(\boldsymbol{r})] = 0, \tag{56}$$

and leads to a general solution of the form

$$\boldsymbol{\chi}^{\boldsymbol{q}}_{ij}(\boldsymbol{r}) = -i\boldsymbol{q}\times\boldsymbol{M}_{ij}(\boldsymbol{r}) + e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\nabla\times\boldsymbol{\kappa}^{\boldsymbol{q}}_{ij}(\boldsymbol{r}), \tag{57}$$

with $\boldsymbol{\kappa}^{\boldsymbol{q}}_{ij}(\boldsymbol{r})$ an arbitrary vector-valued function. Basically, the first term is a special solution that cancels the curl term, and the second term is a general solution of the homogeneous equation $(\nabla + i\boldsymbol{q})\cdot\boldsymbol{\chi}^{\boldsymbol{q}}_{ij}(\boldsymbol{r}) = 0$.

With the above results, let us come back to Eq. (48), and since we care about the static susceptibility, we assume zero field (inclusion of external field will not affect the result here

up to linear order) and

$$\begin{aligned}
\delta \boldsymbol{J}_i^{eq}(\boldsymbol{r}) &= \boldsymbol{J}_i^{leq}(\boldsymbol{r}) - \boldsymbol{J}_i^{eq}(\boldsymbol{r}) \\
&= \sum_{j=1}^{2} \int \mathrm{d}\boldsymbol{r}' \ \boldsymbol{\chi}_{ij}(\boldsymbol{r}, \boldsymbol{r}') x_j(\boldsymbol{r}') \\
&= \int \mathrm{d}\boldsymbol{r}' \ \left[\boldsymbol{\chi}_{i1}(\boldsymbol{r}, \boldsymbol{r}') \delta\mu(\boldsymbol{r}') - \boldsymbol{\chi}_{i2}(\boldsymbol{r}, \boldsymbol{r}') T \delta \left(\frac{1}{T(\boldsymbol{r}')} \right) \right] \\
&\overset{\text{U.P.}}{=\!=\!=} \int \mathrm{d}\boldsymbol{r}' \ \left[\boldsymbol{\chi}_{i1}(\boldsymbol{r}, \boldsymbol{r}') \delta\mu - \boldsymbol{\chi}_{i2}(\boldsymbol{r}, \boldsymbol{r}') T \delta \left(\frac{1}{T} \right) \right] \\
&= \boldsymbol{\chi}_{i1}^{\boldsymbol{q}=0}(\boldsymbol{r}) \delta\mu + \boldsymbol{\chi}_{i2}^{\boldsymbol{q}=0}(\boldsymbol{r}) \delta T/T \\
&= \nabla \times \boldsymbol{\kappa}_{i1}^{\boldsymbol{q}=0}(\boldsymbol{r}) \delta\mu + \nabla \times \boldsymbol{\kappa}_{i2}^{\boldsymbol{q}=0}(\boldsymbol{r}) \delta T/T,
\end{aligned}$$

where U.P. refers to a uniform perturbation of force variation. On the other hand, the variation of current density can be expressed with variation of local magnetization as

$$\begin{aligned}
\delta \boldsymbol{J}_c^{eq}(\boldsymbol{r}) &= \mathrm{Tr}\Big(\hat{\boldsymbol{J}}_c(\boldsymbol{r}) \delta\hat{\rho}^{eq} \Big) = \mathrm{Tr}\Big(\hat{\boldsymbol{J}}_c(\boldsymbol{r}) (\hat{\rho}^{leq} - \hat{\rho}^{eq}) \Big) \\
&= \boldsymbol{J}_c^{leq}(\boldsymbol{r}) - \boldsymbol{J}_c^{eq}(\boldsymbol{r}) = \nabla \times (\boldsymbol{M}_c^{leq}(\boldsymbol{r}) - \boldsymbol{M}_c(\boldsymbol{r})) \\
&= \nabla \times \delta \boldsymbol{M}_c(\boldsymbol{r}), \\
\delta \boldsymbol{J}_Q^{eq}(\boldsymbol{r}) &= \mathrm{Tr}\Big(\hat{\boldsymbol{J}}_Q(\boldsymbol{r}) \delta\hat{\rho}^{eq} \Big) = \mathrm{Tr}\Big((\hat{\boldsymbol{J}}_e(\boldsymbol{r}) - \mu \hat{\boldsymbol{J}}_c(\boldsymbol{r})) \delta\hat{\rho}^{eq} \Big) \\
&= \mathrm{Tr}\Big(\hat{\boldsymbol{J}}_e(\boldsymbol{r}) \delta\hat{\rho}^{eq} \Big) - \mu \, \mathrm{Tr}\Big(\hat{\boldsymbol{J}}_c(\boldsymbol{r}) \delta\hat{\rho}^{eq} \Big) \\
&= \nabla \times \delta \boldsymbol{M}_e(\boldsymbol{r}) - \mu \nabla \times \delta \boldsymbol{M}_c(\boldsymbol{r}),
\end{aligned}$$

thus a comparison of the two expressions leads to

$$\begin{cases}
\boldsymbol{\kappa}_{11}^{\boldsymbol{q}=0}(\boldsymbol{r}) \delta\mu + \boldsymbol{\kappa}_{12}^{\boldsymbol{q}=0}(\boldsymbol{r}) \delta T/T = \delta \boldsymbol{M}_c(\boldsymbol{r}) \\
\boldsymbol{\kappa}_{21}^{\boldsymbol{q}=0}(\boldsymbol{r}) \delta\mu + \boldsymbol{\kappa}_{22}^{\boldsymbol{q}=0}(\boldsymbol{r}) \delta T/T = \delta \boldsymbol{M}_e(\boldsymbol{r}) - \mu \delta \boldsymbol{M}_c(\boldsymbol{r}) = \delta \boldsymbol{M}_Q(\boldsymbol{r}) + \boldsymbol{M}_c(\boldsymbol{r}) \delta\mu
\end{cases},$$

and the thermodynamic variation leads to

$$\boldsymbol{\kappa}_{11}^{\boldsymbol{q}=0}(\boldsymbol{r}) = \left(\frac{\partial \boldsymbol{M}_c(\boldsymbol{r})}{\partial \mu} \right)_T, \tag{58a}$$

$$\boldsymbol{\kappa}_{12}^{\boldsymbol{q}=0}(\boldsymbol{r}) = T \left(\frac{\partial \boldsymbol{M}_c(\boldsymbol{r})}{\partial T} \right)_\mu, \tag{58b}$$

$$\boldsymbol{\kappa}_{21}^{\boldsymbol{q}=0}(\boldsymbol{r}) = \left(\frac{\partial \boldsymbol{M}_Q(\boldsymbol{r})}{\partial \mu} \right)_T + \boldsymbol{M}_c(\boldsymbol{r}), \tag{58c}$$

$$\boldsymbol{\kappa}_{22}^{\boldsymbol{q}=0}(\boldsymbol{r}) = T \left(\frac{\partial \boldsymbol{M}_Q(\boldsymbol{r})}{\partial T} \right)_\mu. \tag{58d}$$

At the same time, we refer to Eq. (57) as

$$\begin{aligned}\boldsymbol{\chi}_{ij}^{\boldsymbol{q}}(\boldsymbol{r}) &= -i\boldsymbol{q}\times\boldsymbol{M}_{ij}(\boldsymbol{r}) + e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\nabla\times\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r}) \\ &= -i\boldsymbol{q}\times\boldsymbol{M}_{ij}(\boldsymbol{r}) + \nabla\times\left(e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r})\right) - (\nabla e^{-i\boldsymbol{q}\cdot\boldsymbol{r}})\times\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r}) \\ &= -i\boldsymbol{q}\times\boldsymbol{M}_{ij}(\boldsymbol{r}) + \nabla\times\left(e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r})\right) + i\boldsymbol{q}e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\times\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r}) \\ &= -i\boldsymbol{q}\times\left[\boldsymbol{M}_{ij}(\boldsymbol{r}) - \boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r})\right] + \nabla\times\left(e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r})\right)\end{aligned}$$

and by applying a momentum curl, we have

$$\nabla_{\boldsymbol{q}}\times\boldsymbol{\chi}_{ij}^{\boldsymbol{q}}(\boldsymbol{r}) = -i\nabla_{\boldsymbol{q}}\times\left(\boldsymbol{q}\times\left[\boldsymbol{M}_{ij}(\boldsymbol{r}) - \boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r})\right]\right) + \nabla_{\boldsymbol{q}}\times\nabla\times\left(e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r})\right).$$

We treat the first term as $-i\boldsymbol{A}$, and

$$\begin{aligned}A_a &= \epsilon_{abc}\partial_{q_b}(\boldsymbol{q}\times(\boldsymbol{M}_{ij}(\boldsymbol{r}) - \boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r})))_c \\ &= \epsilon_{abc}\partial_{q_b}\epsilon_{cmn}q_m(M_{ij,n}(\boldsymbol{r}) - \kappa_{ij,n}^{\boldsymbol{q}}(\boldsymbol{r})) \\ &= (\delta_{am}\delta_{bn} - \delta_{an}\delta_{bm})\left[\delta_{bm}(M_{ij,n}(\boldsymbol{r}) - \kappa_{ij,n}^{\boldsymbol{q}}(\boldsymbol{r})) - q_m\partial_{q_b}\kappa_{ij,n}^{\boldsymbol{q}}(\boldsymbol{r})\right] \\ &= (1-\delta_{bm}^2)(M_{ij,a}(\boldsymbol{r}) - \kappa_{ij,a}^{\boldsymbol{q}}(\boldsymbol{r})) - q_a(\nabla_{\boldsymbol{q}}\cdot\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r}) + (\boldsymbol{q}\cdot\nabla_{\boldsymbol{q}})\kappa_{ij,a}^{\boldsymbol{q}}(\boldsymbol{r})) \\ &= -2(M_{ij,a}(\boldsymbol{r}) - \kappa_{ij,a}^{\boldsymbol{q}}(\boldsymbol{r})) - q_a(\nabla_{\boldsymbol{q}}\cdot\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r}) + (\boldsymbol{q}\cdot\nabla_{\boldsymbol{q}})\kappa_{ij,a}^{\boldsymbol{q}}(\boldsymbol{r})),\end{aligned}$$

thus, by taking the $\boldsymbol{q}\to 0$ limit, and let

$$\boldsymbol{U}_{ij}(\boldsymbol{r}) = \frac{i}{2}\nabla_{\boldsymbol{q}}\times\left(e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r})\right), \tag{59}$$

we obtain that

$$\frac{i}{2}\nabla_{\boldsymbol{q}}\times\boldsymbol{\chi}_{ij}^{\boldsymbol{q}}(\boldsymbol{r})\big|_{\boldsymbol{q}\to 0} = -\boldsymbol{M}_{ij}(\boldsymbol{r}) + \boldsymbol{\kappa}_{ij}^{\boldsymbol{q}=0}(\boldsymbol{r}) - \nabla\times\boldsymbol{U}_{ij}(\boldsymbol{r}). \tag{60}$$

We integrate the above equation over the whole position space, and since

$$\begin{aligned}\int \mathrm{d}\boldsymbol{r}\ \boldsymbol{\chi}_{ij}^{\boldsymbol{q}}(\boldsymbol{r}) &= \int \mathrm{d}\boldsymbol{r}\mathrm{d}\boldsymbol{r}'\ \boldsymbol{\chi}_{ij}(\boldsymbol{r},\boldsymbol{r}')e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')} \\ &= \beta\int \mathrm{d}\boldsymbol{r}\mathrm{d}\boldsymbol{r}'\ e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\left\langle\hat{n}_j(\boldsymbol{r}');\hat{\boldsymbol{J}}_i(\boldsymbol{r})\right\rangle \\ &= \beta\left\langle\hat{n}_{j,-\boldsymbol{q}};\hat{\boldsymbol{J}}_{i,\boldsymbol{q}}\right\rangle,\end{aligned}$$

and by assumption that $\int \mathrm{d}\boldsymbol{r}\ \nabla\times\boldsymbol{U}_{ij}(\boldsymbol{r}) = 0$, we obtain that

$$\frac{i}{2}\nabla_{\boldsymbol{q}}\times\beta\left\langle\hat{n}_{j,-\boldsymbol{q}};\hat{\boldsymbol{J}}_{i,\boldsymbol{q}}\right\rangle\big|_{\boldsymbol{q}\to 0} = -\boldsymbol{M}_{ij} + \boldsymbol{\kappa}_{ij}^{\boldsymbol{q}=0}, \tag{61}$$

where $\boldsymbol{M}_{ij}$ is the total/bulk magnetization, and $\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}=0}$ is the static part of the susceptibility. Bring the Kubo canonical correlation form of $\boldsymbol{\kappa}$ into the expression, we have

$$-\left(\frac{\partial \boldsymbol{M}_c}{\partial \mu}\right)_T = \frac{\beta}{2i}\nabla_{\boldsymbol{q}} \times \langle \hat{N}_{-\boldsymbol{q}}; \hat{\boldsymbol{J}}_{c,\boldsymbol{q}} \rangle \Big|_{\boldsymbol{q}\to 0} \tag{62a}$$

$$\boldsymbol{M}_c - T\left(\frac{\partial \boldsymbol{M}_c}{\partial T}\right)_\mu = \frac{\beta}{2i}\nabla_{\boldsymbol{q}} \times \langle \hat{K}_{-\boldsymbol{q}}; \hat{\boldsymbol{J}}_{c,\boldsymbol{q}} \rangle \Big|_{\boldsymbol{q}\to 0} \tag{62b}$$

$$-\left(\frac{\partial \boldsymbol{M}_Q}{\partial \mu}\right)_T = \frac{\beta}{2i}\nabla_{\boldsymbol{q}} \times \langle \hat{N}_{-\boldsymbol{q}}; \hat{\boldsymbol{J}}_{Q,\boldsymbol{q}} \rangle \Big|_{\boldsymbol{q}\to 0}, \tag{62c}$$

$$2\boldsymbol{M}_Q - T\left(\frac{\partial \boldsymbol{M}_Q}{\partial T}\right)_\mu = \frac{\beta}{2i}\nabla_{\boldsymbol{q}} \times \langle \hat{K}_{-\boldsymbol{q}}; \hat{\boldsymbol{J}}_{Q,\boldsymbol{q}} \rangle \Big|_{\boldsymbol{q}\to 0}. \tag{62d}$$

We can re-interpret the right-hand side with the Maxwell relations by introducing

$$U = TS + \mu N - \boldsymbol{M}_c \cdot \boldsymbol{B} - \boldsymbol{M}_s \cdot \boldsymbol{B}_s,$$

and

$$\begin{cases} \Omega = U - TS - \mu N \\ K = U - \mu N \end{cases} \Rightarrow \begin{cases} \mathrm{d}\Omega = -S\mathrm{d}T - N\mathrm{d}\mu - \boldsymbol{M}_c \cdot \mathrm{d}\boldsymbol{B} - \boldsymbol{M}_s \cdot \mathrm{d}\boldsymbol{B}_s \\ \mathrm{d}K = T\mathrm{d}S - N\mathrm{d}\mu - \boldsymbol{M}_c \cdot \mathrm{d}\boldsymbol{B} - \boldsymbol{M}_s \cdot \mathrm{d}\boldsymbol{B}_s \end{cases},$$

where the heat magnetization is defined as $\boldsymbol{M}_s = \boldsymbol{M}_Q/T$ with $\boldsymbol{B}_s$ the heat magnetic field. We thus have

$$\boldsymbol{M}_c = -\left(\frac{\partial \Omega}{\partial \boldsymbol{B}}\right)_{T,\mu,\boldsymbol{B}_s},\ \boldsymbol{M}_s = -\left(\frac{\partial \Omega}{\partial \boldsymbol{B}_s}\right)_{T,\mu,\boldsymbol{B}},$$

and thus

$$\begin{aligned}
\frac{\partial \boldsymbol{M}_c}{\partial \mu} &= -\frac{\partial}{\partial \mu}\frac{\partial \Omega}{\partial \boldsymbol{B}} = -\frac{\partial}{\partial \boldsymbol{B}}\frac{\partial \Omega}{\partial \mu} = \frac{\partial N}{\partial \boldsymbol{B}}, \\
\frac{\partial \boldsymbol{M}_c}{\partial T} &= -\frac{\partial}{\partial T}\frac{\partial \Omega}{\partial \boldsymbol{B}} = -\frac{\partial}{\partial \boldsymbol{B}}\frac{\partial \Omega}{\partial T} = \frac{\partial S}{\partial \boldsymbol{B}}, \\
\frac{\partial \boldsymbol{M}_s}{\partial \mu} &= -\frac{\partial}{\partial \mu}\frac{\partial \Omega}{\partial \boldsymbol{B}_s} = -\frac{\partial}{\partial \boldsymbol{B}_s}\frac{\partial \Omega}{\partial \mu} = \frac{\partial N}{\partial \boldsymbol{B}_s}, \\
\frac{\partial \boldsymbol{M}_s}{\partial T} &= -\frac{\partial}{\partial T}\frac{\partial \Omega}{\partial \boldsymbol{B}_s} = -\frac{\partial}{\partial \boldsymbol{B}_s}\frac{\partial \Omega}{\partial T} = \frac{\partial S}{\partial \boldsymbol{B}_s}.
\end{aligned}$$

The first equation directly gives us the first equation in Eq. (62), and the second equation can be re-organized as

$$\boldsymbol{M}_c - T\frac{\partial \boldsymbol{M}_c}{\partial T} = \boldsymbol{M}_c - T\left(\frac{\partial S}{\partial \boldsymbol{B}}\right)_{T,\mu,\boldsymbol{B}_s} = -\left(\frac{\partial K}{\partial \boldsymbol{B}}\right)_{T,\mu,\boldsymbol{B}_s},$$

while for the third equation, we note

$$\left(\frac{\partial \boldsymbol{M}_Q}{\partial \mu}\right)_{T,\boldsymbol{B},\boldsymbol{B}_s} = T\left(\frac{\partial \boldsymbol{M}_s}{\partial \mu}\right)_{T,\boldsymbol{B},\boldsymbol{B}_s} = T\left(\frac{\partial N}{\partial \boldsymbol{B}_s}\right)_{T,\mu,\boldsymbol{B}},$$

and for the last equation, we have

$$\begin{aligned}\boldsymbol{M}_s - T\frac{\partial \boldsymbol{M}_s}{\partial T} &= \boldsymbol{M}_s - T\left(\frac{\partial S}{\partial \boldsymbol{B}_s}\right)_{T,\mu,\boldsymbol{B}} = -\left(\frac{\partial K}{\partial \boldsymbol{B}_s}\right)_{T,\mu,\boldsymbol{B}}, \\ &= \frac{\boldsymbol{M}_Q}{T} - \left(\frac{\partial \boldsymbol{M}_Q}{\partial T} - \frac{1}{T}\boldsymbol{M}_Q\right),\end{aligned}$$

and thus

$$2\boldsymbol{M}_Q - T\left(\frac{\partial \boldsymbol{M}_Q}{\partial T}\right)_{\mu,\boldsymbol{B},\boldsymbol{B}_s} = -T\left(\frac{\partial K}{\partial \boldsymbol{B}_s}\right)_{T,\mu,\boldsymbol{B}}.$$

In short, under the re-interpretation of the right-hand side of Eq. (62), we have

$$-\left(\frac{\partial N}{\partial \boldsymbol{B}}\right)_{T,\mu} = \frac{\beta}{2i}\nabla_{\boldsymbol{q}} \times \langle \hat{N}_{-\boldsymbol{q}}; \hat{\boldsymbol{J}}_{c,\boldsymbol{q}}\rangle \big|_{\boldsymbol{q}\to 0}, \tag{63a}$$

$$-\left(\frac{\partial K}{\partial \boldsymbol{B}}\right)_{T,\mu} = \frac{\beta}{2i}\nabla_{\boldsymbol{q}} \times \langle \hat{K}_{-\boldsymbol{q}}; \hat{\boldsymbol{J}}_{c,\boldsymbol{q}}\rangle \big|_{\boldsymbol{q}\to 0}, \tag{63b}$$

$$-T\left(\frac{\partial N}{\partial \boldsymbol{B}_s}\right)_{T,\mu} = \frac{\beta}{2i}\nabla_{\boldsymbol{q}} \times \langle \hat{N}_{-\boldsymbol{q}}; \hat{\boldsymbol{J}}_{Q,\boldsymbol{q}}\rangle \big|_{\boldsymbol{q}\to 0}, \tag{63c}$$

$$-T\left(\frac{\partial K}{\partial \boldsymbol{B}_s}\right)_{T,\mu} = \frac{\beta}{2i}\nabla_{\boldsymbol{q}} \times \langle \hat{K}_{-\boldsymbol{q}}; \hat{\boldsymbol{J}}_{Q,\boldsymbol{q}}\rangle \big|_{\boldsymbol{q}\to 0}. \tag{63d}$$

G. Linear Transport Response

Now we consider the linear transport response of the currents in conjugate with the forces by gradients of the chemical potential and temperature.

We firstly list the quantities of interest

$$\begin{aligned}\boldsymbol{\chi}^{\boldsymbol{q}}_{ij}(\boldsymbol{r}) &= -i\boldsymbol{q} \times \boldsymbol{M}_{ij}(\boldsymbol{r}) + e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\nabla \times \boldsymbol{\kappa}^{\boldsymbol{q}}_{ij}(\boldsymbol{r}), \\ \boldsymbol{\chi}_{ij}(\boldsymbol{r},\boldsymbol{r}') &= \int \frac{\mathrm{d}\boldsymbol{q}}{(2\pi)^d}\, \boldsymbol{\chi}^{\boldsymbol{q}}_{ij}(\boldsymbol{r}) e^{i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}, \\ \hat{\boldsymbol{J}}^{\phi,\psi}_c(\boldsymbol{r}) &= [1+\psi(\boldsymbol{r})]\hat{\boldsymbol{J}}_c(\boldsymbol{r}), \\ \hat{\boldsymbol{J}}^{\phi,\psi}_e(\boldsymbol{r}) &= [1+\psi(\boldsymbol{r})]^2[\hat{\boldsymbol{J}}_e(\boldsymbol{r}) + \phi(\boldsymbol{r})\hat{\boldsymbol{J}}_c(\boldsymbol{r})], \\ \hat{\boldsymbol{J}}^{\phi,\psi}_Q(\boldsymbol{r}) &= \hat{\boldsymbol{J}}^{\phi,\psi}_e(\boldsymbol{r}) - \alpha(\boldsymbol{r})\hat{\boldsymbol{J}}^{\phi,\psi}_c(\boldsymbol{r}), \\ \boldsymbol{J}^{eq}_{c/e}(\boldsymbol{r}) &= \nabla \times \boldsymbol{M}_{c/e}(\boldsymbol{r}),\end{aligned}$$

with generalized inverse temperature and chemical potential

$$\begin{cases}\xi(\boldsymbol{r}) = (1+\psi(\boldsymbol{r}))^{-1}\beta(\boldsymbol{r}), \\ \alpha(\boldsymbol{r}) = (1+\psi(\boldsymbol{r}))(\phi(\boldsymbol{r}) + \mu(\boldsymbol{r})).\end{cases}$$

From the scaling relations and definition of zero-field magnetization, we have

$$\boldsymbol{J}_c^{eq,\phi,\psi}(\boldsymbol{r}) = (1+\psi(\boldsymbol{r}))\nabla\times\boldsymbol{M}_c(\boldsymbol{r}), \tag{64a}$$

$$\boldsymbol{J}_e^{eq,\phi,\psi}(\boldsymbol{r}) = (1+\psi(\boldsymbol{r}))^2\left[\nabla\times\boldsymbol{M}_e(\boldsymbol{r}) + \phi(\boldsymbol{r})\nabla\times\boldsymbol{M}_c(\boldsymbol{r})\right], \tag{64b}$$

$$\boldsymbol{J}_Q^{eq,\phi,\psi}(\boldsymbol{r}) = (1+\psi(\boldsymbol{r}))^2\left[\nabla\times\boldsymbol{M}_e(\boldsymbol{r}) - \mu(\boldsymbol{r})\nabla\times\boldsymbol{M}_c(\boldsymbol{r})\right]. \tag{64c}$$

Then, we refer to the static linear response Eq. (48), and have

$$\begin{aligned}
\boldsymbol{J}_i^{leq,\phi,\psi}(\boldsymbol{r}) &= \boldsymbol{J}_i^{eq,\phi,\psi}(\boldsymbol{r}) + \sum_{j=1}^{2}\int \mathrm{d}\boldsymbol{r}'\ \boldsymbol{\chi}_{ij}(\boldsymbol{r},\boldsymbol{r}')x_j(\boldsymbol{r}') \\
&= \boldsymbol{J}_i^{eq,\phi,\psi}(\boldsymbol{r}) + \sum_{j=1}^{2}\int \mathrm{d}\boldsymbol{r}'\frac{\mathrm{d}\boldsymbol{q}}{(2\pi)^d}\ e^{i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\left(-i\boldsymbol{q}\times\boldsymbol{M}_{ij}(\boldsymbol{r}) + e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\nabla\times\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r})\right)x_j(\boldsymbol{r}') \\
&= \boldsymbol{J}_i^{eq,\phi,\psi}(\boldsymbol{r}) + \sum_{j=1}^{2}\int \mathrm{d}\boldsymbol{r}'\frac{\mathrm{d}\boldsymbol{q}}{(2\pi)^d}\ \left[\left(\nabla_{\boldsymbol{r}'}e^{i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\right)\times\boldsymbol{M}_{ij}(\boldsymbol{r}) + e^{-i\boldsymbol{q}\cdot\boldsymbol{r}'}\nabla\times\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r})\right]x_j(\boldsymbol{r}') \\
&= \boldsymbol{J}_i^{eq,\phi,\psi}(\boldsymbol{r}) + \sum_{j=1}^{2}\int \mathrm{d}\boldsymbol{r}'\frac{\mathrm{d}\boldsymbol{q}}{(2\pi)^d}\ \left[\left(\nabla_{\boldsymbol{r}'}e^{i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}x_j(\boldsymbol{r}')\right) - e^{i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\left(\nabla_{\boldsymbol{r}'}x_j(\boldsymbol{r}')\right)\right]\times\boldsymbol{M}_{ij}(\boldsymbol{r}) \\
&\qquad + \sum_{j=1}^{2}\nabla\times\int \mathrm{d}\boldsymbol{r}'\ \left[\int\frac{\mathrm{d}\boldsymbol{q}}{(2\pi)^d}\ e^{-i\boldsymbol{q}\cdot\boldsymbol{r}'}\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r})\right]x_j(\boldsymbol{r}') \\
&= \boldsymbol{J}_i^{eq,\phi,\psi}(\boldsymbol{r}) + \sum_{j=1}^{2}\left[\boldsymbol{M}_{ij}(\boldsymbol{r})\times\nabla x_j(\boldsymbol{r}) + \nabla\times\int \mathrm{d}\boldsymbol{r}'\ \boldsymbol{\kappa}_{ij}(\boldsymbol{r},\boldsymbol{r}')x_j(\boldsymbol{r}')\right],
\end{aligned} \tag{65}$$

where we have used integral by parts, and defined

$$\boldsymbol{\kappa}_{ij}(\boldsymbol{r},\boldsymbol{r}') = \int\frac{\mathrm{d}\boldsymbol{q}}{(2\pi)^d}\ e^{-i\boldsymbol{q}\cdot\boldsymbol{r}'}\boldsymbol{\kappa}_{ij}^{\boldsymbol{q}}(\boldsymbol{r}). \tag{66}$$

Now we can evaluate each current in term of magnetization, generalized force and static response function. To emphasize the global equilibrium state, we label its thermodynamic quantity with $_0$.

$$\begin{aligned}
\boldsymbol{J}_1^{leq,\phi,\psi}(\boldsymbol{r}) &= \boldsymbol{J}_c^{leq,\phi,\psi}(\boldsymbol{r}) \\
&= (1+\psi(\boldsymbol{r}))\nabla\times\boldsymbol{M}_c(\boldsymbol{r}) - \boldsymbol{M}_c(\boldsymbol{r})\times T_0\nabla\frac{1}{T(\boldsymbol{r})} + \sum_{j=1}^{2}\nabla\times\int \mathrm{d}\boldsymbol{r}'\ \boldsymbol{\kappa}_{1j}(\boldsymbol{r},\boldsymbol{r}')x_j(\boldsymbol{r}') \\
&= \nabla\times[(1+\psi(\boldsymbol{r}))\boldsymbol{M}_c(\boldsymbol{r})] - \boldsymbol{M}_c(\boldsymbol{r})\times\left(-\nabla\psi(\boldsymbol{r}) + T_0\nabla\frac{1}{T(\boldsymbol{r})}\right) + \cdots \\
&= \nabla\times\boldsymbol{M}_c^{\phi,\psi}(\boldsymbol{r}) - \frac{1}{\xi(\boldsymbol{r})}\boldsymbol{M}_c(\boldsymbol{r})\times\boldsymbol{X}_2(\boldsymbol{r}) + O(\delta^2),
\end{aligned}$$

where we have defined the generalized force by temperature gradient as

$$\begin{aligned}\boldsymbol{X}_2(\boldsymbol{r}) &= \nabla\xi(\boldsymbol{r}) = \nabla\frac{1}{(1+\psi(\boldsymbol{r}))k_BT(\boldsymbol{r})}\\ &= -\frac{\nabla\psi(\boldsymbol{r})}{(1+\psi(\boldsymbol{r}))^2k_BT(\boldsymbol{r})} + \frac{1}{(1+\psi(\boldsymbol{r}))k_B}\nabla\frac{1}{T(\boldsymbol{r})}\\ &= \xi(\boldsymbol{r})\left(-\frac{\nabla\psi(\boldsymbol{r})}{1+\psi(\boldsymbol{r})} + T(\boldsymbol{r})\nabla\frac{1}{T(\boldsymbol{r})}\right)\\ &= \xi(\boldsymbol{r})\left(-\nabla\psi(\boldsymbol{r}) + T_0\nabla\frac{1}{T(\boldsymbol{r})}\right) + O(\delta^2),\end{aligned} \tag{67}$$

and the charge magnetization under field in local equilibrium as

$$\begin{aligned}\boldsymbol{M}_c^{\phi,\psi}(\boldsymbol{r}) &\equiv \boldsymbol{M}_c^{\phi,\psi}(\boldsymbol{r}, T(\boldsymbol{r}), \mu(\boldsymbol{r}))\\ &= \boldsymbol{M}_c^{\phi,\psi}(\boldsymbol{r}, T_0, \mu_0) + \delta\boldsymbol{M}_c(\boldsymbol{r}, \delta T(\boldsymbol{r}), \delta\mu(\boldsymbol{r})) + O(\delta^2)\\ &= (1+\psi(\boldsymbol{r}))\boldsymbol{M}_c(\boldsymbol{r}) + \sum_{j=1}^{2}\nabla\times\int \mathrm{d}\boldsymbol{r}'\ \boldsymbol{\kappa}_{1j}(\boldsymbol{r},\boldsymbol{r}')x_j(\boldsymbol{r}') + O(\delta^2).\end{aligned} \tag{68}$$

Similarly but more complicatedly, we give the derivation of the second current, the heat current, and we will add or remove higher order terms at will.

$$\begin{aligned}\boldsymbol{J}_1^{leq,\phi,\psi}(\boldsymbol{r}) &= \boldsymbol{J}_Q^{leq,\phi,\psi}(\boldsymbol{r})\\ &= (1+\psi(\boldsymbol{r}))^2\left[\nabla\times\boldsymbol{M}_e(\boldsymbol{r}) - \mu(\boldsymbol{r})\nabla\times\boldsymbol{M}_c(\boldsymbol{r})\right]\\ &+ \boldsymbol{M}_c(\boldsymbol{r})\times\nabla\mu(\boldsymbol{r}) - 2\boldsymbol{M}_Q(\boldsymbol{r})\times T_0\nabla\frac{1}{T(\boldsymbol{r})} + \sum_{j=1}^{2}\nabla\times\int \mathrm{d}\boldsymbol{r}'\ \boldsymbol{\kappa}_{2j}(\boldsymbol{r},\boldsymbol{r}')x_j(\boldsymbol{r}')\\ &= \nabla\times[(1+\psi)^2\boldsymbol{M}_e] - 2(1+\psi)(\nabla\psi)\times\boldsymbol{M}_e - (1+\psi)^2\mu\nabla\times\boldsymbol{M}_c\\ &+ \nabla\times[(1+\psi)^2\phi\boldsymbol{M}_c] - (\nabla\phi)\times\boldsymbol{M}_c - \phi\nabla\times\boldsymbol{M}_c\\ &+ \boldsymbol{M}_c\times\nabla\mu - 2\boldsymbol{M}_Q\times T_0\nabla\frac{1}{T} + \nabla\times\boldsymbol{\kappa}_{2j}x_j + O(\delta^2)\\ &= \nabla\times\left[(1+\psi)^2(\boldsymbol{M}_e+\phi\boldsymbol{M}_c)\right] - (1+\psi)(\mu+\phi)\nabla\times[(1+\psi)\boldsymbol{M}_c] + \alpha(\nabla\psi)\times\boldsymbol{M}_c\\ &- 2\boldsymbol{M}_Q\times\left[-\nabla\psi + T_0\nabla\frac{1}{T}\right] + 2\mu\boldsymbol{M}_c\times\nabla\psi + \boldsymbol{M}_c\times[\nabla\phi+\nabla\mu] + \nabla\times\boldsymbol{\kappa}_{2j}x_j + O(\delta^2)\\ &= \nabla\times\left[(1+\psi)^2(\boldsymbol{M}_e+\phi\boldsymbol{M}_c)\right] - \alpha\nabla\times[(1+\psi)\boldsymbol{M}_c] - \frac{2}{\xi}\boldsymbol{M}_Q\times\boldsymbol{X}_2\\ &+ \boldsymbol{M}_c\times[\mu\nabla\psi + \nabla\phi + \nabla\mu] + \nabla\times\boldsymbol{\kappa}_{2j}x_j + O(\delta^2)\\ &= \nabla\times\boldsymbol{M}_e^{\phi,\psi}(\boldsymbol{r}) - \alpha(\boldsymbol{r})\nabla\times\boldsymbol{M}_c^{\phi,\psi}(\boldsymbol{r}) - \frac{1}{\xi(\boldsymbol{r})}\boldsymbol{M}_c(\boldsymbol{r})\times\boldsymbol{X}_1 - \frac{2}{\xi(\boldsymbol{r})}\boldsymbol{M}_Q(\boldsymbol{r})\times\boldsymbol{X}_2,\end{aligned}$$

where we have defined the generalized force by chemical potential gradient as

$$\begin{aligned}\boldsymbol{X}_1(\boldsymbol{r}) &= -\xi(\boldsymbol{r})\nabla\alpha(\boldsymbol{r}) \\ &= -\xi(\boldsymbol{r})\left[(\phi+\mu)\nabla\psi + (1+\psi)(\nabla\phi+\nabla\mu)\right] \\ &= -\xi(\boldsymbol{r})(\mu\nabla\psi + \nabla\phi + \nabla\mu) + O(\delta^2),\end{aligned} \tag{69}$$

and the energy magnetization under field in local equilibrium as

$$\begin{aligned}\boldsymbol{M}_e^{\phi,\psi}(\boldsymbol{r}) &\equiv \boldsymbol{M}_e^{\phi,\psi}(\boldsymbol{r}, T(\boldsymbol{r}), \mu(\boldsymbol{r})) \\ &= \boldsymbol{M}_e^{\phi,\psi}(\boldsymbol{r}, T_0, \mu_0) + \delta\boldsymbol{M}_e(\boldsymbol{r}, \delta T(\boldsymbol{r}), \delta\mu(\boldsymbol{r})) + O(\delta^2) \\ &= (1+\psi(\boldsymbol{r}))^2(\boldsymbol{M}_e(\boldsymbol{r}) + \phi(\boldsymbol{r})\boldsymbol{M}_c(\boldsymbol{r})) + \sum_{j=1}^{2}\nabla\times\int \mathrm{d}\boldsymbol{r}'\ \boldsymbol{\kappa}'_{2j}(\boldsymbol{r},\boldsymbol{r}')x_j(\boldsymbol{r}') + O(\delta^2),\end{aligned} \tag{70}$$

where the static response function for energy current is defined as

$$\boldsymbol{\kappa}'_{2j}(\boldsymbol{r},\boldsymbol{r}') = \boldsymbol{\kappa}_{2j}(\boldsymbol{r},\boldsymbol{r}') + \mu_0\boldsymbol{\kappa}_{1j}(\boldsymbol{r},\boldsymbol{r}'), \tag{71}$$

in accordance with the relation between the energy and heat currents.

In short, the static response of currents reads

$$\boldsymbol{J}_1^{leq,\phi,\psi}(\boldsymbol{r}) = \nabla\times\boldsymbol{M}_c^{\phi,\psi}(\boldsymbol{r}) - \frac{1}{\xi(\boldsymbol{r})}\boldsymbol{M}_c(\boldsymbol{r})\times\boldsymbol{X}_2(\boldsymbol{r}), \tag{72a}$$

$$\boldsymbol{J}_1^{leq,\phi,\psi}(\boldsymbol{r}) = \nabla\times\boldsymbol{M}_e^{\phi,\psi}(\boldsymbol{r}) - \alpha(\boldsymbol{r})\nabla\times\boldsymbol{M}_c^{\phi,\psi}(\boldsymbol{r}) - \frac{1}{\xi(\boldsymbol{r})}\boldsymbol{M}_c(\boldsymbol{r})\times\boldsymbol{X}_1 - \frac{2}{\xi(\boldsymbol{r})}\boldsymbol{M}_Q(\boldsymbol{r})\times\boldsymbol{X}_2, \tag{72b}$$

to linear order in the external fields and generalized forces. However, we see that there are terms not proportional to the generalized forces, but directly proportional to the magnetization. These terms are the so-called *magnetization currents* that do not contribute to the transport response, and we thus define the transport currents operator from static response as

$$\hat{\boldsymbol{J}}_{c/e}^{tr,\phi,\psi}(\boldsymbol{r}) = \hat{\boldsymbol{J}}_{c/e}^{\phi,\psi}(\boldsymbol{r}) - \nabla\times\boldsymbol{M}_{c/e}^{\phi,\psi}(\boldsymbol{r}), \tag{73}$$

which continues to the zero-field case and gives zero local transport current. This definition directly leads to the transport currents in local equilibrium with external fields as

$$\boldsymbol{J}_1^{tr,leq,\phi,\psi}(\boldsymbol{r}) = -\frac{1}{\xi(\boldsymbol{r})}\boldsymbol{M}_c(\boldsymbol{r})\times\boldsymbol{X}_2(\boldsymbol{r}), \tag{74a}$$

$$\boldsymbol{J}_1^{tr,leq,\phi,\psi}(\boldsymbol{r}) = -\frac{1}{\xi(\boldsymbol{r})}\boldsymbol{M}_c(\boldsymbol{r})\times\boldsymbol{X}_1 - \frac{2}{\xi(\boldsymbol{r})}\boldsymbol{M}_Q(\boldsymbol{r})\times\boldsymbol{X}_2. \tag{74b}$$

Thus, in combination with the time-dependent response from Kubo formula (assume local equilibrium thus no time dependence)

$$\begin{aligned}\boldsymbol{J}_i^{\mathrm{Kubo},\phi,\psi}(\boldsymbol{r}) &= \mathrm{Tr}\Big(\hat{\rho}_1(t)\hat{\boldsymbol{J}}_i^{\phi,\psi}(\boldsymbol{r})\Big) \\ &= \sum_{j=1}^{2}\frac{1}{i\hbar}\int \mathrm{d}\boldsymbol{r}'\int_0^{+\infty}\mathrm{d}t'\ e^{-\epsilon^+ t'}\left\langle[\hat{\boldsymbol{J}}_{I,i}(\boldsymbol{r},t'),\hat{\boldsymbol{J}}_j(\boldsymbol{r}')]\right\rangle_0\cdot\boldsymbol{X}_j(\boldsymbol{r}'),\end{aligned} \tag{75}$$

to the first order in the generalized forces (thus no field dependence). Thus, with

$$\boldsymbol{J}_i^{tr,\phi,\psi}(\boldsymbol{r}) = \boldsymbol{J}_i^{tr,leq,\phi,\psi}(\boldsymbol{r}) + \boldsymbol{J}_i^{\mathrm{Kubo},\phi,\psi}(\boldsymbol{r}) = \overset{\leftrightarrow}{\mathbf{L}}_{ij}\cdot\boldsymbol{X}_j(\boldsymbol{r}), \tag{76}$$

we arrive at the linear transport response of the currents as

$$\begin{pmatrix}\boldsymbol{J}_1^{tr}\\ \boldsymbol{J}_2^{tr}\end{pmatrix} = \begin{pmatrix}\overset{\leftrightarrow}{\boldsymbol{L}}^{(11)} & \overset{\leftrightarrow}{\boldsymbol{L}}^{(12)} - \dfrac{\boldsymbol{M}_c}{\xi V}\times \\ \overset{\leftrightarrow}{\boldsymbol{L}}^{(21)} - \dfrac{\boldsymbol{M}_c}{\xi V}\times & \overset{\leftrightarrow}{\boldsymbol{L}}^{(22)} - \dfrac{2\boldsymbol{M}_Q}{\xi V}\times\end{pmatrix}\begin{pmatrix}\boldsymbol{X}_1\\ \boldsymbol{X}_2\end{pmatrix}, \tag{77}$$

where quantities are integrated over the volume

$$\boldsymbol{J}_i^{tr} = \frac{1}{V}\int \mathrm{d}\boldsymbol{r}\ \boldsymbol{J}_i^{tr,\phi,\psi}(\boldsymbol{r}). \tag{78}$$

H. Formulation of Current Operators

Let us consider the Hamiltonian

$$\hat{h}_{\phi\psi}(\boldsymbol{r}) = (1+\psi(\boldsymbol{r}))\left(\frac{m}{2}[\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r})]^\dagger\cdot[\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r})] + \hat{\varphi}^\dagger(\boldsymbol{r})[V(\boldsymbol{r})+\phi(\boldsymbol{r})]\varphi(\boldsymbol{r})\right), \tag{79}$$

where the velocity operator is given by

$$\hat{\boldsymbol{v}} = \frac{1}{m}\left(-i\hbar\nabla + \overset{\leftrightarrow}{\boldsymbol{A}}(\boldsymbol{r})\right), \tag{80}$$

with $\overset{\leftrightarrow}{\boldsymbol{A}}(\boldsymbol{r})$ being the Hermitian non-Abelian gauge field describing the spin-orbital coupling, the Zeeman coupling and so on. The operator $\hat{\varphi}(\boldsymbol{r})$ is the fermionic field spinor, and $V(\boldsymbol{r})$ is the periodic lattice potential.

Since we want to derive the energy current operator, we need to consider the time derivative of the energy density operator — the Hamiltonian density operator above. To calculate that, we firstly derive the Heisenberg equation of motion for the fermion operator. Here, we need two commutators for fermions:

$$[\hat{\varphi}_a(\boldsymbol{r}),\hat{\varphi}_b^\dagger(\boldsymbol{r}')X_b(\boldsymbol{r}')] = \delta(\boldsymbol{r}-\boldsymbol{r}')X_a(\boldsymbol{r}), \tag{81a}$$

$$[\hat{\varphi}_a(\boldsymbol{r}), (\partial_i \hat{\varphi}_b^\dagger(\boldsymbol{r}'))X_b(\boldsymbol{r}')] = -\delta(\boldsymbol{r}-\boldsymbol{r}')\partial_i X_a(\boldsymbol{r}), \tag{81b}$$

where $X_b(\boldsymbol{r})$ is some operator that anticommutates with the fermion operator $\hat{\varphi}_a(\boldsymbol{r})$. An Einstein summation is implied over the repeated index b. The proof is straightforward,

$$\begin{aligned}
[\hat{\varphi}_a(\boldsymbol{r}), \hat{\varphi}_b^\dagger(\boldsymbol{r}')X_b(\boldsymbol{r}')] &= \hat{\varphi}_a(\boldsymbol{r})\hat{\varphi}_b^\dagger(\boldsymbol{r}')X_b(\boldsymbol{r}') - \hat{\varphi}_b^\dagger(\boldsymbol{r}')X_b(\boldsymbol{r}')\hat{\varphi}_a(\boldsymbol{r}) \\
&= (\hat{\varphi}_a(\boldsymbol{r})\hat{\varphi}_b^\dagger(\boldsymbol{r}') + \hat{\varphi}_b^\dagger(\boldsymbol{r}')\hat{\varphi}_a(\boldsymbol{r}))X_b(\boldsymbol{r}') \\
&= \delta_{ab}\delta(\boldsymbol{r}-\boldsymbol{r}')X_b(\boldsymbol{r}') = \delta(\boldsymbol{r}-\boldsymbol{r}')X_a(\boldsymbol{r}), \\
[\hat{\varphi}_a(\boldsymbol{r}), (\partial_i\hat{\varphi}_b^\dagger(\boldsymbol{r}'))X_b(\boldsymbol{r}')] &= [\hat{\varphi}_a(\boldsymbol{r}), \partial_i(\hat{\varphi}_b^\dagger(\boldsymbol{r}')X_b(\boldsymbol{r}')) - \hat{\varphi}_b^\dagger(\boldsymbol{r}')(\partial_i X_b(\boldsymbol{r}'))] \\
&= -[\hat{\varphi}_a(\boldsymbol{r}), \hat{\varphi}_b^\dagger(\boldsymbol{r}')(\partial_i X_b(\boldsymbol{r}'))] \\
&= -\delta(\boldsymbol{r}-\boldsymbol{r}')\partial_i X_a(\boldsymbol{r}),
\end{aligned}$$

where in the second proof we have dropped a total derivative term that does not contribute to the commutator by considering full space integration.

Now by explicitly expanding the Hamiltonian density operator, we have

$$\begin{aligned}
\hat{h}_{\phi\psi}(\boldsymbol{r}) &= (1+\psi(\boldsymbol{r}))\left(\frac{m}{2}[\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r})]^\dagger \cdot [\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r})] + \hat{\varphi}^\dagger(\boldsymbol{r})[V(\boldsymbol{r})+\phi(\boldsymbol{r})]\varphi(\boldsymbol{r})\right) \\
&= \frac{1+\psi}{2m}\left(i\hbar\partial_i\hat{\varphi}_a^\dagger + \hat{\varphi}_b^\dagger A_{ba}^i\right)\left(-i\hbar\partial_i\hat{\varphi}_a + A_{ac}^i\hat{\varphi}_c\right) + (1+\psi)\hat{\varphi}^\dagger(V+\phi)\hat{\varphi} \\
&= \frac{1+\psi}{2m}\left(\hbar^2\partial_i\hat{\varphi}_a^\dagger\partial_i\hat{\varphi}_a + i\hbar\left[(\partial_i\hat{\varphi}_a^\dagger)A_{ac}^i\hat{\varphi}_c - \hat{\varphi}_b^\dagger A_{ba}^i\partial_i\hat{\varphi}_a\right] + \hat{\varphi}_b^\dagger(A^i)_{bc}^2\hat{\varphi}_c\right) \\
&+ (1+\psi)\hat{\varphi}_a^\dagger(V+\phi)\hat{\varphi}_a,
\end{aligned} \tag{82}$$

where we have used the Hermitian property of the gauge field and Einstein summation over the repeated indices. Utilizing the commutators above, the Heisenberg equation of motion for the fermion operator is given by

$$\begin{aligned}
i\hbar\partial_t\hat{\varphi}_a(\boldsymbol{r}) &= [\hat{\varphi}_a(\boldsymbol{r}), \hat{H}_{\phi\psi}] = [\hat{\varphi}_a(\boldsymbol{r}), \hat{h}_{\phi\psi}(\boldsymbol{r})] \\
&= -\frac{\hbar^2}{2m}\partial_i\left[(1+\psi)\partial_i\hat{\varphi}_a\right] - \frac{i\hbar}{2m}\left(\partial_i\left[(1+\psi)A_{ac}^i\hat{\varphi}_c\right] + (1+\psi)A_{ac}^i\partial_i\hat{\varphi}_c\right) \\
&+ \frac{1+\psi}{2m}(A^i)_{ac}^2\hat{\varphi}_c + (1+\psi)(V+\phi)\hat{\varphi}_a,
\end{aligned}$$

which means that

$$\begin{aligned}
i\hbar\partial_t\hat{\varphi}(\boldsymbol{r}) &= \frac{1}{2m}\left[-\hbar^2\partial_i(1+\psi)\partial_i - i\hbar\partial_i(1+\psi)A^i - (1+\psi)A^i\partial_i + (1+\psi)A^iA^i\right]\hat{\varphi} \\
&\quad + (1+\psi)(V+\phi)\hat{\varphi} \\
&= \frac{1}{2m}(-i\hbar\nabla + \overset{\leftrightarrow}{\boldsymbol{A}})\cdot(1+\psi)(-i\hbar\nabla + \overset{\leftrightarrow}{\boldsymbol{A}})\hat{\varphi} + (1+\psi)(V+\phi)\hat{\varphi} \\
&= \left[\frac{m}{2}\hat{\boldsymbol{v}}\cdot(1+\psi(\boldsymbol{r}))\hat{\boldsymbol{v}} + (1+\psi(\boldsymbol{r}))(V(\boldsymbol{r})+\phi(\boldsymbol{r}))\right]\hat{\varphi}(\boldsymbol{r}) \\
&= \hat{\mathcal{H}}_{\phi\psi}(\boldsymbol{r})\hat{\varphi}(\boldsymbol{r}),
\end{aligned}$$

where the effective Hamiltonian density operator is given by

$$\hat{\mathcal{H}}_{\phi\psi}(\boldsymbol{r}) = \frac{m}{2}\hat{\boldsymbol{v}}\cdot(1+\psi(\boldsymbol{r}))\hat{\boldsymbol{v}} + (1+\psi(\boldsymbol{r}))(V(\boldsymbol{r})+\phi(\boldsymbol{r})). \tag{83}$$

With this, we can calculate the time derivative of the energy density operator, which is given by

$$\begin{aligned}
i\hbar\partial_t\hat{h}_{\phi\psi}(\boldsymbol{r}) &= (1+\psi)\left[\frac{m}{2}(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\cdot(\hat{\boldsymbol{v}}\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi}) + \hat{\varphi}^\dagger(V+\phi)(\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi}) - \text{h.c.}\right] \\
&= (1+\psi)\left\{\frac{1}{2}(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\cdot(-i\hbar\nabla + \overset{\leftrightarrow}{\boldsymbol{A}})(\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi}) - \frac{1}{2}\left[i\hbar\nabla(\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi})^\dagger + (\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi})^\dagger\overset{\leftrightarrow}{\boldsymbol{A}}\right]\right. \\
&\quad \left. + \hat{\varphi}^\dagger(V+\phi)(\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi}) + (\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi})^\dagger(V+\phi)\hat{\varphi}\right\} \\
&= -\frac{i\hbar}{2}\nabla\cdot\left\{(1+\psi)\left[(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger(\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi}) + (\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi})^\dagger\hat{\boldsymbol{v}}\hat{\varphi}\right]\right\} \\
&\quad + \frac{i\hbar}{2}\left\{\left[\nabla\cdot(1+\psi)(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\right]\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi} + (\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi})^\dagger\nabla\cdot(1+\psi)\hat{\boldsymbol{v}}\hat{\varphi}\right\} \\
&\quad + (1+\psi)\left\{\left[\frac{1}{2}(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\cdot\overset{\leftrightarrow}{\boldsymbol{A}} + \hat{\varphi}^\dagger(V+\phi)\right]\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi} - (\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi})^\dagger\left[\frac{1}{2}\overset{\leftrightarrow}{\boldsymbol{A}}\cdot\hat{\boldsymbol{v}} + V + \phi\right]\hat{\varphi}\right\} \\
&= -\frac{i\hbar}{2}\nabla\cdot\left\{(1+\psi)\left[(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger(\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi}) + (\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi})^\dagger\hat{\boldsymbol{v}}\hat{\varphi}\right]\right\} \\
&\quad + \left[\frac{1}{2}(i\hbar\nabla + \overset{\leftrightarrow}{\boldsymbol{A}})\cdot(1+\psi)(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger + \hat{\varphi}^\dagger(1+\psi)(V+\phi)\right]\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi} \\
&\quad - (\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi})^\dagger\left[\frac{1}{2}(-i\hbar\nabla + \overset{\leftrightarrow}{\boldsymbol{A}})\cdot(1+\psi)\hat{\boldsymbol{v}} + (1+\psi(\boldsymbol{r}))(V+\phi)\right]\hat{\varphi} \\
&= -\frac{i\hbar}{2}\nabla\cdot\left\{(1+\psi)\left[(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger(\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi}) + (\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi})^\dagger\hat{\boldsymbol{v}}\hat{\varphi}\right]\right\} \\
&\quad + \left\{\left[\frac{m}{2}\hat{\boldsymbol{v}}\cdot(1+\psi)\hat{\boldsymbol{v}} + (1+\psi)(V+\phi)\right]\hat{\varphi}\right\}^\dagger\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi} \\
&\quad - \hat{\mathcal{H}}_{\phi\psi}\hat{\varphi}\left[\frac{m}{2}\hat{\boldsymbol{v}}\cdot(1+\psi)\hat{\boldsymbol{v}} + (1+\psi)(V+\phi)\right]\hat{\varphi} \\
&= -i\hbar\nabla\cdot\hat{\boldsymbol{J}}_e^{\phi\psi} + (\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi})^\dagger\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi} - (\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi})^\dagger\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi} \\
&= -i\hbar\nabla\cdot\hat{\boldsymbol{J}}_e^{\phi\psi}(\boldsymbol{r}),
\end{aligned} \tag{84}$$

where we have defined the energy current operator

$$\hat{\boldsymbol{J}}_e^{\phi\psi}(\boldsymbol{r}) = \frac{1}{2}(1+\psi(\boldsymbol{r}))\left[(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{\mathcal{H}}_{\phi\psi}(\boldsymbol{r})\hat{\varphi}(\boldsymbol{r})) + (\hat{\mathcal{H}}_{\phi\psi}(\boldsymbol{r})\hat{\varphi}(\boldsymbol{r}))^\dagger\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r})\right]. \tag{85}$$

Note that the minus sign in the first line is due to conjugate of Schrödinger equation.

We now still need to verify the scaling relation of the energy current operator. And since we know that it contains charge current operator, it is wise to also calculate it. Consider the charge density operator $\hat{n}(\boldsymbol{r}) = \hat{\varphi}^\dagger(\boldsymbol{r})\hat{\varphi}(\boldsymbol{r})$

$$\begin{aligned}
i\hbar\partial_t\hat{n}(\boldsymbol{r}) &= i\hbar\left[-(\partial_t\hat{\varphi}^\dagger)\hat{\varphi} + \hat{\varphi}^\dagger\partial_t\hat{\varphi}\right] \\
&= -(\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi})^\dagger\hat{\varphi} + \hat{\varphi}^\dagger\hat{\mathcal{H}}_{\phi\psi}\hat{\varphi} \\
&= -\frac{m}{2}\left(\hat{\boldsymbol{v}}\cdot(1+\psi)\hat{\boldsymbol{v}}\hat{\varphi}\right)^\dagger\hat{\varphi} - (1+\psi)(V+\phi)\hat{\varphi}^\dagger\hat{\varphi} \\
&+ \frac{m}{2}\hat{\varphi}^\dagger\left(\hat{\boldsymbol{v}}\cdot(1+\psi)\hat{\boldsymbol{v}}\hat{\varphi}\right) + (1+\psi)(V+\phi)\hat{\varphi}^\dagger\hat{\varphi} \\
&= -\frac{1}{2}\left[i\hbar\left(\nabla\cdot(1+\psi)(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\right)\hat{\varphi} + \overset{\leftrightarrow}{\boldsymbol{A}}\cdot(1+\psi)(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\hat{\varphi}\right] \\
&+ \frac{1}{2}\left[\hat{\varphi}^\dagger(-i\hbar\nabla)\cdot(1+\psi)(\hat{\boldsymbol{v}}\hat{\varphi}) + \hat{\varphi}^\dagger\overset{\leftrightarrow}{\boldsymbol{A}}\cdot(1+\psi)(\hat{\boldsymbol{v}}\hat{\varphi})\right] \\
&= -\frac{i\hbar}{2}\nabla\cdot\left\{(1+\psi)\left[(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\hat{\varphi} + \hat{\varphi}^\dagger(\hat{\boldsymbol{v}}\hat{\varphi})\right]\right\} \\
&+ \frac{i\hbar}{2}(1+\psi)\left[(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\cdot\nabla\hat{\varphi} + (\nabla\hat{\varphi}^\dagger)\cdot\hat{\boldsymbol{v}}\hat{\varphi}\right] \\
&- \frac{1}{2}\overset{\leftrightarrow}{\boldsymbol{A}}\cdot(1+\psi)\left[(\hat{\boldsymbol{v}}\hat{\varphi})\hat{\varphi} + \hat{\varphi}^\dagger(\hat{\boldsymbol{v}}\hat{\varphi})\right] \\
&= -\frac{i\hbar}{2}\nabla\cdot\left\{(1+\psi)\left[(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\hat{\varphi} + \hat{\varphi}^\dagger(\hat{\boldsymbol{v}}\hat{\varphi})\right]\right\} \\
&+ \frac{1+\psi}{2}\left\{-(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger(-i\hbar\nabla + \overset{\leftrightarrow}{\boldsymbol{A}})\hat{\varphi} + [(i\hbar\nabla + \overset{\leftrightarrow}{\boldsymbol{A}})\hat{\varphi}^\dagger](\hat{\boldsymbol{v}}\hat{\varphi})\right\} \\
&= -i\hbar\nabla\cdot\hat{\boldsymbol{J}}_c^{\phi\psi} + \frac{1+\psi}{2}\left\{-(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger(\hat{\boldsymbol{v}}\hat{\varphi}) + (\hat{\boldsymbol{v}}\hat{\varphi})^\dagger(\hat{\boldsymbol{v}}\hat{\varphi})\right\} \\
&= -i\hbar\nabla\cdot\hat{\boldsymbol{J}}_c^{\phi\psi}(\boldsymbol{r}),
\end{aligned} \tag{86}$$

where we have defined the charge current operator

$$\hat{\boldsymbol{J}}_c^{\phi\psi}(\boldsymbol{r}) = \frac{1+\psi(\boldsymbol{r})}{2}\left[(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))^\dagger\hat{\varphi}(\boldsymbol{r}) + \hat{\varphi}^\dagger(\boldsymbol{r})(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))\right], \tag{87}$$

which naturally satisfies the scaling relation

$$\hat{\boldsymbol{J}}_c^{\phi\psi}(\boldsymbol{r}) = (1+\psi(\boldsymbol{r}))\hat{\boldsymbol{J}}_c(\boldsymbol{r}),\ \hat{\boldsymbol{J}}_c(\boldsymbol{r}) = \frac{1}{2}\left[(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))^\dagger\hat{\varphi}(\boldsymbol{r}) + \hat{\varphi}^\dagger(\boldsymbol{r})(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))\right]. \tag{88}$$

To verify the scaling relation of the energy current operator, let us firstly denote

$$\hat{H}_0 = \hat{\mathcal{H}}_{00} = \frac{m}{2}\hat{\boldsymbol{v}}^2 + V(\boldsymbol{r}), \tag{89}$$

and assuming

$$\begin{aligned}\hat{\mathcal{H}}_{\phi\psi} &= (1+\psi(\boldsymbol{r}))(\hat{H}_0+\phi(\boldsymbol{r})) - \hat{A} \\ &= (1+\psi(\boldsymbol{r}))\frac{m}{2}\hat{\boldsymbol{v}}^2 + (1+\psi(\boldsymbol{r}))(V(\boldsymbol{r})+\phi(\boldsymbol{r})) - \hat{A},\end{aligned}$$

thus leading to

$$\hat{A} = (1+\psi(\boldsymbol{r}))\frac{m}{2}\hat{\boldsymbol{v}}^2 - \frac{m}{2}\hat{\boldsymbol{v}}\cdot(1+\psi(\boldsymbol{r}))\hat{\boldsymbol{v}} = \frac{i\hbar}{2}\nabla\psi(\boldsymbol{r})\cdot\hat{\boldsymbol{v}},$$

and suggests us to write down

$$\hat{\mathcal{H}}_{\phi\psi} = (1+\psi(\boldsymbol{r}))(\hat{H}_0+\phi(\boldsymbol{r})) - \frac{i\hbar}{2}\nabla\psi(\boldsymbol{r})\cdot\hat{\boldsymbol{v}}. \tag{90}$$

In this way, we can rewrite the energy current operator as

$$\begin{aligned}\hat{\boldsymbol{J}}_e^{\phi\psi}(\boldsymbol{r}) &= \frac{1}{2}(1+\psi(\boldsymbol{r}))\left[(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{\mathcal{H}}_{\phi\psi}(\boldsymbol{r})\hat{\varphi}(\boldsymbol{r})) + (\hat{\mathcal{H}}_{\phi\psi}(\boldsymbol{r})\hat{\varphi}(\boldsymbol{r}))^\dagger\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r})\right] \\ &= \frac{1+\psi}{2}\left\{(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\left[(1+\psi)(\hat{H}_0+\phi)\hat{\varphi} - \frac{i\hbar}{2}\nabla\psi\cdot\hat{\boldsymbol{v}}\hat{\varphi}\right]\right. \\ &\left. + \left[(1+\psi)\left((\hat{H}_0+\phi)\hat{\varphi}\right)^\dagger + \frac{i\hbar}{2}(\nabla\psi)\cdot(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\right](\hat{\boldsymbol{v}}\hat{\varphi})\right\} \\ &= \frac{(1+\psi)^2}{2}\left\{(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger(\hat{H}_0\hat{\varphi}) + (\hat{H}_0\hat{\varphi})^\dagger(\hat{\boldsymbol{v}}\hat{\varphi}) + \phi\left[(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\hat{\varphi} + \hat{\varphi}^\dagger(\hat{\boldsymbol{v}}\hat{\varphi})\right]\right\} \\ &- \frac{i\hbar}{4}(1+\psi)\left[(\nabla\psi\cdot(\hat{\boldsymbol{v}}\hat{\varphi}))(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger - \left(\nabla\psi\cdot(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\right)(\hat{\boldsymbol{v}}\hat{\varphi})\right] \\ &= (1+\psi(\boldsymbol{r}))^2\left(\hat{\boldsymbol{J}}_e(\boldsymbol{r}) + \phi(\boldsymbol{r})\hat{\boldsymbol{J}}_c(\boldsymbol{r})\right) + \left[\nabla(1+\psi(\boldsymbol{r}))^2\right]\times\hat{\boldsymbol{\Lambda}}(\boldsymbol{r}),\end{aligned} \tag{91}$$

where we have defined

$$\hat{\boldsymbol{J}}_e(\boldsymbol{r}) = \hat{\boldsymbol{J}}_e^{00}(\boldsymbol{r}) = \frac{1}{2}\left[(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger(\hat{H}_0\hat{\varphi}) + (\hat{H}_0\hat{\varphi})^\dagger(\hat{\boldsymbol{v}}\hat{\varphi})\right], \tag{92a}$$

$$\hat{\boldsymbol{\Lambda}}(\boldsymbol{r}) = -\frac{i\hbar}{8}(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\times(\hat{\boldsymbol{v}}\hat{\varphi}). \tag{92b}$$

It is thus clear that there exists an extra term that prohibits the scaling relation of the energy current operator, and we need to redefine the energy current operator satisfying the scaling relation as

$$\hat{\boldsymbol{J}}_e^s(\boldsymbol{r}) = \hat{\boldsymbol{J}}_e(\boldsymbol{r}) - \nabla\times\hat{\boldsymbol{\Lambda}}(\boldsymbol{r}), \tag{93a}$$

$$\hat{\boldsymbol{J}}_e^{\phi\psi,s}(\boldsymbol{r}) = \hat{\boldsymbol{J}}_e^{\phi\psi}(\boldsymbol{r}) - \nabla\times[(1+\psi)^2\hat{\boldsymbol{\Lambda}}(\boldsymbol{r})], \tag{93b}$$

such that

$$\begin{aligned}\hat{\boldsymbol{J}}_e^{\phi\psi,s}(\boldsymbol{r}) &= (1+\psi(\boldsymbol{r}))^2\left(\hat{\boldsymbol{J}}_e(\boldsymbol{r})+\phi(\boldsymbol{r})\hat{\boldsymbol{J}}_c(\boldsymbol{r})\right)+\left[\nabla(1+\psi(\boldsymbol{r}))^2\right]\times\hat{\boldsymbol{\Lambda}}(\boldsymbol{r})-\nabla\times[(1+\psi)^2\hat{\boldsymbol{\Lambda}}(\boldsymbol{r})]\\ &= (1+\psi(\boldsymbol{r}))^2\left(\hat{\boldsymbol{J}}_e(\boldsymbol{r})+\phi(\boldsymbol{r})\hat{\boldsymbol{J}}_c(\boldsymbol{r})\right)-(1+\psi(\boldsymbol{r}))^2\nabla\times\hat{\boldsymbol{\Lambda}}(\boldsymbol{r})\\ &= (1+\psi(\boldsymbol{r}))^2\left(\hat{\boldsymbol{J}}_e^s(\boldsymbol{r})+\phi(\boldsymbol{r})\hat{\boldsymbol{J}}_c(\boldsymbol{r})\right).\end{aligned}$$

Note that the curl term vanishes in the divergence of the energy current operator, thus does not affect the continuity equation.

As a summary, we write

$$\hat{\boldsymbol{J}}_c(\boldsymbol{r}) = \frac{1}{2}\left[(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))^\dagger\hat{\varphi}(\boldsymbol{r})+\hat{\varphi}^\dagger(\boldsymbol{r})(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))\right], \tag{94a}$$

$$\hat{\boldsymbol{J}}_e^s(\boldsymbol{r}) = \frac{1}{2}\left[(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{H}_0\hat{\varphi}(\boldsymbol{r}))+(\hat{H}_0\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))\right]-\nabla\times\hat{\boldsymbol{\Lambda}}(\boldsymbol{r}), \tag{94b}$$

$$\hat{\boldsymbol{J}}_c^{\phi\psi}(\boldsymbol{r}) = (1+\psi(\boldsymbol{r}))\hat{\boldsymbol{J}}_c(\boldsymbol{r}), \tag{94c}$$

$$\hat{\boldsymbol{J}}_e^{\phi\psi,s}(\boldsymbol{r}) = (1+\psi(\boldsymbol{r}))^2\left(\hat{\boldsymbol{J}}_e^s(\boldsymbol{r})+\phi(\boldsymbol{r})\hat{\boldsymbol{J}}_c(\boldsymbol{r})\right), \tag{94d}$$

$$\hat{\boldsymbol{J}}_Q^{\phi\psi}(\boldsymbol{r}) = (1+\psi(\boldsymbol{r}))^2\left(\hat{\boldsymbol{J}}_e^s(\boldsymbol{r})-\mu(\boldsymbol{r})\hat{\boldsymbol{J}}_c(\boldsymbol{r})\right), \tag{94e}$$

where $\hat{\boldsymbol{\Lambda}}(\boldsymbol{r}) = -\frac{i\hbar}{8}(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\times(\hat{\boldsymbol{v}}\hat{\varphi})$, and $\hat{H}_0$ is the unperturbed Hamiltonian density operator, which can be general. The quasi-particle velocity operator can be generally given by

$$\hat{\boldsymbol{v}} = \dot{\hat{\boldsymbol{r}}} = \frac{1}{i\hbar}[\hat{\boldsymbol{r}},\hat{H}_0]\longrightarrow\frac{1}{\hbar}\nabla_{\boldsymbol{k}}\hat{H}_0. \tag{95}$$

I. Thermal-Hall Coefficients from Microscopic States

With the formulas and current operators defined in the previous sections, we can now compute the thermal-Hall coefficients from microscopic states, and we will see how the magnetization helps to recover the Wiedemann-Franz law. At zero field, apply a y-directional temperature gradient, the current-force relation from Eq. (77) tells that

$$J_Q^x = J_{2,x}^{tr} = L_{xy}^{(22)}X_{2,y}-\frac{2(\boldsymbol{M}_Q\times\boldsymbol{X}_2)_x}{\xi V} = \left(L_{xy}^{(22)}+\frac{2M_Q^z}{\beta V}\right)X_{2,y}. \tag{96}$$

From the above deduction, we have the contribution from Kubo formula as

$$L_{\alpha\beta} = \frac{\hbar}{iV}\sum_{mnk}\frac{f_m(k)-f_n(k)}{(\varepsilon_{mk}-\varepsilon_{nk})^2}\langle mk|\hat{j}_\beta(k)|nk\rangle\langle nk|\hat{j}_\alpha(k)|mk\rangle, \tag{97}$$

indicating that $J_\alpha^{\rm Kubo} = L_{\alpha\beta}(-\partial_\beta\psi)$. However, in expression of $\boldsymbol{X}_2$, we see the combination $-\nabla\psi(\boldsymbol{r}) + T_0\nabla\frac{1}{T(\boldsymbol{r})}$, thus to arrive at the thermal Hall coefficient given by $\boldsymbol{J} = -\kappa\nabla T$, we equal the two sides by

$$L_{\alpha\beta}(-\partial_\beta\psi) = L_{\alpha\beta}T\partial_\beta\frac{1}{T} = -L_{\alpha\beta}\frac{1}{T}\partial_\beta T = -\kappa\partial_\beta T, \tag{98}$$

thus correct the Kubo formula by

$$\kappa_{\alpha\beta}^{\rm Kubo} = \frac{1}{T}L_{\alpha\beta} = \frac{\hbar}{iVT}\sum_{mn\boldsymbol{k}}\frac{f_m(\boldsymbol{k}) - f_n(\boldsymbol{k})}{(\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}})^2}\left\langle\psi_{m\boldsymbol{k}}|\hat{J}_Q^\beta(\boldsymbol{k})|\psi_{n\boldsymbol{k}}\right\rangle\left\langle\psi_{n\boldsymbol{k}}|\hat{J}_Q^\alpha(\boldsymbol{k})|\psi_{m\boldsymbol{k}}\right\rangle, \tag{99}$$

with $|\psi_{m\boldsymbol{k}}\rangle$ the Bloch states.

Now we write down the thermal current operator at zero fields and constant potentials,

$$\begin{aligned}
\hat{\boldsymbol{J}}_Q(\boldsymbol{r}) &= \hat{\boldsymbol{J}}_e^s(\boldsymbol{r}) - \mu\hat{\boldsymbol{J}}_c(\boldsymbol{r}) \\
&= \frac{1}{2}\left[(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{K}\hat{\varphi}(\boldsymbol{r})) + (\hat{K}\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))\right] - \nabla\times\hat{\boldsymbol{\Lambda}}(\boldsymbol{r}) \\
&= \frac{(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{K}\hat{\varphi}(\boldsymbol{r})) + (\hat{K}\hat{\varphi}(\boldsymbol{v}))^\dagger(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))}{2} + \frac{i\hbar}{8}\partial_j\left[(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{v}_j\hat{\varphi}(\boldsymbol{r})) - (\hat{v}_j\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))\right],
\end{aligned} \tag{100}$$

where we have calculated

$$\begin{aligned}
\nabla\times\hat{\boldsymbol{\Lambda}} &= -\frac{i\hbar}{8}\nabla\times\left[(\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\times(\hat{\boldsymbol{v}}\hat{\varphi})\right] \\
&= -\frac{i\hbar}{8}\hat{e}_i\varepsilon_{ijk}\partial_j\left((\hat{\boldsymbol{v}}\hat{\varphi})^\dagger\times(\hat{\boldsymbol{v}}\hat{\varphi})\right)_k \\
&= -\frac{i\hbar}{8}\hat{e}_i\varepsilon_{ijk}\varepsilon_{kmn}\partial_j\left[(\hat{v}_m\hat{\varphi})^\dagger(\hat{v}_n\hat{\varphi})\right] \\
&= -\frac{i\hbar}{8}\hat{e}_i(\delta_{im}\delta_{jn} - \delta_{in}\delta_{jm})\partial_j\left[(\hat{v}_m\hat{\varphi})^\dagger(\hat{v}_n\hat{\varphi})\right] \\
&= -\frac{i\hbar}{8}\hat{e}_i\partial_j\left[(\hat{v}_i\hat{\varphi})^\dagger(\hat{v}_j\hat{\varphi}) - (\hat{v}_j\hat{\varphi})^\dagger(\hat{v}_i\hat{\varphi})\right].
\end{aligned}$$

With this formula, we need to calculate its matrix element over Bloch states, and we need to find its $\boldsymbol{k}$-space representation. To this, let us introduce the Fourier transform of the field operator,

$$\hat{\varphi}(\boldsymbol{r}) = \frac{1}{\sqrt{V}}\sum_{\boldsymbol{k}}e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}}, \tag{101}$$

and the first term in the current operator becomes

$$
\begin{aligned}
&\int \mathrm{d}\boldsymbol{r}\frac{(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{K}\hat{\varphi}(\boldsymbol{r}))+(\hat{K}\hat{\varphi}(\boldsymbol{v}))^\dagger(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))}{2}\\
=&\frac{1}{V}\int \mathrm{d}\boldsymbol{r}\frac{(\hat{\boldsymbol{v}}\sum_{\boldsymbol{k}}e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}})^\dagger(\hat{K}\sum_{\boldsymbol{k}'}e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}'})+(\hat{K}\sum_{\boldsymbol{k}}e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}})^\dagger(\hat{\boldsymbol{v}}\sum_{\boldsymbol{k}'}e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}'})}{2}\\
=&\sum_{\boldsymbol{k}}\sum_{\boldsymbol{k}'}\hat{\varphi}^\dagger_{\boldsymbol{k}}\frac{1}{V}\int\frac{\hat{\boldsymbol{v}}_{\boldsymbol{k}}e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\hat{K}(\boldsymbol{r})e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}+e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\hat{K}(\boldsymbol{r})e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\hat{\boldsymbol{v}}_{\boldsymbol{k}'}}{2}\hat{\varphi}_{\boldsymbol{k}'}\\
=&\sum_{\boldsymbol{k}}\sum_{\boldsymbol{k}'}\hat{\varphi}^\dagger_{\boldsymbol{k}}\frac{\hat{\boldsymbol{v}}_{\boldsymbol{k}}\hat{K}(\boldsymbol{k})+\hat{K}(\boldsymbol{k})\hat{\boldsymbol{v}}_{\boldsymbol{k}'}}{2}\delta_{\boldsymbol{k},\boldsymbol{k}'}\hat{\varphi}_{\boldsymbol{k}'}\\
=&\sum_{\boldsymbol{k}}\hat{\varphi}^\dagger_{\boldsymbol{k}}\frac{1}{2}\{\hat{\boldsymbol{v}}_{\boldsymbol{k}},\hat{K}(\boldsymbol{k})\}\hat{\varphi}_{\boldsymbol{k}},
\end{aligned}
$$

where we have utilized $\hat{\boldsymbol{v}}(-i\nabla)e^{i\boldsymbol{k}\cdot\boldsymbol{r}}=\hat{\boldsymbol{v}}(\boldsymbol{k})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$, and relied on the periodicity of grand potential. The second term in the current operator can be calculated similarly,

$$
\begin{aligned}
\int \mathrm{d}\boldsymbol{r}\nabla\times\hat{\boldsymbol{\Lambda}}&=-\frac{i\hbar}{8}\hat{e}_i\int\mathrm{d}\boldsymbol{r}\partial_j\left[(\hat{v}_i\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{v}_j\hat{\varphi}(\boldsymbol{r}))-(\hat{v}_j\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{v}_i\hat{\varphi}(\boldsymbol{r}))\right]\\
&=-\frac{i\hbar}{8V}\hat{e}_i\int\mathrm{d}\boldsymbol{r}\partial_j\left[(\hat{v}_i\sum_{\boldsymbol{k}}e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}})^\dagger(\hat{v}_j\sum_{\boldsymbol{k}'}e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}'})-(\hat{v}_j\sum_{\boldsymbol{k}}e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}})^\dagger(\hat{v}_i\sum_{\boldsymbol{k}'}e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}'})\right]\\
&=-\frac{i\hbar}{8V}\hat{e}_i\sum_{\boldsymbol{k}}\sum_{\boldsymbol{k}'}\hat{\varphi}^\dagger_{\boldsymbol{k}}\int\mathrm{d}\boldsymbol{r}\partial_j\left[\hat{v}^i_{\boldsymbol{k}}e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\hat{v}^j_{\boldsymbol{k}'}-\hat{v}^j_{\boldsymbol{k}}e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\hat{v}^i_{\boldsymbol{k}'}\right]\hat{\varphi}_{\boldsymbol{k}'}\\
&=-\frac{\hbar}{8V}\hat{e}_i\sum_{\boldsymbol{k}}\sum_{\boldsymbol{k}'}\hat{\varphi}^\dagger_{\boldsymbol{k}}(k_j-k'_j)\int\mathrm{d}\boldsymbol{r}e^{-i(\boldsymbol{k}-\boldsymbol{k}')\cdot\boldsymbol{r}}\left[\hat{v}^i_{\boldsymbol{k}}\hat{v}^j_{\boldsymbol{k}'}-\hat{v}^j_{\boldsymbol{k}}\hat{v}^i_{\boldsymbol{k}'}\right]\hat{\varphi}_{\boldsymbol{k}'}\\
&=-\frac{\hbar}{8}\hat{e}_i\sum_{\boldsymbol{k}}\sum_{\boldsymbol{k}'}\hat{\varphi}^\dagger_{\boldsymbol{k}}(k_j-k'_j)\delta_{\boldsymbol{k}\boldsymbol{k}'}[\hat{v}^i_{\boldsymbol{k}},\hat{v}^j_{\boldsymbol{k}}]\hat{\varphi}_{\boldsymbol{k}'}=0,
\end{aligned}
$$

which can also be seen since the term is a total derivative and thus does not contribute to global response. Thus the $\boldsymbol{k}$-space representation of the thermal current operator is

$$
\hat{\boldsymbol{J}}_Q(\boldsymbol{k})=\frac{1}{2}\{\hat{\boldsymbol{v}}_{\boldsymbol{k}},\hat{K}(\boldsymbol{k})\}. \tag{102}
$$

Bring this into the Kubo formula, and we can replace Bloch states with its periodic one,

thus we have

$$\begin{aligned}
\kappa_{\alpha\beta}^{\mathrm{Kubo}} &= \frac{\hbar}{iVT}\sum_{mn\boldsymbol{k}}\frac{f_m(\boldsymbol{k})-f_n(\boldsymbol{k})}{(\varepsilon_{m\boldsymbol{k}}-\varepsilon_{n\boldsymbol{k}})^2}\langle u_{m\boldsymbol{k}}|\hat{J}_Q^{\beta}(\boldsymbol{k})|u_{n\boldsymbol{k}}\rangle\langle u_{n\boldsymbol{k}}|\hat{J}_Q^{\alpha}(\boldsymbol{k})|u_{m\boldsymbol{k}}\rangle \\
&= \frac{\hbar}{iVT}\sum_{mn\boldsymbol{k}}\frac{f_m(\boldsymbol{k})-f_n(\boldsymbol{k})(\varepsilon_{m\boldsymbol{k}}+\varepsilon_{n\boldsymbol{k}}-2\mu)^2}{4(\varepsilon_{m\boldsymbol{k}}-\varepsilon_{n\boldsymbol{k}})^2}\langle u_{m\boldsymbol{k}}|\hat{v}_{\boldsymbol{k}}^{\beta}|u_{n\boldsymbol{k}}\rangle\langle u_{n\boldsymbol{k}}|\hat{v}_{\boldsymbol{k}}^{\alpha}|u_{m\boldsymbol{k}}\rangle \\
&= \frac{\mathrm{Im}}{2\hbar VT}\sum_{mn\boldsymbol{k}}\frac{f_m(\boldsymbol{k})(\varepsilon_{m\boldsymbol{k}}+\varepsilon_{n\boldsymbol{k}}-2\mu)^2}{(\varepsilon_{m\boldsymbol{k}}-\varepsilon_{n\boldsymbol{k}})^2}\langle u_{m\boldsymbol{k}}|\partial_\beta h_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle\langle u_{n\boldsymbol{k}}|\partial_\alpha h_{\boldsymbol{k}}|u_{m\boldsymbol{k}}\rangle \\
&= \frac{\mathrm{Im}}{2\hbar VT}\sum_{mn\boldsymbol{k}}f_m(\boldsymbol{k})(\varepsilon_{m\boldsymbol{k}}+\varepsilon_{n\boldsymbol{k}}-2\mu)^2\langle\partial_\beta u_{m\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle\langle u_{n\boldsymbol{k}}|\partial_\alpha u_{m\boldsymbol{k}}\rangle \\
&= \frac{\mathrm{Im}}{2\hbar VT}\sum_{mn\boldsymbol{k}}f_m(\boldsymbol{k})\langle\partial_\beta u_{m\boldsymbol{k}}|(\varepsilon_{m\boldsymbol{k}}+h_{\boldsymbol{k}}-2\mu)^2|u_{n\boldsymbol{k}}\rangle\langle u_{n\boldsymbol{k}}|\partial_\alpha u_{m\boldsymbol{k}}\rangle \\
&= \frac{\mathrm{Im}}{2\hbar VT}\sum_{m\boldsymbol{k}}f_m(\boldsymbol{k})\langle\partial_\beta u_{m\boldsymbol{k}}|(\varepsilon_{m\boldsymbol{k}}+h_{\boldsymbol{k}}-2\mu)^2|\partial_\alpha u_{m\boldsymbol{k}}\rangle,
\end{aligned} \tag{103}$$

where in the forth equality we have used the equality by considering

$$\begin{aligned}
&\partial_\alpha(h_{\boldsymbol{k}}|u_{m\boldsymbol{k}}\rangle) = (\partial_\alpha h_{\boldsymbol{k}})|u_{m\boldsymbol{k}}\rangle + h_{\boldsymbol{k}}\partial_\alpha|u_{m\boldsymbol{k}}\rangle \\
=&\partial_\alpha(\varepsilon_{m\boldsymbol{k}}|u_{m\boldsymbol{k}}\rangle) = (\partial_\alpha\varepsilon_{m\boldsymbol{k}})|u_{m\boldsymbol{k}}\rangle + \varepsilon_{m\boldsymbol{k}}\partial_\alpha|u_{m\boldsymbol{k}}\rangle,
\end{aligned}$$

and a left action of $\langle u_{n\boldsymbol{k}}|$ with $n\neq m$ gives

$$\langle u_{n\boldsymbol{k}}|\partial_\alpha h_{\boldsymbol{k}}|u_{m\boldsymbol{k}}\rangle = (\varepsilon_{m\boldsymbol{k}}-\varepsilon_{n\boldsymbol{k}})\langle u_{n\boldsymbol{k}}|\partial_\alpha u_{m\boldsymbol{k}}\rangle, \tag{104}$$

and with its conjugate, we arrive at the forth equality. Note that if we left act with $\langle u_{m\boldsymbol{k}}|$, we will arrive at the Feynman-Hellman theorem,

$$\langle u_{m\boldsymbol{k}}|\partial_\alpha h_{\boldsymbol{k}}|u_{m\boldsymbol{k}}\rangle = \partial_\alpha\varepsilon_{m\boldsymbol{k}}. \tag{105}$$

For later convenience, we write the expression in a way such that the Fermi-Dirac distribution does not directly depend on the specific state, by

$$m_2^{\alpha\beta}(\varepsilon) = \frac{1}{\hbar}\,\mathrm{Im}\sum_{n\boldsymbol{k}}\langle\partial_\beta u_{n\boldsymbol{k}}|(\hat{h}_{\boldsymbol{k}}-\varepsilon)^2|\partial_\alpha u_{n\boldsymbol{k}}\rangle\,\delta(\varepsilon-\varepsilon_{n\boldsymbol{k}}), \tag{106a}$$

$$m_1^{\alpha\beta}(\varepsilon) = \frac{1}{\hbar}\,\mathrm{Im}\sum_{n\boldsymbol{k}}\langle\partial_\beta u_{n\boldsymbol{k}}|(\hat{h}_{\boldsymbol{k}}-\varepsilon)|\partial_\alpha u_{n\boldsymbol{k}}\rangle\,\delta(\varepsilon-\varepsilon_{n\boldsymbol{k}}), \tag{106b}$$

$$\omega^{\alpha\beta}(\varepsilon) = \frac{2}{\hbar}\,\mathrm{Im}\sum_{n\boldsymbol{k}}\langle\partial_\beta u_{n\boldsymbol{k}}|\partial_\alpha u_{n\boldsymbol{k}}\rangle\,\delta(\varepsilon-\varepsilon_{n\boldsymbol{k}}), \tag{106c}$$

thus writing

$$\kappa_{\alpha\beta}^{\mathrm{Kubo}} = \frac{1}{2VT}\int\mathrm{d}\varepsilon\; f(\varepsilon)\left(m_2^{\alpha\beta}(\varepsilon)+4(\varepsilon-\mu)m_1^{\alpha\beta}(\varepsilon)+2(\varepsilon-\mu)^2\omega^{\alpha\beta}(\varepsilon)\right). \tag{107}$$

Next, we need to calculate the magnetization contribution, which is given by

$$2\boldsymbol{M}_Q - T\left(\frac{\partial \boldsymbol{M}_Q}{\partial T}\right)_\mu = \frac{\beta}{2i}\nabla_{\boldsymbol{q}} \times \langle \hat{K}_{-\boldsymbol{q}}; \hat{\boldsymbol{J}}_{Q,\boldsymbol{q}} \rangle \big|_{\boldsymbol{q}\to 0},$$

and by denoting $\tilde{\boldsymbol{M}}_Q = 2\boldsymbol{M}_Q - T\left(\frac{\partial \boldsymbol{M}_Q}{\partial T}\right)_\mu$, together with considering the thermal Hall coefficient from M_Q^z, we calculate

$$\tilde{M}_Q^z = \frac{\beta}{2i}\left(\partial_{q_x}\langle \hat{K}_{-\boldsymbol{q}}; \hat{J}_{Q,\boldsymbol{q}}^y\rangle - \partial_{q_y}\langle \hat{K}_{-\boldsymbol{q}}; \hat{J}_{Q,\boldsymbol{q}}^x\rangle\right)\Big|_{\boldsymbol{q}\to 0}. \tag{108}$$

To do this, let us generally consider

$$\begin{aligned}
\partial_{q_\alpha}\langle \hat{K}_{-\boldsymbol{q}}; \hat{J}_{Q,\boldsymbol{q}}^\beta\rangle &= \partial_{q_\alpha}\int \mathrm{d}\boldsymbol{r}\mathrm{d}\boldsymbol{r}'\ e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\langle \hat{K}(\boldsymbol{r}'); \hat{J}_Q^\beta(\boldsymbol{r})\rangle \\
&= \frac{1}{\beta}\partial_{q_\alpha}\int \mathrm{d}\boldsymbol{r}\mathrm{d}\boldsymbol{r}'\ e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\,\mathrm{Tr}\left(\hat{\rho}^{eq}\int_0^\beta \mathrm{d}\lambda\ e^{\lambda\hat{H}_0}\hat{K}(\boldsymbol{r}')e^{-\lambda\hat{H}_0}\hat{J}_Q^\beta(\boldsymbol{r})\right) \\
&= \frac{1}{\beta}\partial_{q_\alpha}\,\mathrm{Tr}\left(\hat{\rho}^{eq}\int_0^\beta \mathrm{d}\lambda\ e^{\lambda\hat{H}_0}\left[\int \mathrm{d}\boldsymbol{r}'\frac{\{e^{i\boldsymbol{q}\cdot\boldsymbol{r}'},\hat{K}(\boldsymbol{r}')\}}{2}\right]e^{-\lambda\hat{H}_0}\left[\int \mathrm{d}\boldsymbol{r}e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\hat{J}_Q^\beta(\boldsymbol{r})\right]\right) \\
&= \frac{1}{\beta}\partial_{q_\alpha}\sum_{mn\boldsymbol{k}\boldsymbol{k}'}\frac{f_m(\boldsymbol{k})-f_n(\boldsymbol{k}')}{\varepsilon_{m\boldsymbol{k}}-\varepsilon_{n\boldsymbol{k}'}}\langle m\boldsymbol{k}|\hat{K}_{-\boldsymbol{q}}|n\boldsymbol{k}'\rangle\langle n\boldsymbol{k}'|\hat{J}_{Q,\boldsymbol{q}}^\beta|m\boldsymbol{k}\rangle,
\end{aligned} \tag{109}$$

where we have re-labelled

$$\hat{K}_{-\boldsymbol{q}} = \int \mathrm{d}\boldsymbol{r}'\frac{\{e^{i\boldsymbol{q}\cdot\boldsymbol{r}'},\hat{K}(\boldsymbol{r}')\}}{2}, \tag{110a}$$

$$\hat{J}_{Q,\boldsymbol{q}}^\beta = \int \mathrm{d}\boldsymbol{r}e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\hat{J}_Q^\beta(\boldsymbol{r}), \tag{110b}$$

where the grand potential is symmetrized for Hermitian (it is an out-act operator while the current is a closed one). In the last line, we have used the deduced result before.

Let us calculate

$$\begin{aligned}
\int \mathrm{d}\boldsymbol{r}\{e^{i\boldsymbol{q}\cdot\boldsymbol{r}},\hat{K}(\boldsymbol{r})\} &= \int \mathrm{d}\boldsymbol{r}\ \left(e^{i\boldsymbol{q}\cdot\boldsymbol{r}}\hat{K}(\boldsymbol{r}) + \hat{K}(\boldsymbol{r})e^{i\boldsymbol{q}\cdot\boldsymbol{r}}\right) \\
&= \int \mathrm{d}\boldsymbol{r}\ \left(e^{i\boldsymbol{q}\cdot\boldsymbol{r}}\hat{c}_{\boldsymbol{r}}^\dagger K(\boldsymbol{r})\hat{c}_{\boldsymbol{r}} + \hat{c}_{\boldsymbol{r}}^\dagger K(\boldsymbol{r})\hat{c}_{\boldsymbol{r}}e^{i\boldsymbol{q}\cdot\boldsymbol{r}}\right) \\
&= \frac{1}{V}\int \mathrm{d}\boldsymbol{r}\ \sum_{\boldsymbol{k},\boldsymbol{k}'}\left(e^{i\boldsymbol{q}\cdot\boldsymbol{r}}e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\hat{c}_{\boldsymbol{k}}^\dagger K(\boldsymbol{r})e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\hat{c}_{\boldsymbol{k}'} + e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\hat{c}_{\boldsymbol{k}}^\dagger K(\boldsymbol{r})e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}e^{i\boldsymbol{q}\cdot\boldsymbol{r}}\hat{c}_{\boldsymbol{k}'}\right) \\
&= \sum_{\boldsymbol{k},\boldsymbol{k}'}\frac{1}{V}\int \mathrm{d}\boldsymbol{r}\ e^{i(\boldsymbol{q}-\boldsymbol{k}+\boldsymbol{k}')\cdot\boldsymbol{r}}\hat{c}_{\boldsymbol{k}}^\dagger\left(K_{\boldsymbol{k}'}+K_{\boldsymbol{k}'+\boldsymbol{q}}\right)\hat{c}_{\boldsymbol{k}'} \\
&= \sum_{\boldsymbol{k},\boldsymbol{k}'}\delta_{\boldsymbol{k},\boldsymbol{k}'+\boldsymbol{q}}\hat{c}_{\boldsymbol{k}}^\dagger\left(K_{\boldsymbol{k}'}+K_{\boldsymbol{k}'+\boldsymbol{q}}\right)\hat{c}_{\boldsymbol{k}'} \\
&= \sum_{\boldsymbol{k}}\hat{c}_{\boldsymbol{k}}^\dagger(K_{\boldsymbol{k}-\boldsymbol{q}}+K_{\boldsymbol{k}})\hat{c}_{\boldsymbol{k}-\boldsymbol{q}},
\end{aligned}$$

$$
\begin{aligned}
&\int \mathrm{d}\boldsymbol{r} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}} \frac{(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{K}\hat{\varphi}(\boldsymbol{r})) + (\hat{K}\hat{\varphi}(\boldsymbol{v}))^\dagger(\hat{\boldsymbol{v}}\hat{\varphi}(\boldsymbol{r}))}{2} \\
=&\frac{1}{V}\int \mathrm{d}\boldsymbol{r} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}} \frac{(\hat{\boldsymbol{v}}\sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}})^\dagger(\hat{K}\sum_{\boldsymbol{k}'} e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}'}) + (\hat{K}\sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}})^\dagger(\hat{\boldsymbol{v}}\sum_{\boldsymbol{k}'} e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}'})}{2} \\
=&\sum_{\boldsymbol{k}}\sum_{\boldsymbol{k}'}\hat{\varphi}^\dagger_{\boldsymbol{k}}\frac{1}{V}\int \frac{\hat{\boldsymbol{v}}_{\boldsymbol{k}}e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\hat{K}(\boldsymbol{r})e^{i\boldsymbol{k}'\cdot\boldsymbol{r}} + e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\left(\hat{K}(\boldsymbol{r})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\right)^\dagger\hat{\boldsymbol{v}}_{\boldsymbol{k}'}}{2}\hat{\varphi}_{\boldsymbol{k}'} \\
=&\sum_{\boldsymbol{k}}\sum_{\boldsymbol{k}'}\hat{\varphi}^\dagger_{\boldsymbol{k}}\frac{\hat{\boldsymbol{v}}_{\boldsymbol{k}}\hat{K}_{\boldsymbol{k}'} + \hat{K}_{\boldsymbol{k}}\hat{\boldsymbol{v}}_{\boldsymbol{k}'}}{2}\delta_{\boldsymbol{k}+\boldsymbol{q},\boldsymbol{k}'}\hat{\varphi}_{\boldsymbol{k}'} \\
=&\sum_{\boldsymbol{k}}\hat{\varphi}^\dagger_{\boldsymbol{k}}\frac{\hat{\boldsymbol{v}}_{\boldsymbol{k}}\hat{K}_{\boldsymbol{k}+\boldsymbol{q}} + \hat{K}_{\boldsymbol{k}}\hat{\boldsymbol{v}}_{\boldsymbol{k}+\boldsymbol{q}}}{2}\hat{\varphi}_{\boldsymbol{k}+\boldsymbol{q}},
\end{aligned}
$$

$$
\begin{aligned}
\int \mathrm{d}\boldsymbol{r} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\nabla\times\hat{\boldsymbol{\Lambda}} &= -\frac{i\hbar}{8}\hat{e}_i\int \mathrm{d}\boldsymbol{r} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\partial_j\left[(\hat{v}_i\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{v}_j\hat{\varphi}(\boldsymbol{r})) - (\hat{v}_j\hat{\varphi}(\boldsymbol{r}))^\dagger(\hat{v}_i\hat{\varphi}(\boldsymbol{r}))\right] \\
&= -\frac{i\hbar}{8V}\hat{e}_i\int \mathrm{d}\boldsymbol{r} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\partial_j\left[(\hat{v}_i\sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}})^\dagger(\hat{v}_j\sum_{\boldsymbol{k}'} e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\hat{\varphi}_{\boldsymbol{k}'}) - \text{h.c.}\right] \\
&= -\frac{i\hbar}{8V}\hat{e}_i\sum_{\boldsymbol{k},\boldsymbol{k}'}\hat{\varphi}^\dagger_{\boldsymbol{k}}\int \mathrm{d}\boldsymbol{r} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\partial_j\left[\hat{v}^i_{\boldsymbol{k}}e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\hat{v}^j_{\boldsymbol{k}'} - \hat{v}^j_{\boldsymbol{k}}e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}e^{i\boldsymbol{k}'\cdot\boldsymbol{r}}\hat{v}^i_{\boldsymbol{k}'}\right]\hat{\varphi}_{\boldsymbol{k}'} \\
&= -\frac{\hbar}{8V}\hat{e}_i\sum_{\boldsymbol{k},\boldsymbol{k}'}\hat{\varphi}^\dagger_{\boldsymbol{k}}(k_j - k'_j)\int \mathrm{d}\boldsymbol{r} e^{-i(\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{k}')\cdot\boldsymbol{r}}\left[\hat{v}^i_{\boldsymbol{k}}\hat{v}^j_{\boldsymbol{k}'} - \hat{v}^j_{\boldsymbol{k}}\hat{v}^i_{\boldsymbol{k}'}\right]\hat{\varphi}_{\boldsymbol{k}'} \\
&= -\frac{\hbar}{8}\hat{e}_i\sum_{\boldsymbol{k},\boldsymbol{k}'}\hat{\varphi}^\dagger_{\boldsymbol{k}}(k_j - k'_j)\left(\hat{v}^i_{\boldsymbol{k}}\hat{v}^j_{\boldsymbol{k}'} - \hat{v}^j_{\boldsymbol{k}}\hat{v}^i_{\boldsymbol{k}'}\right)\delta_{\boldsymbol{k}+\boldsymbol{q},\boldsymbol{k}'}\hat{\varphi}_{\boldsymbol{k}'} \\
&= \frac{\hbar}{8}\hat{e}_i\sum_{\boldsymbol{k}}\hat{\varphi}^\dagger_{\boldsymbol{k}}q_j\left[\hat{v}^i_{\boldsymbol{k}}\hat{v}^j_{\boldsymbol{k}+\boldsymbol{q}} - \hat{v}^j_{\boldsymbol{k}}\hat{v}^i_{\boldsymbol{k}+\boldsymbol{q}}\right]\hat{\varphi}_{\boldsymbol{k}+\boldsymbol{q}}.
\end{aligned}
$$

We thus obtain that

$$
\begin{aligned}
\partial_{q_\alpha}\langle\hat{K}_{-\boldsymbol{q}};\hat{J}^\beta_{Q,\boldsymbol{q}}\rangle &= \frac{1}{\beta}\partial_{q_\alpha}\sum_{mn\boldsymbol{k}\boldsymbol{k}'}\frac{f_m(\boldsymbol{k}) - f_n(\boldsymbol{k}')}{\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}'}}\langle m\boldsymbol{k}|\hat{K}_{-\boldsymbol{q}}|n\boldsymbol{k}'\rangle\langle n\boldsymbol{k}'|\hat{J}^\beta_{Q,\boldsymbol{q}}|m\boldsymbol{k}\rangle \\
&= \frac{1}{\beta}\partial_{q_\alpha}\sum_{mn\boldsymbol{k}}\frac{f_m(\boldsymbol{k}) - f_n(\boldsymbol{k}-\boldsymbol{q})}{\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}}}\langle u_{m\boldsymbol{k}}|\frac{\hat{K}_{\boldsymbol{k}} + \hat{K}_{\boldsymbol{k}-\boldsymbol{q}}}{2}|u_{n,\boldsymbol{k}-\boldsymbol{q}}\rangle \qquad (111) \\
&\quad \langle u_{n,\boldsymbol{k}-\boldsymbol{q}}|\frac{\hat{v}^\beta_{\boldsymbol{k}-\boldsymbol{q}}\hat{K}_{\boldsymbol{k}} + \hat{K}_{\boldsymbol{k}-\boldsymbol{q}}\hat{v}^\beta_{\boldsymbol{k}}}{2} - \frac{\hbar\sum_\gamma q_\gamma\left[\hat{v}^\beta_{\boldsymbol{k}-\boldsymbol{q}}\hat{v}^\gamma_{\boldsymbol{k}} - \hat{v}^\gamma_{\boldsymbol{k}-\boldsymbol{q}}\hat{v}^\beta_{\boldsymbol{k}}\right]}{8}|u_{m\boldsymbol{k}}\rangle\,.
\end{aligned}
$$

Now we would like to take the derivative and take the limit $\boldsymbol{q}\to 0$. The first thing we take care is the derivative of the Fermi-Dirac distribution, and by a Taylor expansion, we have

$$
\begin{aligned}
\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}} &= \varepsilon_{n\boldsymbol{k}} - \boldsymbol{q}\cdot\nabla_{\boldsymbol{k}}\varepsilon_{n\boldsymbol{k}} + O(q^2), \\
f_n(\boldsymbol{k}-\boldsymbol{q}) = f(\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}}) &= f(\varepsilon_{n\boldsymbol{k}} - \boldsymbol{q}\cdot\nabla_{\boldsymbol{k}}\varepsilon_{n\boldsymbol{k}} + O(q^2)) \qquad (112) \\
&= f(\varepsilon_{n\boldsymbol{k}}) - \boldsymbol{q}\cdot(\nabla_{\boldsymbol{k}}\varepsilon_{n\boldsymbol{k}})f'_n(\boldsymbol{k}) + O(q^2),
\end{aligned}
$$

where $f_n'(\boldsymbol{k}) \equiv \left.\dfrac{\partial f(\varepsilon)}{\partial \varepsilon}\right|_{\varepsilon=\varepsilon_{n\boldsymbol{k}}}$. In this way, we find that for small $\boldsymbol{q}$,

$$\frac{f_n(\boldsymbol{k})-f_n(\boldsymbol{k}-\boldsymbol{q})}{\varepsilon_{n\boldsymbol{k}}-\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}}}=\frac{\boldsymbol{q}\cdot(\nabla_{\boldsymbol{k}}\varepsilon_{n\boldsymbol{k}})f_n'(\boldsymbol{k})+O(q^2)}{\boldsymbol{q}\cdot\nabla_{\boldsymbol{k}}\varepsilon_{n\boldsymbol{k}}+O(q^2)}=f_n'(\boldsymbol{k})+O(q). \tag{113}$$

On the other hand, by taking the derivative, we have

$$\begin{aligned}
\partial_{q_\alpha}\frac{f_n(\boldsymbol{k})-f_n(\boldsymbol{k}-\boldsymbol{q})}{\varepsilon_{n\boldsymbol{k}}-\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}}}\bigg|_{\boldsymbol{q}\to 0} &= \left.\frac{-(\partial_{q_\alpha}f_n(\boldsymbol{k}-\boldsymbol{q}))(\varepsilon_{n\boldsymbol{k}}-\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}})-(f_n(\boldsymbol{k})-f_n(\boldsymbol{k}-\boldsymbol{q}))(-\partial_{q_\alpha}\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}})}{(\varepsilon_{n\boldsymbol{k}}-\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}})^2}\right|_{\boldsymbol{q}\to 0}\\
&= \left[\partial_{k_\alpha}f_n(\boldsymbol{k})-\lim_{\boldsymbol{q}\to 0}\frac{f_n(\boldsymbol{k})-f_n(\boldsymbol{k}-\boldsymbol{q})}{\varepsilon_{n\boldsymbol{k}}-\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}}}\partial_{k_\alpha}\varepsilon_{n,\boldsymbol{k}}\right]\frac{1}{\varepsilon_{n\boldsymbol{k}}-\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}}}\\
&= \left[\partial_{k_\alpha}f_n(\boldsymbol{k})-f_n'(\boldsymbol{k})\partial_{k_\alpha}\varepsilon_{n,\boldsymbol{k}}\right]\frac{1}{\varepsilon_{n\boldsymbol{k}}-\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}}}\\
&= \left[\partial_{k_\alpha}f_n(\boldsymbol{k})-\partial_{k_\alpha}f_n(\boldsymbol{k})\right]\frac{1}{\varepsilon_{n\boldsymbol{k}}-\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}}}=0.
\end{aligned} \tag{114}$$

The clarification is important since as long as the derivative is not on the $\langle u_{m\boldsymbol{k}}|\frac{\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}-\boldsymbol{q}}}{2}|u_{n,\boldsymbol{k}-\boldsymbol{q}}\rangle$, this term will force $m=n$ by the $\boldsymbol{q}\to 0$ limit, and requires the above disscussion. We thus have

$$\begin{aligned}
&\beta\partial_{q_\alpha}\langle\hat{K}_{-\boldsymbol{q}};\hat{J}^\beta_{Q,\boldsymbol{q}}\rangle\Big|_{\boldsymbol{q}\to 0}\\
=&\sum_{mn\boldsymbol{k}}\left(\partial_{q_\alpha}\frac{f_m(\boldsymbol{k})-f_n(\boldsymbol{k}-\boldsymbol{q})}{\varepsilon_{m\boldsymbol{k}}-\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}}}\right)\Big|_{\boldsymbol{q}\to 0}\langle u_{m\boldsymbol{k}}|\hat{K}_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle\langle u_{n\boldsymbol{k}}|\frac{\hat{v}^\beta_{\boldsymbol{k}}\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}}\hat{v}^\beta_{\boldsymbol{k}}}{2}|u_{m\boldsymbol{k}}\rangle\\
+&\frac{f_m(\boldsymbol{k})-f_n(\boldsymbol{k}-\boldsymbol{q})}{\varepsilon_{m\boldsymbol{k}}-\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}}}\Big|_{\boldsymbol{q}\to 0}\left(\partial_{q_\alpha}\langle u_{m\boldsymbol{k}}|\frac{\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}-\boldsymbol{q}}}{2}|u_{n,\boldsymbol{k}-\boldsymbol{q}}\rangle\right)\Big|_{\boldsymbol{q}\to 0}\langle u_{n\boldsymbol{k}}|\frac{\hat{v}^\beta_{\boldsymbol{k}}\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}}\hat{v}^\beta_{\boldsymbol{k}}}{2}|u_{m\boldsymbol{k}}\rangle\\
+&\frac{f_m(\boldsymbol{k})-f_n(\boldsymbol{k}-\boldsymbol{q})}{\varepsilon_{m\boldsymbol{k}}-\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}}}\Big|_{\boldsymbol{q}\to 0}\langle u_{m\boldsymbol{k}}|\hat{K}_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle\left(\partial_{q_\alpha}\langle u_{n\boldsymbol{k}-\boldsymbol{q}}|\frac{\hat{v}^\beta_{\boldsymbol{k}-\boldsymbol{q}}\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}-\boldsymbol{q}}\hat{v}^\beta_{\boldsymbol{k}}}{2}|u_{m\boldsymbol{k}}\rangle\right)\Big|_{\boldsymbol{q}\to 0}\\
-&\frac{\hbar}{8}\frac{f_m(\boldsymbol{k})-f_n(\boldsymbol{k}-\boldsymbol{q})}{\varepsilon_{m\boldsymbol{k}}-\varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}}}\Big|_{\boldsymbol{q}\to 0}\langle u_{m\boldsymbol{k}}|\hat{K}_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle\left(\partial_{q_\alpha}\langle u_{n\boldsymbol{k}-\boldsymbol{q}}|\sum_\gamma q_\gamma\left[\hat{v}^\beta_{\boldsymbol{k}-\boldsymbol{q}}\hat{v}^\gamma_{\boldsymbol{k}}-\hat{v}^\gamma_{\boldsymbol{k}-\boldsymbol{q}}\hat{v}^\beta_{\boldsymbol{k}}\right]|u_{m\boldsymbol{k}}\rangle\right)\Big|_{\boldsymbol{q}\to 0}\\
=&\sum_{m\neq n,\boldsymbol{k}}\frac{f_m(\boldsymbol{k})-f_n(\boldsymbol{k})}{\varepsilon_{m\boldsymbol{k}}-\varepsilon_{n\boldsymbol{k}}}\left(\partial_{q_\alpha}\langle u_{m\boldsymbol{k}}|\frac{\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}-\boldsymbol{q}}}{2}|u_{n,\boldsymbol{k}-\boldsymbol{q}}\rangle\right)\Big|_{\boldsymbol{q}\to 0}\langle u_{n\boldsymbol{k}}|\frac{\hat{v}^\beta_{\boldsymbol{k}}\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}}\hat{v}^\beta_{\boldsymbol{k}}}{2}|u_{m\boldsymbol{k}}\rangle\\
+&\sum_{n\boldsymbol{k}}f_n'(\boldsymbol{k})\left(\partial_{q_\alpha}\langle u_{n\boldsymbol{k}}|\frac{\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}-\boldsymbol{q}}}{2}|u_{n,\boldsymbol{k}-\boldsymbol{q}}\rangle\right)\Big|_{\boldsymbol{q}\to 0}\langle u_{n\boldsymbol{k}}|\frac{\hat{v}^\beta_{\boldsymbol{k}}\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}}\hat{v}^\beta_{\boldsymbol{k}}}{2}|u_{n\boldsymbol{k}}\rangle\\
+&\sum_{n\boldsymbol{k}}f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\left(\partial_{q_\alpha}\langle u_{n\boldsymbol{k}-\boldsymbol{q}}|\frac{\hat{v}^\beta_{\boldsymbol{k}-\boldsymbol{q}}\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}-\boldsymbol{q}}\hat{v}^\beta_{\boldsymbol{k}}}{2}|u_{n\boldsymbol{k}}\rangle\right)\Big|_{\boldsymbol{q}\to 0}\\
-&\frac{\hbar}{8}\sum_{n\boldsymbol{k}}f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\left(\partial_{q_\alpha}\langle u_{n\boldsymbol{k}-\boldsymbol{q}}|\sum_\gamma q_\gamma\left[\hat{v}^\beta_{\boldsymbol{k}-\boldsymbol{q}}\hat{v}^\gamma_{\boldsymbol{k}}-\hat{v}^\gamma_{\boldsymbol{k}-\boldsymbol{q}}\hat{v}^\beta_{\boldsymbol{k}}\right]|u_{n\boldsymbol{k}}\rangle\right)\Big|_{\boldsymbol{q}\to 0},
\end{aligned} \tag{115}$$

where the first term in the first equality equals zero, and we have split the second term into two parts, one with $m = n$ and the other with $m \neq n$. Labelling the four summation terms in the second equality as $I_1^{\alpha\beta}$, $I_2^{\alpha\beta}$, $I_3^{\alpha\beta}$, and $I_4^{\alpha\beta}$, we have separately as

$$
\begin{aligned}
I_1^{\alpha\beta} &= \sum_{m\neq n,\boldsymbol{k}} \frac{f_m(\boldsymbol{k}) - f_n(\boldsymbol{k})}{\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}} \left(\partial_{q_\alpha} \langle u_{m\boldsymbol{k}} | \frac{\hat{K}_{\boldsymbol{k}} + \hat{K}_{\boldsymbol{k}-\boldsymbol{q}}}{2} | u_{n,\boldsymbol{k}-\boldsymbol{q}} \rangle \right) \Big|_{\boldsymbol{q}\to 0} \langle u_{n\boldsymbol{k}} | \frac{\hat{v}_{\boldsymbol{k}}^\beta \hat{K}_{\boldsymbol{k}} + \hat{K}_{\boldsymbol{k}} \hat{v}_{\boldsymbol{k}}^\beta}{2} | u_{m\boldsymbol{k}} \rangle \\
&= \sum_{m\neq n,\boldsymbol{k}} \frac{f_m(\boldsymbol{k}) - f_n(\boldsymbol{k})}{\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}} \partial_{q_\alpha} \left(\frac{\varepsilon_{m\boldsymbol{k}} + \varepsilon_{n,\boldsymbol{k}-\boldsymbol{q}} - 2\mu}{2} \langle u_{m\boldsymbol{k}} | u_{n,\boldsymbol{k}-\boldsymbol{q}} \rangle \right) \Big|_{\boldsymbol{q}\to 0} \frac{\varepsilon_{m\boldsymbol{k}} + \varepsilon_{n\boldsymbol{k}} - 2\mu}{2} \langle u_{n\boldsymbol{k}} | \hat{v}_{\boldsymbol{k}}^\beta | u_{m\boldsymbol{k}} \rangle \\
&= -\sum_{m\neq n,\boldsymbol{k}} \frac{f_m(\boldsymbol{k}) - f_n(\boldsymbol{k})}{\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}} \frac{(\varepsilon_{m\boldsymbol{k}} + \varepsilon_{n\boldsymbol{k}} - 2\mu)^2}{4} \langle u_{m\boldsymbol{k}} | \partial_{k_\alpha} u_{n\boldsymbol{k}} \rangle \langle u_{n\boldsymbol{k}} | \hat{v}_{\boldsymbol{k}}^\beta | u_{m\boldsymbol{k}} \rangle \\
&= \frac{1}{4\hbar} \sum_{m\neq n,\boldsymbol{k}} (f_m(\boldsymbol{k}) - f_n(\boldsymbol{k}))(\varepsilon_{m\boldsymbol{k}} + \varepsilon_{n\boldsymbol{k}} - 2\mu)^2 \langle u_{m\boldsymbol{k}} | \partial_{k_\alpha} u_{n\boldsymbol{k}} \rangle \langle \partial_{k_\beta} u_{n\boldsymbol{k}} | u_{m\boldsymbol{k}} \rangle \\
&= \frac{1}{4\hbar} \sum_{mn\boldsymbol{k}} (f_m(\boldsymbol{k}) - f_n(\boldsymbol{k}))(\varepsilon_{m\boldsymbol{k}} + \varepsilon_{n\boldsymbol{k}} - 2\mu)^2 \langle u_{m\boldsymbol{k}} | \partial_{k_\alpha} u_{n\boldsymbol{k}} \rangle \langle \partial_{k_\beta} u_{n\boldsymbol{k}} | u_{m\boldsymbol{k}} \rangle \\
&= -\frac{1}{2\hbar} \sum_{mn\boldsymbol{k}} f_n(\boldsymbol{k})(\varepsilon_{m\boldsymbol{k}} + \varepsilon_{n\boldsymbol{k}} - 2\mu)^2 \langle u_{m\boldsymbol{k}} | \partial_{k_\alpha} u_{n\boldsymbol{k}} \rangle \langle \partial_{k_\beta} u_{n\boldsymbol{k}} | u_{m\boldsymbol{k}} \rangle \\
&= -\frac{1}{2\hbar} \sum_{n\boldsymbol{k}} f_n(\boldsymbol{k}) \langle \partial_{k_\beta} u_{n\boldsymbol{k}} | (\hat{h}_{\boldsymbol{k}} + \varepsilon_{n\boldsymbol{k}} - 2\mu)^2 \sum_m | u_{m\boldsymbol{k}} \rangle \langle u_{m\boldsymbol{k}} | \partial_{k_\alpha} u_{n\boldsymbol{k}} \rangle \\
&= -\frac{1}{2\hbar} \sum_{n\boldsymbol{k}} f_n(\boldsymbol{k}) \langle \partial_{k_\beta} u_{n\boldsymbol{k}} | (\hat{h}_{\boldsymbol{k}} + \varepsilon_{n\boldsymbol{k}} - 2\mu)^2 | \partial_{k_\alpha} u_{n\boldsymbol{k}} \rangle \,,
\end{aligned}
\tag{116}
$$

$$
\begin{aligned}
I_2^{\alpha\beta} &= \sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k}) \left(\partial_{q_\alpha} \langle u_{n\boldsymbol{k}} | \frac{\hat{K}_{\boldsymbol{k}} + \hat{K}_{\boldsymbol{k}-\boldsymbol{q}}}{2} | u_{n,\boldsymbol{k}-\boldsymbol{q}} \rangle \right) \Big|_{\boldsymbol{q}\to 0} \langle u_{n\boldsymbol{k}} | \frac{\hat{v}_{\boldsymbol{k}}^\beta \hat{K}_{\boldsymbol{k}} + \hat{K}_{\boldsymbol{k}} \hat{v}_{\boldsymbol{k}}^\beta}{2} | u_{n\boldsymbol{k}} \rangle \\
&= -\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k}) \langle u_{n\boldsymbol{k}} | \left((\partial_{k_\alpha} \hat{K}_{\boldsymbol{k}})/2 \, | u_{n\boldsymbol{k}} \rangle + \hat{K}_{\boldsymbol{k}} \, | \partial_{k_\alpha} u_{n\boldsymbol{k}} \rangle \right) (\varepsilon_{n\boldsymbol{k}} - \mu) \langle u_{n\boldsymbol{k}} | \hat{v}_{\boldsymbol{k}}^\beta | u_{n\boldsymbol{k}} \rangle \\
&= -\frac{\hbar}{2} \sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k}) (\varepsilon_{n\boldsymbol{k}} - \mu) \langle u_{n\boldsymbol{k}} | \hat{v}_{\boldsymbol{k}}^\alpha | u_{n\boldsymbol{k}} \rangle \langle u_{n\boldsymbol{k}} | \hat{v}_{\boldsymbol{k}}^\beta | u_{n\boldsymbol{k}} \rangle \\
&\quad - \sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k}) (\varepsilon_{n\boldsymbol{k}} - \mu)^2 \langle u_{n\boldsymbol{k}} | \partial_{k_\alpha} u_{n\boldsymbol{k}} \rangle \langle u_{n\boldsymbol{k}} | \hat{v}_{\boldsymbol{k}}^\beta | u_{n\boldsymbol{k}} \rangle \,,
\end{aligned}
\tag{117}
$$

$$
\begin{aligned}
I_3^{\alpha\beta} &= \sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\left(\partial_{q_\alpha}\langle u_{n\boldsymbol{k}-\boldsymbol{q}}|\frac{\hat{v}^\beta_{\boldsymbol{k}-\boldsymbol{q}}\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}-\boldsymbol{q}}\hat{v}^\beta_{\boldsymbol{k}}}{2}|u_{n\boldsymbol{k}}\rangle\right)\Big|_{\boldsymbol{q}\to 0} \\
&= -\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\langle\partial_{k_\alpha}u_{n\boldsymbol{k}}|\frac{\hat{v}^\beta_{\boldsymbol{k}}\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}}\hat{v}^\beta_{\boldsymbol{k}}}{2}|u_{n\boldsymbol{k}}\rangle \\
&\quad - f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\langle u_{n\boldsymbol{k}}|\frac{(\partial_{k_\alpha}\hat{v}^\beta_{\boldsymbol{k}})\hat{K}_{\boldsymbol{k}}+(\partial_{k_\alpha}\hat{K}_{\boldsymbol{k}})\hat{v}^\beta_{\boldsymbol{k}}}{2}|u_{n\boldsymbol{k}}\rangle \\
&= -\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\langle\partial_{k_\alpha}u_{n\boldsymbol{k}}|\frac{\hat{v}^\beta_{\boldsymbol{k}}\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}}\hat{v}^\beta_{\boldsymbol{k}}}{2}|u_{n\boldsymbol{k}}\rangle \\
&\quad - \frac{1}{2\hbar}f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)^2\langle u_{n\boldsymbol{k}}|\partial_{k_\alpha}\partial_{k_\beta}\hat{K}_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle \\
&\quad - \frac{\hbar}{2}f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\langle u_{n\boldsymbol{k}}|\hat{v}^\alpha_{\boldsymbol{k}}\hat{v}^\beta_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle\,,
\end{aligned}
\tag{118}
$$

$$
\begin{aligned}
I_4^{\alpha\beta} &= -\frac{\hbar}{8}\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\left(\partial_{q_\alpha}\langle u_{n\boldsymbol{k}-\boldsymbol{q}}|\sum_\gamma q_\gamma\left[\hat{v}^\beta_{\boldsymbol{k}-\boldsymbol{q}}\hat{v}^\gamma_{\boldsymbol{k}}-\hat{v}^\gamma_{\boldsymbol{k}-\boldsymbol{q}}\hat{v}^\beta_{\boldsymbol{k}}\right]|u_{n\boldsymbol{k}}\rangle\right)\Big|_{\boldsymbol{q}\to 0} \\
&= -\frac{\hbar}{8}\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\langle u_{n\boldsymbol{k}}|\hat{v}^\beta_{\boldsymbol{k}}\hat{v}^\alpha_{\boldsymbol{k}}-\hat{v}^\alpha_{\boldsymbol{k}}\hat{v}^\beta_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle\,,
\end{aligned}
\tag{119}
$$

where in I_4 we note that the derivative must be taken on the linear term in $\boldsymbol{q}$. We thus finally obtain that

$$
\begin{aligned}
\beta\partial_{q_\alpha}\langle\hat{K}_{-\boldsymbol{q}};\hat{J}^\beta_{Q,\boldsymbol{q}}\rangle\Big|_{\boldsymbol{q}\to 0} &= I_1^{\alpha\beta}+I_2^{\alpha\beta}+I_3^{\alpha\beta}+I_4^{\alpha\beta} \\
&= -\frac{1}{2\hbar}\sum_{n\boldsymbol{k}} f_n(\boldsymbol{k})\langle\partial_{k_\beta}u_{n\boldsymbol{k}}|(\hat{h}_{\boldsymbol{k}}+\varepsilon_{n\boldsymbol{k}}-2\mu)^2|\partial_{k_\alpha}u_{n\boldsymbol{k}}\rangle \\
&\quad - \frac{\hbar}{2}\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\langle u_{n\boldsymbol{k}}|\hat{v}^\alpha_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle\langle u_{n\boldsymbol{k}}|\hat{v}^\beta_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle \\
&\quad - \sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)^2\langle u_{n\boldsymbol{k}}|\partial_{k_\alpha}u_{n\boldsymbol{k}}\rangle\langle u_{n\boldsymbol{k}}|\hat{v}^\beta_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle \\
&\quad - \sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\langle\partial_{k_\alpha}u_{n\boldsymbol{k}}|\frac{\hat{v}^\beta_{\boldsymbol{k}}\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}}\hat{v}^\beta_{\boldsymbol{k}}}{2}|u_{n\boldsymbol{k}}\rangle \\
&\quad - \frac{1}{2\hbar}\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)^2\langle u_{n\boldsymbol{k}}|\partial_{k_\alpha}\partial_{k_\beta}\hat{K}_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle \\
&\quad - \frac{\hbar}{8}\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\langle u_{n\boldsymbol{k}}|\hat{v}^\beta_{\boldsymbol{k}}\hat{v}^\alpha_{\boldsymbol{k}}+3\hat{v}^\alpha_{\boldsymbol{k}}\hat{v}^\beta_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle\,.
\end{aligned}
\tag{120}
$$

Thus, as $\tilde{M}^z_Q = \frac{\beta}{2i}\left(\partial_{q_x}\langle\hat{K}_{-\boldsymbol{q}};\hat{J}^y_{Q,\boldsymbol{q}}\rangle - \partial_{q_y}\langle\hat{K}_{-\boldsymbol{q}};\hat{J}^x_{Q,\boldsymbol{q}}\rangle\right)\Big|_{\boldsymbol{q}\to 0}$, we can drop the symmetric terms

(the second and fifth terms), and combine the third and fourth terms by noting that

$$
\begin{aligned}
&(\varepsilon_{n\boldsymbol{k}}-\mu)\,\langle u_{n\boldsymbol{k}}|\partial_{k_\alpha}u_{n\boldsymbol{k}}\rangle\,\langle u_{n\boldsymbol{k}}|\hat{v}^\beta_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle+\langle\partial_{k_\alpha}u_{n\boldsymbol{k}}|\frac{\hat{v}^\beta_{\boldsymbol{k}}\hat{K}_{\boldsymbol{k}}+\hat{K}_{\boldsymbol{k}}\hat{v}^\beta_{\boldsymbol{k}}}{2}|u_{n\boldsymbol{k}}\rangle\\
=&\frac{1}{\hbar}\,\langle u_{n\boldsymbol{k}}|\partial_{k_\alpha}u_{n\boldsymbol{k}}\rangle\,(\varepsilon_{n\boldsymbol{k}}-\mu)\partial_{k_\beta}\varepsilon_{n\boldsymbol{k}}+\frac{1}{2\hbar}\,\langle\partial_{k_\alpha}u_{n\boldsymbol{k}}|(\partial_{k_\beta}\hat{K}^2_{\boldsymbol{k}})|u_{n\boldsymbol{k}}\rangle\\
=&-\frac{1}{2\hbar}(\partial_{k_\beta}\xi_{n\boldsymbol{k}})^2\,\langle\partial_{k_\alpha}u_{n\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle+\frac{1}{2\hbar}\Big[(\partial_{k_\beta}\xi^2_{n\boldsymbol{k}})\,\langle\partial_{k_\alpha}u_{n\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle+\langle\partial_{k_\alpha}u_{n\boldsymbol{k}}|\xi^2_{n\boldsymbol{k}}-\hat{K}^2_{\boldsymbol{k}}|\partial_{k_\beta}u_{n\boldsymbol{k}}\rangle\Big]\\
=&\frac{1}{2\hbar}\,\langle\partial_{k_\alpha}u_{n\boldsymbol{k}}|(\varepsilon_{n\boldsymbol{k}}-\mu)^2-\hat{K}^2_{\boldsymbol{k}}|\partial_{k_\beta}u_{n\boldsymbol{k}}\rangle\,,
\end{aligned}
$$

where in the intermediate step we defined $\xi_{n\boldsymbol{k}}=\varepsilon_{n\boldsymbol{k}}-\mu$. We have used the Feynman-Hellmann theorem and its variation by noting that

$$
\begin{aligned}
\partial_{k_\beta}(\hat{K}^2_{n\boldsymbol{k}}\,|u_{n\boldsymbol{k}}\rangle)&=(\partial_{k_\beta}\hat{K}^2_{n\boldsymbol{k}})\,|u_{n\boldsymbol{k}}\rangle+\hat{K}^2_{n\boldsymbol{k}}\,|\partial_{k_\beta}u_{n\boldsymbol{k}}\rangle\\
=\partial_{k_\beta}(\xi^2_{n\boldsymbol{k}}\,|u_{n\boldsymbol{k}}\rangle)&=(\partial_{k_\beta}\xi^2_{n\boldsymbol{k}})\,|u_{n\boldsymbol{k}}\rangle+\xi^2_{n\boldsymbol{k}}\,|\partial_{k_\beta}u_{n\boldsymbol{k}}\rangle\,,
\end{aligned}
$$

and a left action of $\langle\partial_{k_\alpha}u_{n\boldsymbol{k}}|$ leads to

$$
\langle\partial_{k_\alpha}u_{n\boldsymbol{k}}|(\partial_{k_\beta}\hat{K}^2_{\boldsymbol{k}})|u_{n\boldsymbol{k}}\rangle=\langle\partial_{k_\alpha}u_{n\boldsymbol{k}}|\xi^2_{n\boldsymbol{k}}-\hat{K}^2_{\boldsymbol{k}}|\partial_{k_\beta}u_{n\boldsymbol{k}}\rangle+(\partial_{k_\beta}\xi^2_{n\boldsymbol{k}})\,\langle\partial_{k_\alpha}u_{n\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle\,.
$$

We also used that

$$
\partial_k\,\langle m|n\rangle=0\Rightarrow\langle\partial_k m|n\rangle=-\,\langle m|\partial_k n\rangle\,.
$$

We thus obtain that

$$
\begin{aligned}
\tilde{M}^z_Q=&-\frac{\mathrm{Im}}{2\hbar}\sum_{n\boldsymbol{k}}f_n(\boldsymbol{k})\,\langle\partial_{k_y}u_{n\boldsymbol{k}}|(\hat{h}_{\boldsymbol{k}}+\varepsilon_{n\boldsymbol{k}}-2\mu)^2|\partial_{k_x}u_{n\boldsymbol{k}}\rangle\\
&-\frac{\mathrm{Im}}{2\hbar}\sum_{n\boldsymbol{k}}f'_n(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\,\langle\partial_{k_x}u_{n\boldsymbol{k}}|(\varepsilon_{n\boldsymbol{k}}-\mu)^2-\hat{K}^2_{\boldsymbol{k}}|\partial_{k_y}u_{n\boldsymbol{k}}\rangle\\
&-\frac{\hbar\,\mathrm{Im}}{4}\sum_{n\boldsymbol{k}}f'_n(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\,\langle u_{n\boldsymbol{k}}|\hat{v}^x_{\boldsymbol{k}}\hat{v}^y_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle\,.
\end{aligned}
\tag{121}
$$

We again note that

$$
\begin{aligned}
&\mathrm{Im}\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\,\langle u_{n\boldsymbol{k}}|\hat{v}^x_{\boldsymbol{k}}\hat{v}^y_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle \\
=&\,\mathrm{Im}\sum_{mn\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\,\langle u_{n\boldsymbol{k}}|\hat{v}^x_{\boldsymbol{k}}|u_{m\boldsymbol{k}}\rangle\,\langle u_{m\boldsymbol{k}}|\hat{v}^y_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle \\
=&\frac{\mathrm{Im}}{\hbar^2}\sum_{m\neq n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)(\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m\boldsymbol{k}})^2\,\langle \partial_{k_x}u_{n\boldsymbol{k}}|u_{m\boldsymbol{k}}\rangle\,\langle u_{m\boldsymbol{k}}|\partial_{k_y}u_{n\boldsymbol{k}}\rangle \\
+&\frac{\mathrm{Im}}{\hbar^2}\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\,\langle u_{n\boldsymbol{k}}|\hat{v}^x_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle\,\langle u_{n\boldsymbol{k}}|\hat{v}^y_{\boldsymbol{k}}|u_{n\boldsymbol{k}}\rangle \\
=&\frac{\mathrm{Im}}{\hbar^2}\sum_{mn\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)(\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m\boldsymbol{k}})^2\,\langle \partial_{k_x}u_{n\boldsymbol{k}}|u_{m\boldsymbol{k}}\rangle\,\langle u_{m\boldsymbol{k}}|\partial_{k_y}u_{n\boldsymbol{k}}\rangle \\
=&\frac{\mathrm{Im}}{\hbar^2}\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\,\langle \partial_{k_x}u_{n\boldsymbol{k}}|\sum_m(\varepsilon_{n\boldsymbol{k}}-\hat{h}_{\boldsymbol{k}})^2|u_{m\boldsymbol{k}}\rangle\,\langle u_{m\boldsymbol{k}}|\partial_{k_y}u_{n\boldsymbol{k}}\rangle \\
=&\frac{\mathrm{Im}}{\hbar^2}\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\,\langle \partial_{k_x}u_{n\boldsymbol{k}}|(\varepsilon_{n\boldsymbol{k}}-\hat{h}_{\boldsymbol{k}})^2|\partial_{k_y}u_{n\boldsymbol{k}}\rangle\,,
\end{aligned}
$$

and we can thus write the final expression as

$$
\begin{aligned}
\tilde{M}^z_Q=&-\frac{\mathrm{Im}}{2\hbar}\sum_{n\boldsymbol{k}} f_n(\boldsymbol{k})\,\langle \partial_{k_y}u_{n\boldsymbol{k}}|(\hat{h}_{\boldsymbol{k}}+\varepsilon_{n\boldsymbol{k}}-2\mu)^2|\partial_{k_x}u_{n\boldsymbol{k}}\rangle \\
&+\frac{\mathrm{Im}}{4\hbar}\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\,\langle \partial_{k_y}u_{n\boldsymbol{k}}|2(\varepsilon_{n\boldsymbol{k}}-\mu)^2-2\hat{K}^2_{\boldsymbol{k}}+(\varepsilon_{n\boldsymbol{k}}-\hat{h}_{\boldsymbol{k}})^2|\partial_{k_x}u_{n\boldsymbol{k}}\rangle\,,
\end{aligned}
\tag{122}
$$

and by noting that

$$
\begin{aligned}
&2(\varepsilon_{n\boldsymbol{k}}-\mu)^2-2\hat{K}^2_{\boldsymbol{k}}+(\varepsilon_{n\boldsymbol{k}}-h_{\boldsymbol{k}})^2 \\
=&2(\varepsilon_{n\boldsymbol{k}}+\hat{h}_{\boldsymbol{k}}-2\mu)(\varepsilon_{n\boldsymbol{k}}-\hat{h}_{\boldsymbol{k}})+(\varepsilon_{n\boldsymbol{k}}-h_{\boldsymbol{k}})^2 \\
=&2(\hat{h}_{\boldsymbol{k}}-\varepsilon_{n\boldsymbol{k}})(\varepsilon_{n\boldsymbol{k}}-\hat{h}_{\boldsymbol{k}})+4(\varepsilon_{n\boldsymbol{k}}-\mu)(\varepsilon_{n\boldsymbol{k}}-\hat{h}_{\boldsymbol{k}})+(\varepsilon_{n\boldsymbol{k}}-\hat{h}_{\boldsymbol{k}})^2 \\
=&4(\varepsilon_{n\boldsymbol{k}}-\mu)(\varepsilon_{n\boldsymbol{k}}-\hat{h}_{\boldsymbol{k}})-(\varepsilon_{n\boldsymbol{k}}-\hat{h}_{\boldsymbol{k}})^2,
\end{aligned}
$$

we can write

$$
\begin{aligned}
\tilde{M}^z_Q=&-\frac{\mathrm{Im}}{2\hbar}\sum_{n\boldsymbol{k}} f_n(\boldsymbol{k})\,\langle \partial_{k_y}u_{n\boldsymbol{k}}|(\hat{h}_{\boldsymbol{k}}+\varepsilon_{n\boldsymbol{k}}-2\mu)^2|\partial_{k_x}u_{n\boldsymbol{k}}\rangle \\
&+\frac{\mathrm{Im}}{4\hbar}\sum_{n\boldsymbol{k}} f_n'(\boldsymbol{k})(\varepsilon_{n\boldsymbol{k}}-\mu)\,\langle \partial_{k_y}u_{n\boldsymbol{k}}|4(\varepsilon_{n\boldsymbol{k}}-\mu)(\varepsilon_{n\boldsymbol{k}}-\hat{h}_{\boldsymbol{k}})-(\varepsilon_{n\boldsymbol{k}}-\hat{h}_{\boldsymbol{k}})^2|\partial_{k_x}u_{n\boldsymbol{k}}\rangle\,.
\end{aligned}
\tag{123}
$$

Now we solve $\tilde{\boldsymbol{M}}_Q = 2\boldsymbol{M}_Q - T\left(\dfrac{\partial \boldsymbol{M}_Q}{\partial T}\right)_\mu$, or equivalently

$$
2M^z_Q+\beta\frac{\partial M^z_Q}{\partial\beta}=\tilde{M}^z_Q, \tag{124}
$$

and we note that

$$f_n'(\boldsymbol{k}) = \left.\frac{\partial f}{\partial \varepsilon}\right|_{\varepsilon=\varepsilon_{n\boldsymbol{k}}} = \frac{\beta}{\varepsilon_{n\boldsymbol{k}} - \mu}\frac{\partial f_n(\boldsymbol{k})}{\partial \beta}. \tag{125}$$

Now to solve the equation, we note that as $n\boldsymbol{k}$ are independent labels, we can formally solve the equation at each point, by writing

$$\begin{aligned}
2M + \beta\frac{\partial M}{\partial \beta} &= -\frac{1}{2}f(\xi)(\xi+\hat{k})^2 + \frac{1}{4}\beta\frac{\partial f}{\partial \beta}\left[4\xi(\xi-\hat{k}) - (\xi-\hat{k})^2\right] \\
&= -\frac{1}{2}f(\xi)(\hat{k}^2 + 2\xi\hat{k} + \xi^2) \\
&\quad -\frac{1}{4}\beta\frac{\partial f}{\partial \beta}(\hat{k}^2 + 2\xi\hat{k} - 3\xi^2).
\end{aligned}$$

Thus, we see that the difference lies in the ξ terms, with quadratic dependence on ξ and linear in f, and we can guess that the solution is of the form

$$M = -\frac{1}{4}f(\xi)(\hat{k}^2 + 2\xi\hat{k}) + A\int_0^\xi \mathrm{d}x\ xf(x) + Bf(\xi)\xi^2, \tag{126}$$

with A and B being constants to be determined, and its derivative with respect to β is

$$\begin{aligned}
\frac{\partial M}{\partial \beta} &= -\frac{1}{4}\frac{\partial f}{\partial \beta}(\hat{k}^2 + 2\xi\hat{k}) + A\int_0^\xi \mathrm{d}x\ x\frac{\partial f}{\partial \beta} + B\frac{\partial f}{\partial \beta}\xi^2 \\
&= -\frac{1}{4}\frac{\partial f}{\partial \beta}(\hat{k}^2 + 2\xi\hat{k}) + \frac{A}{\beta}\int_0^\xi \mathrm{d}x\ x^2\frac{\partial f}{\partial x} + B\frac{\partial f}{\partial \beta}\xi^2 \\
&= -\frac{1}{4}\frac{\partial f}{\partial \beta}(\hat{k}^2 + 2\xi\hat{k}) + \frac{A}{\beta}\left(x^2 f(x)\Big|_0^\xi - 2\int_0^\xi \mathrm{d}x\ xf(x)\right) + B\frac{\partial f}{\partial \beta}\xi^2 \\
&= -\frac{1}{4}\frac{\partial f}{\partial \beta}(\hat{k}^2 + 2\xi\hat{k}) + \frac{A}{\beta}\xi^2 f(\xi) - \frac{2A}{\beta}\int_0^\xi \mathrm{d}x\ xf(x) + B\frac{\partial f}{\partial \beta}\xi^2.
\end{aligned}$$

In this sense, we have

$$\begin{aligned}
2M + \beta\frac{\partial M}{\partial \beta} &= -\frac{1}{2}f(\xi)(\hat{k}^2 + 2\xi\hat{k}) + 2A\int_0^\xi \mathrm{d}x\ xf(x) + 2Bf(\xi)\xi^2 \\
&\quad -\frac{1}{4}\beta\frac{\partial f}{\partial \beta}(\hat{k}^2 + 2\xi\hat{k}) + A\xi^2 f(\xi) - 2A\int_0^\xi \mathrm{d}x\ xf(x) + B\beta\frac{\partial f}{\partial \beta}\xi^2 \\
&= -\frac{1}{2}f(\xi)\left[\hat{k}^2 + 2\xi\hat{k} - 2(A+2B)\xi^2\right] \\
&\quad -\frac{1}{4}\beta\frac{\partial f}{\partial \beta}(\hat{k}^2 + 2\xi\hat{k} - 4B\xi^2),
\end{aligned}$$

which requires that

$$\begin{cases} -2(A+2B) = 1 \\ -4B = -3 \end{cases} \Longrightarrow \begin{cases} A = -2 \\ B = \dfrac{3}{4} \end{cases}. \tag{127}$$

Thus, we finally obtain that

$$M = -\frac{1}{4}f(\xi)(\hat{k}^2 + 2\xi\hat{k} - 3\xi^2) - 2\int_0^\xi \mathrm{d}x\ xf(x), \tag{128}$$

which translates to

$$\begin{aligned} M_Q^z &= -\frac{\mathrm{Im}}{4\hbar}\sum_{n\boldsymbol{k}} f_n(\boldsymbol{k})\,\langle\partial_{k_y}u_{n\boldsymbol{k}}|\hat{K}_{\boldsymbol{k}}^2 + 2\xi_{n\boldsymbol{k}}\hat{K}_{\boldsymbol{k}} - 3\xi_{n\boldsymbol{k}}^2|\partial_{k_x}u_{n\boldsymbol{k}}\rangle \\ &\quad - \frac{2\,\mathrm{Im}}{\hbar}\sum_{n\boldsymbol{k}}\langle\partial_{k_y}u_{n\boldsymbol{k}}|\partial_{k_x}u_{n\boldsymbol{k}}\rangle\int_0^{\xi_{n\boldsymbol{k}}}\mathrm{d}x\ xf(x) \\ &= -\frac{1}{4}\int\mathrm{d}\varepsilon\left[m_2^{xy}(\varepsilon)f(\varepsilon-\mu) + 4(\varepsilon-\mu)m_1^{xy}(\varepsilon)f(\varepsilon-\mu) + 4\omega^{xy}(\varepsilon)\int_0^{\varepsilon-\mu}\mathrm{d}x\ xf(x)\right]. \end{aligned} \tag{129}$$

Again we note that the current-force relation $J_Q^{x,\mathrm{static}} = \frac{2M_Q^z}{\beta V}X_{2,y}$ with $X_2 = \beta T\nabla 1/T = -\beta/T\nabla T$ tells that

$$\begin{aligned} \kappa_{xy}^{\mathrm{static}} &= \frac{\beta}{T}\frac{2M_Q^z}{\beta V} \\ &= -\frac{1}{2VT}\int\mathrm{d}\xi\left[m_2^{xy}(\xi)f(\xi) + 4\xi m_1^{xy}(\xi)f(\xi) + 4\omega^{xy}(\xi)\int_0^\xi\mathrm{d}x\ xf(x)\right]. \end{aligned} \tag{130}$$

The final expression for total thermal Hall conductivity is thus

$$\begin{aligned} \kappa_{xy} &= \kappa_{xy}^{\mathrm{Kubo}} + \kappa_{xy}^{\mathrm{static}} \\ &= \frac{1}{VT}\int\mathrm{d}\xi\ \omega^{xy}(\xi)\left(f(\xi)\xi^2 - 2\int_0^\xi\mathrm{d}x\ xf(x)\right). \end{aligned} \tag{131}$$

To simplify the expression, we note that by definition

$$\omega^{\alpha\beta}(\xi) = \frac{2}{\hbar}\,\mathrm{Im}\sum_{n\boldsymbol{k}}\langle\partial_\beta u_{n\boldsymbol{k}}|\partial_\alpha u_{n\boldsymbol{k}}\rangle\,\delta(\xi-\xi_{n\boldsymbol{k}}),$$

and we can thus define the dimensionless quasi-particle anomalous Hall conductivity as

$$\begin{aligned} N(\xi) &= \frac{1}{2\pi}\int\mathrm{d}^2k\sum_n\Omega_{n\boldsymbol{k}}^{xy}\Theta(\xi-\xi_{n\boldsymbol{k}}) \\ &= \frac{4\pi}{V}\,\mathrm{Im}\sum_{n\boldsymbol{k}}\langle\partial_x u_{n\boldsymbol{k}}|\partial_y u_{n\boldsymbol{k}}\rangle\,\Theta(\xi-\xi_{n\boldsymbol{k}}), \end{aligned} \tag{132}$$

such that

$$\omega^{xy}(\xi) = -\frac{V}{h}\frac{\partial N}{\partial\xi}. \tag{133}$$

To proceed, let us also define the integral function

$$p(\xi) = \int_{-\infty}^\xi\mathrm{d}x\ xf(x), \tag{134}$$

such that

$$\frac{\partial p}{\partial \xi} = \xi f(\xi), \tag{135}$$

and

$$\int_0^{\xi} \mathrm{d}x\ x f(x) = p(\xi) - C,\ C = \int_{-\infty}^{0} \mathrm{d}x\ x f(x). \tag{136}$$

The deduction holds essentially because that due to finite bandwidth, the constant C is finite by a physical cut-off in distribution function. Thus, we have

$$\begin{aligned}
\kappa_{xy} &= -\frac{1}{hT}\int \mathrm{d}\xi\ \frac{\partial N}{\partial \xi}\left[f(\xi)\xi^2 - 2(p(\xi) - C)\right] \\
&= -\frac{1}{hT}\left\{N(\xi)\left[f(\xi)\xi^2 - 2(p(\xi) - C)\right]\Big|_{-\infty}^{+\infty} - \int \mathrm{d}\xi\ N(\xi)\left[\xi^2 f'(\xi) + 2\xi f(\xi) - 2\xi f(\xi)\right]\right\} \\
&= \frac{1}{hT}\int \mathrm{d}\xi\ N(\xi)\xi^2 f'(\xi),
\end{aligned}$$

where we have used that $N(\xi)$ vanishes at $\pm\infty$. Back in energy ε, it is

$$\kappa_{xy} = \frac{1}{hT}\int \mathrm{d}\varepsilon\ (\varepsilon - \mu)^2 N(\varepsilon) f'(\varepsilon - \mu), \tag{137}$$

with

$$N(\varepsilon) = \frac{4\pi}{V}\,\mathrm{Im}\sum_{n\boldsymbol{k}} \langle \partial_x u_{n\boldsymbol{k}} | \partial_y u_{n\boldsymbol{k}} \rangle\, \Theta(\varepsilon - \varepsilon_{n\boldsymbol{k}}),\ f(\varepsilon - \mu) = \frac{1}{e^{\beta(\varepsilon-\mu)} + 1}.$$

The same deduction can be done for Bosonic system with replacement of distribution function, and can also be done for superconducting system using BdG Hamiltonian, where the degree of freedom is artificially doubled and should be reduced by one-half. Thus for fermionic system, the final expression for thermal Hall conductivity is

$$\kappa_{xy} = \omega\frac{1}{hT}\int \mathrm{d}\varepsilon\ (\varepsilon - \mu)^2 N(\varepsilon) f'(\varepsilon - \mu), \tag{138}$$

where $\omega = 1$ for normal state and $\omega = 1/2$ for superconductor.

The last is to give a brief discussion on the relation between the thermal Hall conductivity and the anomalous Hall conductivity, which requires a Sommerfeld expansion of the thermal Hall conductivity. By integration by parts, we rewrite

$$\kappa_{xy} = -\frac{\omega}{hT}\int \mathrm{d}\varepsilon\ f(\varepsilon - \mu)\frac{\mathrm{d}(\varepsilon - \mu)^2 N(\varepsilon)}{\mathrm{d}\varepsilon},$$

thus identify

$$H(\varepsilon) = \frac{\mathrm{d}(\varepsilon - \mu)^2 N(\varepsilon)}{\mathrm{d}\varepsilon} = 2(\varepsilon - \mu)N(\varepsilon) + (\varepsilon - \mu)^2 N'(\varepsilon),$$

and

$$H'(\varepsilon) = 2N(\varepsilon) + 2(\varepsilon-\mu)N'(\varepsilon) + (\varepsilon-\mu)^2 N''(\varepsilon)$$
$$H'''(\varepsilon) = 12N''(\varepsilon) + 8(\varepsilon-\mu)N'''(\varepsilon) + (\varepsilon-\mu)^2 N^{(4)}(\varepsilon).$$

Thus, we can expand the thermal Hall conductivity as

$$\begin{aligned}\kappa_{xy} &= -\frac{\omega}{hT}\left[\int_{-\infty}^{\mu} \mathrm{d}\varepsilon\ \frac{\mathrm{d}(\varepsilon-\mu)^2 N(\varepsilon)}{\mathrm{d}\varepsilon} + \frac{\pi^2}{6\beta^2}H'(\mu) + \frac{7\pi^4}{360\beta^4}H'''(\mu) + \cdots\right] \\ &= -\omega\frac{\pi^2 k_B^2}{3h}TN(\mu) - \omega\frac{7\pi^4 k_B^4}{30h}T^3 N''(\mu) + \cdots,\end{aligned} \tag{139}$$

which recovers **the Wiedemann-Franz law** at low temperature as

$$\frac{\kappa_{xy}}{\sigma_{xy}} = \frac{-\frac{\pi^2 k_B^2}{3h}TN(\mu)}{-\frac{e^2}{h}N(\mu)} = \frac{\pi^2 k_B^2}{3e^2}T, \tag{140}$$

for the normal state with $\omega = 1$.

J. Thermoelectric-Hall Coefficients as a Direct Analogy

Here we directly give the thermoelectric Hall coefficient/transverse Seebeck coefficient as a direct analogy to the thermal Hall coefficient, which is defined as defined as

$$J_c^x = -\alpha_{xy}\partial_y T, \tag{141}$$

and is thus given by

$$J_{1,x}^{tr} = L_{xy}^{(12)}X_{2,y} - \frac{(\boldsymbol{M}_c \times \boldsymbol{X}_2)_x}{\xi V} = \left(L_{xy}^{(12)} + \frac{M_c^z}{\beta V}\right)X_{2,y}, \tag{142}$$

and leads to

$$\alpha_{xy} = \frac{L_{xy}^{(12)}}{k_B T^2} + \frac{M_c^z}{TV}. \tag{143}$$

This form is much similar to the thermal Hall coefficient, and in the same spirit, we compare the ingredients.

We note that compared with $L^{(22)}$ that is about heat current (basically velocity times Hamiltonian) correlation, $L^{(12)}$ reduces to electric current - heat current correlation, thus lacks of a factor of grand potential, which thus should lead to

$$\begin{aligned}\alpha_{\alpha\beta}^{\mathrm{Kubo}} &= \frac{\mathrm{Im}}{\hbar VT}\sum_{m\boldsymbol{k}} f_m(\boldsymbol{k})\left\langle\partial_\beta u_{m\boldsymbol{k}}\right|\left(\varepsilon_{m\boldsymbol{k}} + h_{\boldsymbol{k}} - 2\mu\right)\left|\partial_\alpha u_{m\boldsymbol{k}}\right\rangle \\ &= \frac{1}{VT}\int \mathrm{d}\varepsilon\ f(\varepsilon)\left(m_1^{\alpha\beta}(\varepsilon) + (\varepsilon-\mu)\omega^{\alpha\beta}(\varepsilon)\right).\end{aligned} \tag{144}$$

And in the same spirit, only the term related to Berry curvature contributes, and shall similarly finally lead to

$$\begin{aligned}\alpha_{xy} &= \omega\frac{e}{hT}\int \mathrm{d}\varepsilon\ (\varepsilon-\mu)N(\varepsilon)f'(\varepsilon-\mu)\\ &= -\omega\frac{e\pi^2k_B^2}{3h}TN'(\mu) - \omega\frac{7e\pi^4k_B^4}{90h}T^3N'''(\mu)+\cdots,\end{aligned} \tag{145}$$

which recovers **the Mott relation** at low temperature as

$$\frac{\alpha_{xy}}{\sigma_{xy}} = \frac{-\frac{e\pi^2k_B^2}{3h}TN'(\mu)}{-\frac{e^2}{h}N(\mu)} = \frac{\pi^2k_B^2T}{3e}\frac{\sigma'(\mu)}{\sigma(\mu)}, \tag{146}$$

for the normal state with $\omega = 1$.

III. Numerical Results

Now we present the numerical results of several representative examples.

A. 2DEG

The model of two-dimensional electron gas (2DEG) reads

$$H_{\mathrm{2DEG}}(\boldsymbol{k}) = \frac{k^2}{2m} + \lambda(\boldsymbol{k}\times\boldsymbol{\sigma})_z + V_z\sigma_z. \tag{147}$$

The thermal Hall conductivity of the system (on lattice) is shown in Fig. 1, with parameters

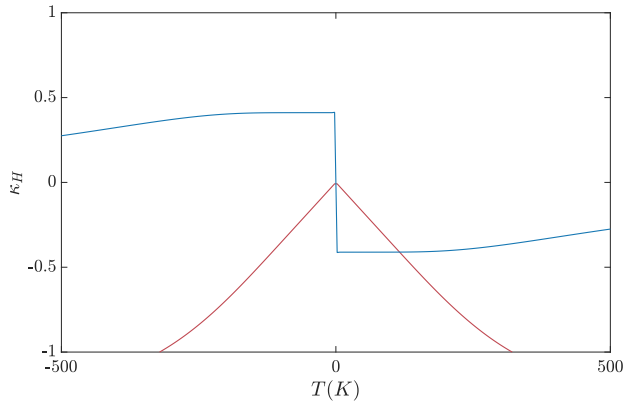


FIG. 1: Thermal Hall conductivity $100\kappa_H/(\pi^2k_B^2/3h)$ (blue) and $\kappa_H/(\pi^2k_B^2T/3h)$ (red) as a function of temperature T for the 2DEG model.

$\lambda = 0.5$, $m = 2$, $V_z = 0.1$. It is observed that there exists a non-half-quantized thermal Hall conductivity plateau at low temperatures.

Adding superconductivity to the system, the Bogoliubov-de Gennes (BdG) Hamiltonian under Nambu basis reads

$$H_{\text{2DEG}}^{\text{BdG}}(\boldsymbol{k}) = \begin{pmatrix} \frac{k^2}{2m} - \mu + \lambda(\boldsymbol{k} \times \boldsymbol{\sigma})_z + V_z\sigma_z & \Delta \\ \Delta^* & -\frac{k^2}{2m} + \mu - \lambda(\boldsymbol{k} \times \boldsymbol{\sigma})_z + V_z\sigma_z \end{pmatrix}. \tag{148}$$

By setting superconducting gap $\Delta = 0.03$, we explore the fictitious quasi-particle anomalous

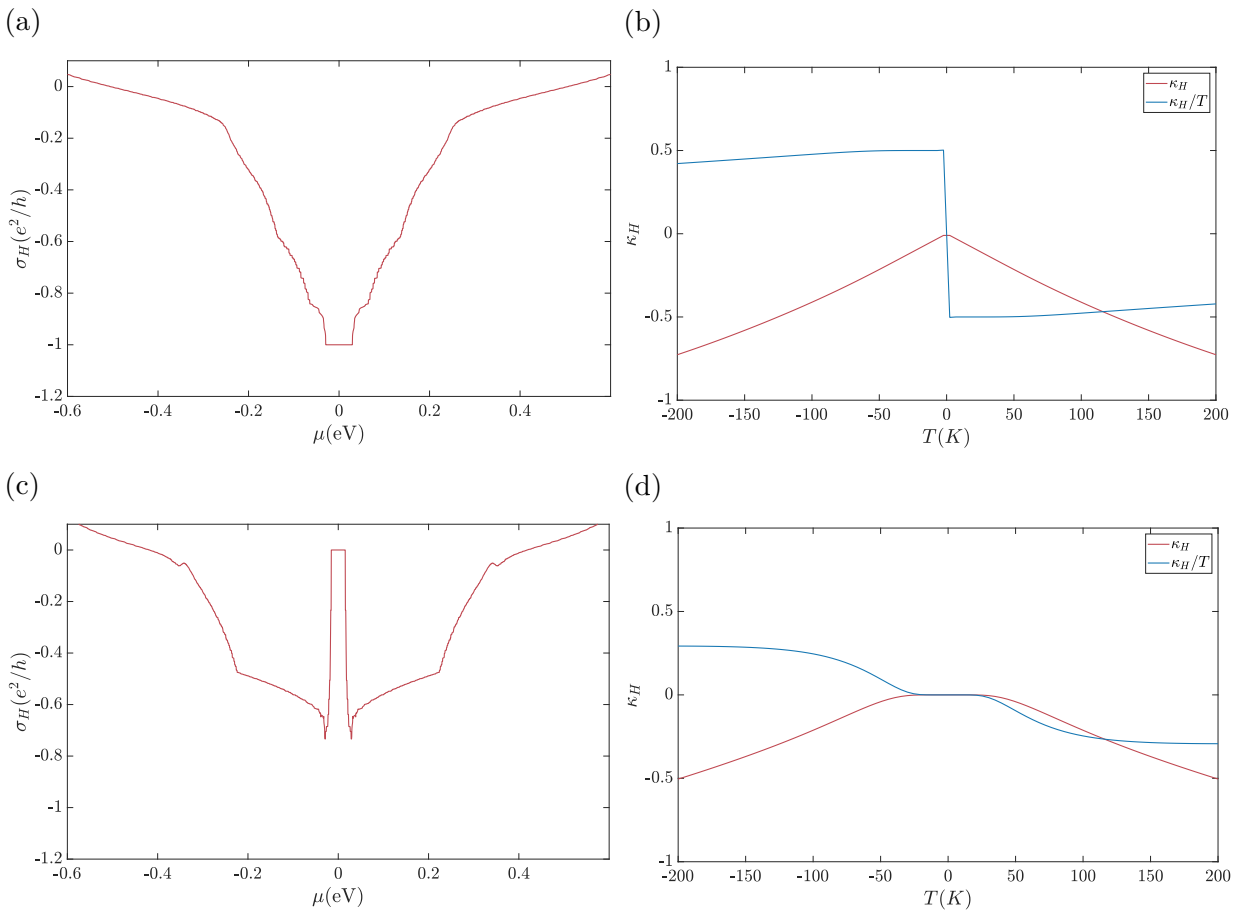


FIG. 2: (a) Fictitious quasi-particle anomalous Hall conductivity in topological superconductor with $\mu = 0.02$. (b) Corresponding thermal coefficient in (a). (c) Fictitious quasi-particle anomalous Hall conductivity in trivial superconductor with $\mu = 0.12$. (d) Corresponding thermal coefficient in (c).

Hall conductivity and thermal Hall conductivity of the system as shown in Fig. 2. In topologically non-trivial state with $\mu = 0.02$, we observe the non-zero Chern number plateau in the anomalous Hall conductivity, and the thermal Hall conductivity shows half-quantization plateau at low temperatures. On the other hand, in trivial state with $\mu = 0.12$,

the anomalous Hall conductivity is zero, and the thermal Hall conductivity is also zero at low temperatures.

This clearly illustrates the unique property of a chiral topological superconductor different from either the normal state or the trivial superconductor. In experiment, by measuring the thermal Hall conductivity while cooling the system down, one is expected to observe a non-half-quantize to half-quantize transition from room temperature to low temperatures, which is a signature of the topological superconductor phase transition.

B. TI Film

IV. Discussion and Conclusion

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